Trigonometry

MATH 123 Fall 2026

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Grand Valley State University

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Chapter 1

Course Syllabus

1.1 TL;DR

- ***This course is not graded the way you are used to!*** Read Assessment, particularly Introduction to Ungrading and Culminating Experience. Also check the Important Dates!
- Make sure to read Course Policies and Expectations, in particular pay attention to Getting More from Class.
- Review Classroom and Campus Safety and Safety Information Locations.

This document is intended to give you the basics of what you need to know to succeed in this course, including not only details of Content, and Course Policies and Expectations, but also the guiding principles behind how I approach teaching, and teaching this course in particular.

I reserve the right to adjust this document as needed and will notify you of any changes through Blackboard.

1.2 Instructor and Course Information

1.2.1 Prerequisites

MTH 122.

1.2.2 Meeting Times

Section 11	Tuesday and Thursday 4:00– 5:15, MAK D1-117
Section 14	Tuesday and Thursday 11:30–12:45, MAK D2-329
Individual Meetings	Scheduled Separately, MAK A2-110 or via Zoom

1.2.3 Contact Information

Table 1 Instructor and Department Information

Item	Value
Instructor	Dr. Clark Wells
Office Location	MAK Room A2-110 MAK
Office Phone Number	(616) 331-2444
Email	wellsc@gvsu.edu
Department Office Location	Room A2-178 MAK
Department Phone Number	(616) 331-2040
Math Tutoring Center	Room A2-601 MAK

Table 2 Student Hours

Day	Time	Location
Tuesday	2:00–2:50	MAK A2-110
Thursday	2:00–2:50	MAK A2-601 (Math Tutoring Center)

Student Hours are subject to change upon discussion with class members.

1.2.4 Withdrawal Date, Safety, and Accommodation

1.2.4.1 Withdrawal Date

The deadline for withdrawing with a 'W' grade is November 7 at 5:00 PM and is the responsibility of the student.

1.2.4.2 Accessibility, and Accommodation

If you need accommodations due to a learning, physical, or other disability you must present a memo to me from Student Accessibility Resources (SAR) indicating the existence of a disability and the suggested reasonable accommodations. If you have not already done so, please contact the Student Accessibility Resource (215 The Blue Connection), by calling 331-2490 or emailing access@gvsu.edu. Please note that I am unable to provide accommodations based upon disability until I have received a copy of the SAR issued memo. For more information visit gvsu.edu/accessibility.

Growth as a person does is certainly not limited to the classroom or academics. If there is an upcoming faith or religious observance that could have an impact on your attendance or work in the class, please submit a request during the first two weeks of the semester, or as soon as is reasonably possible, so that we can plan together accordingly.

1.2.4.3 Classroom and Campus Safety

I consider classroom and campus safety our shared responsibility. Please familiarize yourself with the classroom emergency placard, which contains important information that could prove useful should an emergency arise. Emergency situations could include a fire alarm, tornado warning, medical emergency, or hostile intruder, and our response might consist of evacuating the building in an orderly manner, moving as a group to a different location in the building, rendering assistance a class member, or staying put and locking the classroom door. In all cases, I will provide clear instructions as to what you should do.

Please make sure to have your primary mobile number listed in Banner so that you will automatically receive any campus alerts. Should you receive an emergency notification in class, notify me immediately.

You can find more information about campus safety can be found at the sites shown in Table 3.

Table 3 Safety Information Locations

Information	Resource Location
Directions for enabling additional contact means to the campus alert system	https://www.gvsu.edu/gvsualert
List of campus tornado safety locations	https://www.gvsu.edu/emergency/tornado-safety-location-2.htm
Active shooter response	https://www.gvsu.edu/dps/active-shooter-response-85.htm
Medical emergencies and campus AED locations	https://www.gvsu.edu/emergency/medical-emergency-15.htm

1.3 Introduction and Teaching Philosophy

1.3.1 Introduction to Math 123

1.3.1.1 Introduction

Trigonometry, Math 123 is designed to give you tools to succeed in various programs in science, such as health professions, geology, and biology.

1.3.1.2 Course Description

Math 123 is a study of oscillatory behavior, circular functions, and their applications. In order to understand these things, it is important to see where they arise. This means analyzing data or situations, choosing among appropriate

models, and solving applied problems. Of course, we can't predict every situation you will encounter in your future, so it is also important to develop flexibility and resilience.

More information can be found in Content and Objectives.

1.3.2 Teaching Philosophy

1.3.2.1 Learning is fun!

Learning is about feeding our curiosity, stretching ourselves to be better, finding out, and understanding. Those things are all rewarding, which is another word for *fun*. When learning isn't fun, there is something badly wrong.

You know what isn't fun? Grades. Getting them, assigning them, working for them — none of these things are fun. They are, in many ways, the *opposite* of fun.

A growing body of evidence shows essentially no correlation between grades and success in future endeavors. Anecdotal evidence and educational research support the idea that grades, in fact, create a *dis-incentive* for learning and create incentives for several negative behaviors, including engaging in academic dishonesty ("cheating"). Neuroscience research further shows that dopamine is critical for learning, except in the case of learning with punishment. In other words, *you need fun to learn*. See, for example, `elcenia.com/iamapirate/dopamine.pdf`

One thing we will do in this course is remove the pressure of the grade using a version of specifications grading that is sometimes called *ungrading*. More details are in Assessment.

1.3.2.2 Math is fun!

Many of you may have an experience of mathematics that does not suggest math is fun. Even if you know and appreciate the *usefulness* of math, the subject is often presented in a way that sucks all the fun out of it. Imagine sitting down to play a video game and finding that before you can actually start playing the game, you must first spend hours and hours memorizing cheat codes! "Now we need to master the *up, up, down, down, left, right, left, right, B, A, Start* sequence and its variations. This will make later games easier. Enter this sequence 35 times on the next screen."

This is nothing like what mathematics really is. Yes, sometimes memorizing techniques can make later things easier, but these techniques have deeper meanings and origins. They are not just a random sequence of buttons to push.

Instead, mathematics is about analyzing situations, carefully describing or quantifying them, and looking for (more or less universal) patterns. That may still sound dry, but my personal goal in this class, in addition to giving you useful techniques, is to help you to see the beauty and joy of mathematics.

1.4 Objectives and Content

1.4.1 Content

1.4.1.1 Topics

- Circular functions
- Vectors
- Complex numbers
- Polar Coordinates
- Graphing sinusoidal functions
- Inverse trigonometric functions
- Recognizing and applying trigonometric identities
- Solving equations involving trigonometric functions
- Solving triangles

1.4.1.2 Additional Content

In addition to developing mathematical tools, techniques, and skills, we will seek to further our understanding of the role and importance of mathematics as a not only a tool, but as a field of human endeavor. We will explore how various concepts have developed and evolved, and how various mathematical models are interrelated.

1.4.2 Objectives

The syllabus of record for MTH 123 states:

> After successful completion of the course, students will be able to
>
> 1. Explain the definition of the circular functions.
> 2. Compute the circular functions exactly for common functions.
> 3. Graph sinusoidal functions and determine their equations.
> 4. Explain the definition of the inverse trigonometric functions.
> 5. Perform computations of inverse trigonometric functions.
> 6. Apply trigonometric identities to computations of trigonometric functions.
> 7. Solve equations involving trigonometric functions.
> 8. Solve application problems using trigonometric functions.

Furthermore, Grand Valley State University's mission statement is: "At Grand Valley State University, we empower learners in their pursuits, professions, and purpose. The university enriches society through excellent teaching, active scholarship, advancement of equity, and public service."

This course is a step along the way to you achieving your own professional and personal goals. In light of this, some of my goals for this course are for you

- to develop the skill of thinking mathematically,
- to have a stronger idea of how trigonometry relates to future classes and to the physical world, and
- to have an idea of what mathematics actually is.

1.4.3 Materials and Resources

1.4.3.1 Primary Reference

The main text for this course will be Trigonometry by Ted Sundstrom and Steven Schlicker. This PDF is a free, open educational resource.

1.4.3.2 Additional Materials

We will also use Trigonometry by Katherine Yoshiwara. This is an online, free, open educational resource.

Any required supplementary material used will be put placed on BlackBoard, on reserve in the library, or otherwise made available to you.

1.5 Important Dates

Aug. 31	Classes Begin
Sep. 4	100% Tuition Refund Deadline (before 5:00pm)
Sep. 4	Last Day to Add, Register, or Pay (before 5:00pm)
Sep. 7	Labor Day (No Classes)
Sep. 25	Credit/No Credit Deadline (before 5:00pm)
Sep. 25	75% Tuition Refund Deadline (before 5:00pm)

Sep. 8–18	***Goal Setting Meeting***
Sep. 29	***Progress Check 1***
Oct. 12–16	Mid-term Evaluations
Oct. 13	***Progress Check 2***
Oct. 22	Mid-term Grades Available to Students
Oct. 25–27	Fall Break
Oct. 30–Nov. 12	***Progress Meeting***
Nov. 13	Withdraw with a 'W'Grade (before 5:00pm)
Nov. 25–29	Thanksgiving Day recess
Nov. 16–20	***Progress Check 3***
Dec. 12	Classes End and Commencement
Dec. 12–19	Examination Week
Dec. 14–18	***Final Meetings***
Dec. 19	Semester Ends
Dec. 22	Grades Due
Dec. 24	Grades Available to Students

Chapter 2

Course Documents

2.1 Assessment

2.1.1 Introduction to Ungrading

As mentioned earlier, substantial evidence shows that grades and grading are inherently antithetical to learning. *Ungrading* shifts the goal of a course from achievement to learning.

It is also important to note that the content of this course focuses on developing conceptual and procedural fluency with a set of concepts and tools as well as developing a "mathematical mindset". Thus, it makes sense to have learning targets that students need to satisfy in order to consider the course successful.

Ultimately, though, this is your course. It is you who are the learner, and so you are in the best position to determine what will be of the most interest, what will be the most engaging, and, beyond the basics, what will be of the most value to you. More than that, *you* are the best person to determine how much you have learned. Therefore, you will be an active participant in determining the direction of the course and in your evaluation.

To put things bluntly, I will determine, based on a set of learning targets and a body of evidence that you present to me, whether you have done the work necessary to pass the class. Beyond that, *you* will determine your own grade, based on the evidence you present that you have met your own goals, as outlined in Portfolio and Reader's Guide.

2.1.2 Portfolio and Reader's Guide

During the first week of class, we will discuss as a class what success in this class might look like, and then you will meet individually with me to develop a set of personalized goals. (See Important Dates.)

These goals will not about grades and grading. Instead, you are invited to consider your *learning*, and more generally what *success* in this course will be for you.

Over the course of the semester, you will assemble a comprehensive portfolio that demonstrates that you have met the basic requirements for the course and demonstrates your progress toward your goals. Part of this portfolio will be three *Progress Checks*.

In addition to the Progress Checks, you are expected to regularly submit items to me for focused feedback. Based on this feedback, you will reflect, rework, extend, and revise as necessary to meet basic requirements and show forward progress.

At least one time, mid-semester, you will schedule an individual meeting to discuss your progress, your successes, and any challenges you are facing.

For each individual meeting, you will prepare and share with me an agenda or guide ahead of time with specific items you wish to discuss. This guide should discuss successes, challenges, and progress, and should direct the reader where to find evidence of these items. For example, you might include something along the lines of "*Successes* Since our last meeting, I have reviewed and improved my understanding of difference quotients and average rate of change. Some good examples of my understanding and interpretation can be found in *Driving North/South* and *Filling a Tank*."[1]

Your guide should also address the following questions.

- What are you particularly proud of or excited about?
- What has been challenging?

[1]These titles are completely invented and may not be anything like the titles of activities we actually use.

- What do you plan to do next?, and
- What do you need from me (the instructor) to facilitate your progress toward your goals?

For your final portfolio conference, which will be during the week of final exams, you will submit your final portfolio (with a final version of your reader's guide), again citing items of interest. Then, based on your learning and progress toward your goals and your submitted evidence, you will tell me what grade you have earned for the course.

2.1.3 Required Elements of the Portfolio

The following items are required to be in your portfolio.

2.1.3.1 Reader's Guide and Progress Report

Depending on how you choose to organize your portfolio, this could simply be a table of contents plus a brief discussion of your progress.

For each submission of your portfolio, you will update your report to discuss your overall progress, to highlight progress since the last check-in, and to cite specific items of interest. This will also include a discussion of progress on you group project and of group dynamics.

The goal of this is to help the reader (your instructor) prepare for the meeting with you and to make items easy to find.

One way to think of this is to think of answering the questions,

- What are you particularly proud of or excited about?
- What has been challenging?
- What progress have you made on the group project?
- How has your group been sharing responsibilities?
- What do you plan to do next?, and
- What do you need from me, the instructor, to facilitate your progress toward your goals?

2.1.3.2 Progress Checks

Several times throughout the semester, we will have in-class progress checks. These are opportunities for you to show your understanding of concepts and fluency with calculations.

When your progress checks are returned to you, you are expected to put them in your portfolio and write a reflection on your performance, noting things that went well as well as areas of future growth.

If there were substantial errors or misconceptions, those should also be addressed in the reflection.

2.1.3.3 Catalog of Detailed Examples

Over the course of the semester we will be study functions and their derivatives from graphical, analytic, numerical, and conceptual viewpoints. For each topic, whenever possible, you must include at least one detailed example of the use of that concept in a practical setting. That is, each example must:

- Begin with a practical situation where the numbers have units and meaning.
- Include a discussion of why the particular concept or tool was used (possibly including why other approaches would not be appropriate).
- Show the process of using the concept to understand and solve the problem.
- Include multiple representations, such as graphical, numerical, and algebraic representations.
- Interpret the solution in context, giving a result with practical implications.

2.1.3.4 Overviews of Activities, Techniques, and Applications

Over the course of the semester you will be assigned multiple activities, investigations, and projects dealing with techniques, concepts, and applications of calculus.

For each *topic* we study, not each activity, you are expected to demonstrate understanding of the main mathematical ideas and an ability to apply any associated techniques.

Like everything in the portfolio, these are works in progress, and you are expected to revisit them in light of later activities or readings that may change your understanding of them.

2.1.3.5 Citation of Sources

If you make use of any outside resources for this class, you are required to cite those sources. The particular format of citations is not important. If you need specificity, please follow MLA standards.

2.1.4 Optional Elements of the Portfolio

You can (and are encouraged to) include any items in your portfolio that demonstrate progress toward your individual goals. The following items are some things that previous students have chosen to include.

2.1.4.1 Details of Activities and Investigations

In addition to the basics for an activity, you may choose to include more.

For instance, you might include work to show that you have:

- gained additional depth of understanding of important concepts
- understood the historical or philosophical importance of a topic (existence of solutions versus calculation of solutions), or
- made connections to other topics in this course, in other courses, your major, or to everyday life.

2.1.4.2 Experiments and Explorations

If there is a topic from class that you feel does not go far enough, you are free to investigate more deeply. For example, if you particularly enjoy global optimization, you might look at a more complex optimization, such as minimizing the construction cost for an oil barrel.

2.1.4.3 Summaries of Outside Reading

If you use outside sources to guide your learning, you may want to include more than just citations. You can summarize what you learned from those sources and include the summaries.

2.1.4.4 Other Items

Other items may be included in your Portfolio if you they demonstrate your progress in meeting your course goals. Some examples of what you may choose to include are practice problems, conjectures, or expository papers.

2.1.5 Professionalism

For this course, I define professionalism to mean respect for yourself, your classmates, and your instructor. This includes (but is not limited to)

- consistent forward progress and timely completion of assignments,
- prompt scheduling of and on-time arrival for individual meetings,
- preparation for class, and
- attendance and participation in class.

A pattern of unprofessional behavior during the semester may, at my discretion, result in a course grade penalty of up to two full letter grades (20%).

2.1.6 Culminating Experience

There will be no final exam in this course. Instead, your individual conference with me during the final period will serve as the culminating experience in this course.

2.2 Course Policies and Expectations

This course is subject to the GVSU policies listed at http://www.gvsu.edu/coursepolicies/. The following policies also apply.

2.2.1 Use of Generative AI

Use of Generative AI Large Language Models [LLMs] (such as ChatGPT, Gemini, Copilot, etc.) in this class is subject to a few conditions, namely Awareness, Transparency, and Accountability. Transparency and accountability are fairly straight-forward, and are closely related to Academic Integrity, but awareness is both more subtle and arguably more important.

2.2.1.1 Awareness

If you plan to use generative AI in this class, be aware of what you are using it for and whether it helps you toward your goals or just wastes your time. I find a useful analogy to be the idea of ***taking a forklift to the gym***.

If your goal in going to the gym is to strengthen your muscles by lifting weights, using a forklift to lift the weights defeats the purpose. Sure, you can get in your reps, you can "lift" more than you could without the forklift, and you can do more reps and not get tired, but it doesn't work toward your goal of strengthening your muscles. It is a waste of your time. Worse, it gets in other people's way as they try to reach their strength training goals.

On the other hand, maybe you went to the gym because you work there and the boss wants you to move several large, heavy machines from one part of the building to another. If that's the case, using the forklift makes a lot of sense, because it is in fact helping with your goal of moving the machines.

Education is about strengthening your mental muscles. Be aware of how the use of heavy machinery either helps or hinders your progress.

Awareness also includes awareness of the ethical and practical pitfalls involved in using generative AI. You should be aware that

- AI is trained on the works of other people
- Plausibility, not objective truth, is the training goal of AI. It often makes things up that sound plausible.
- AI is resource intensive. For example, according to what-uses-more.com, a paragraph of AI generated text uses more than 23 times as much electricity and water than a Google search without AI.

2.2.1.2 Transparency

When you use generative AI, be transparent: make note of exactly what you asked the AI to do (your prompts), and exactly what part of the AI results you used in your materials. For example, you might include something like the following in your portfolio.

> "In this problem, we were asked to model the number of hours of daylight as a function of the date. I knew this would be either sine or cosine, but I still didn't understand how to make the right days line up with the peaks and valleys, so I asked Copilot to explain. My prompt was 'Explain how to shift a sine function so that the peaks and valleys line up with data.'"
>
> "Copilot started by giving a formula for a sine function as $y = A\sin\big(B(x - C)\big) + D$ and then gave step-by-step alignment guide. The important part for me was how to adjust the 'phase shift', C. It said to 'Find where a peak (or valley) occurs in your data, and shift the sine wave so its peak matches that point: $C = x$-value of desired peak.' I was then able to align the peak amount of sunlight with June 21 by using $C = 172$ days since the beginning of the year."

2.2.1.3 Accountability

When you use generative AI, you are accountable. What that means is that you must be able to explain all of your work and understand the power and the limitations of using LLMs.

Using the example of fitting a sine wave, a student submitting this should have noticed that ***Copilot got it wrong***. Setting $C = 172$ places the *midline* of the sine function at $x = 172$ days. Using $C = 172$ in the formula $y = A\cos\big(B(x - C)\big) + D$ would place the peak at $x = 172$, or you could find the date of the spring equinox and set C to that value in $y = A\sin\big(B(x-C)\big)+D$.

2.2.2 Initial Drafts of Work in Timed Settings

Initial drafts of work done in class (a setting with limited time) are not expected to be typeset. Instead, high resolution images or digital scans of your initial work can be included in your portfolio.

Reflections on your work, revisions, and corrections, though, are expected to follow the guidelines given for work in un-timed settings.

2.2.3 Submitting Un-Timed Work for Feedback

2.2.3.1 Format of Submissions

All drafts of work done in settings without prescribed time constraints, such as exercises, proofs, expository writing, and so forth,

- must be written using the software capable of high-quality mathematical typesetting or neatly hand-written and scanned;
- must be spell-checked and proofread for grammatical correctness;
- must be submitted electronically, via BlackBoard or shared link; and
- must be available in a form suitable for commenting and editing.

See Mathematical Typesetting, for more information about options for mathematical typesetting software.

2.2.3.2 FERPA and Email

Please note that due to security issues and based on guidance from the Department of Education regarding FERPA, the Family Educational Rights and Privacy Act of 1974, I cannot use e-mail or any unsecured platform to discuss grades.

2.2.4 Time Management

One of my goals for this course is to challenge you to grow intellectually while giving you the tools and environment that will allow you to succeed. If you at any time feel that you are not being challenged to achieve your fullest potential in understanding the concepts we address, or if you have comments about the pace of the class or any other concerns, I encourage you to discuss your concerns with me during my student hours or an appointment with me.

As a university mathematics course, you should expect the course to be challenging. While we will all work to make time spent in class as productive as possible, it is not possible to learn everything using class time alone: much of your learning and understanding will occur as you reflect on what we have done in class, work on problems, and read and discuss course material outside of class.

The most important rule for this course is *do not get behind in your work.* Using a portfolio removes the inherent penalties in a slow start, but the trade off is that it can be tempting to put things off until later. Always stay up to date with the material in this course. Make succeeding in this course a priority!

2.2.5 Getting More from Class

Learning cannot take place solely in the classroom, and you should expect to put in considerable work outside of class. Different students learn in different ways and require different types of support. I am committed to providing you with the support that you require, whether that means being a critical questioner of your processes and results, providing guidance and mentoring, or something else entirely.

This section has some advice and perspectives that I hope will help you make the most of your time and effort in this class.

2.2.5.1 Attend Student Hours

First, never hesitate to come to me during student hours or to make an appointment to see me. My primary job is to facilitate your understanding and learning, so when you come to me you are allowing me to do my job well.

2.2.5.2 Form a Study Group

You may also find that working with a group of classmates is beneficial. I encourage you to form groups to work together outside of class. Working with a group also allows you to see things from different perspectives.

If you or your study group have questions, please come to me and we will work through them together. If my student hours are at times that you cannot make due to other commitments, make an appointment to see me at another time.

2.2.5.3 Make Attendance a Priority

As discussed in Assessment, attendance in class is an important part of this course. You are expected to be present and on time each day we meet. You are responsible for announcements made in class concerning material covered, assignments, changes in the syllabus or due dates, or anything else pertinent to the course. Furthermore, many of the activities from class and reflections on class will become part of your portfolio.

Nevertheless, your health and safety and that of your classmates is important. Therefore *if you are ill, stay at home.* In the event of a University recognized absence, arrangements can be made for you to mitigate the effects of missing class.

2.2.5.4 Come to Class Prepared

It is imperative that you work on a consistent basis. This applies to both the day-to-day work to prepare for class as well as the more long-term work such as the portfolio.

You should keep a well-organized record of your study notes, completed problems, and problems in progress for future reference. Note that these are acceptable as part of your portfolio if you believe they help make your case!

You must understand that a great deal of your learning in this course must occur on your own. It is your responsibility to complete assigned reading, read assignments carefully, complete the problems, be prepared for class, and to seek support as needed.

2.2.5.5 Participate in Class

For many people, their experience has taught them that asking questions or questioning results is frowned on as either disrespectful or inattentive. In this class, I want to cultivate an atmosphere where questioning processes, results, and reasoning is seen as showing respect that the other person's ideas are worth thinking about. This type of respectful questioning (including questioning your own processes, thoughts, and results) is crucial to becoming an effective and flexible thinker.

Asking and answering questions are not the only ways, or even the best ways, to participate in class. While we may think of class participation in terms of speaking up in class, participation can take many forms, such as active listening in groups, writing up or journaling individual or group assignments, drawing graphs or creating visual representations, sharing strategies, questioning results, and many others. In every class meeting, there will be ample opportunity for you to actively participate.

2.2.5.6 Make Full Use of Additional Learning Resources

The idea of "asking for help" can sometimes take on a negative connotation. I prefer to think of it as one way to make use of all of your available resources. If you need to travel from the Allendale campus to the Grand Rapids, you probably don't think that taking the Laker Line instead driving a car is a bad thing, it's simply another effective way to travel. And there are also lots of reasons why some students might not have access to a car, including having a flat tire or not having a license. In a similar way, there are lots of reasons to seek out additional learning resources.

In addition to student hours and individual meetings with me, GVSU provides a variety of tutoring resources, including the Math Center (located in Mackinac Hall). More information about other tutoring services can be found on the GVSU Tutoring website.

Using these resources is *smart*, and can be beneficial to every student.

2.2.6 Academic Integrity

2.2.6.1 Integrity of Scholarship and Grades

Truth and honesty: The principles of truth and honesty are recognized as fundamental to a community of teachers and scholars. The university expects that both faculty members and students will honor these principles and in so doing protect the validity of university grades. This means that all academic work will be done by the student to whom it is assigned without unauthorized aid of any kind. Instructors, for their part, will exercise care in the planning and supervision of academic work, so that honest effort will be positively encouraged. Compliance shall include compliance with the following specific rules:

1. No student shall knowingly, without authorization, procure, provide, or accept any materials which contain questions or answers to any examination or assignment.
2. No student shall, without authorization, complete, in part or in total, any examination or assignment for another person.
3. No student shall, without authorization, allow any examination or assignment to be completed, in part or in total, by another person.
4. No student shall knowingly plagiarize or copy the work of another person and submit it as his or her own.
5. No student shall submit work that has been previously graded or is being submitted concurrently to more than one course without authorization from the instructor(s) of the class(es) to which the student wishes to submit it.

2.2.6.2 Plagiarism

Any ideas or material taken from another source for either written or oral presentation must be fully acknowledged. Offering the work of someone else as one's own is plagiarism. The language or ideas taken from another may range from isolated formulas, sentences, or paragraphs to entire articles copied from books, periodicals, speeches, or the writing of other students. The offering of materials assembled or collected by others in the form of projects or collections without acknowledgment also is considered plagiarism. Any student who fails to give credit in written or oral work for the ideas or materials that have been taken from another is guilty of plagiarism.

Such activity may result in failure of a specific assignment, an entire course, or, if flagrant, dismissal from Grand Valley. For further information see Student Rights and Responsibilities.

2.2.7 Communication and Technology Use

2.2.7.1 Technology Requirements for Online/Hybrid Courses

Although this class face-to-face, there is always the possibility that we may have to move online (synchronous). Thus, GVSU's Technology Requirements for Online/Hybrid Courses apply.

2.2.7.2 BlackBoard

Materials and information for this course will be posted to the course BlackBoard page. If you do not have access to the student network or need help in using BlackBoard, please let me know as soon as possible.

2.2.7.3 E-mail

Please check that the e-mail associated with your BlackBoard account is one that you check, as I will use that e-mail to send messages to the entire class.

2.2.7.4 WeBWorK

Practice problems will be available to you via the WeBWorK homework system. You are encouraged to make use of WeBWorK to practice and check your understanding. Although the feedback from WeBWorK is not as detailed as that from the instructor, it has the advantage of being essentially instantaneous!

2.2.7.5 Computer Algebra Systems

We will often make use of Desmos Graphing Calculator and Amplify.

We will also sometimes use iPython interactive notebooks (such as Google Colaboratory). It is an interface to the python programming language, but ***you are not expected to learn to program***.

2.2.7.6 Mathematical Typesetting

All material submitted for your portfolio is expected to be written using using the software capable of high-quality mathematical typesetting.

Options for mathematical typesetting include markdown notebooks, such as Google Colab or Jupyter Notebooks, or Google Docs.

2.2.7.7 Recording of Classes

In the event that we meet virtually via Zoom, class meetings will be recorded and will be available to students enrolled in this course as a study aid. Students who do not want their likeness during class participation included in the recorded class session may elect to turn off the camera and change their name in the Zoom window. Recordings will be available for viewing subject to the following:

- Only students enrolled in the class may view the recording.
- Students may not post or use the recordings in any other setting (e.g., social media) for any purpose. Students who violate this will be subject to student discipline, up to and including expulsion.
- If faculty have any plan to use the recording for a different class in the future, the faculty member will need written FERPA consent from those students in the current class who are identifiable in any of the recordings. A FERPA consent form signed by all students in the course will also be needed if the recordings are made available to others beyond the classroom on a non-secure digital platform.

Chapter 3

Course Notes

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

3.1 Week 1

3.1.1 Tuesday, Sept. 1

Introduction, highlights from the syllabus, and an introduction to vectors using "Turtle Graphics" in a Google Colaboratory Worksheet.

3.1.1.1 Main Ideas

The main take-away ideas from the introduction to vectors were:

- Vectors describe quantities that have not only a **magnitude**, or size, but also a **direction**.
- Vectors are often represented graphically using arrows, with a **tail**, where the vector starts, and a **head**, where the vector ends.
- A vector can be interpreted as a change, and graphically, only the direction and size, not the specific location, of a vector is important. So, for example, an arrow from $(0, 0)$ to $(30, -60)$ and an arrow from $(-10, 30)$ to $(20, -30)$ are different representation of the exact same vector, because in each case the x-coordinate changes by $+30$ and the y-coordinate changes by -60.
- Two ways to represent vectors algebraically are using **angle brackets**, such as $\langle 30, -60 \rangle$, with the **elements** of the vector listed horizontally, and using **square brackets**, such as $\begin{bmatrix} 30 \\ -60 \end{bmatrix}$, with the elements listed vertically.
- Because the angle bracket and square bracket representations of vectors take a lot of space, we will often "name" vectors by assigning them to a variable, usually written either as a bold letter or an italic letter with an arrow over it. So, we might say something like "let $\vec{v} = \langle 30, -60 \rangle$ and let $\vec{w} = \langle 20, 20 \rangle$."
- Vectors can be combined by *adding* them. Writing $\vec{u} = \vec{v} + \vec{w}$ indicates that if we add $\vec{v}$ and $\vec{w}$ the **resultant vector** is $\vec{u}$. Graphically, to find $\vec{u} = \vec{v} + \vec{w}$ when we are given $\vec{v}$ and $\vec{w}$, we place the tail of $\vec{w}$ at the head of $\vec{v}$, and $\vec{u}$ is the vector from the tail of $\vec{v}$ to the head of $\vec{w}$. (See example below.)

Example 4 Addition of Vectors. Let $\vec{v} = \langle 30, -60 \rangle$ and let $\vec{w} = \langle 20, 20 \rangle$. Let $\vec{u} = \vec{v} + \vec{w}$. We will represent the addition of $\vec{v}$ and $\vec{w}$ graphically and algebraically.

To add $\vec{v}$ and $\vec{w}$ graphically, since we have no information about where the tail of $\vec{v}$ should go, we will draw $\vec{v}$ with its tail at the origin (and so its head is at $(30, -60)$). Then, we draw $\vec{w}$ with its tail at the head of $\vec{v}$. Then, the vector $\vec{u} = \vec{v} + \vec{w}$ will have its tail at the tail of $\vec{v}$ and its head at the head of $\vec{w}$. So, for this example, the tail of $\vec{u}$ is at the origin and the head of $\vec{u}$ is at $(50, -40)$. ***Note:*** We have included a grid and tick marks on the graph, but they are not necessary. You can add vectors graphically without knowing their algebraic values!

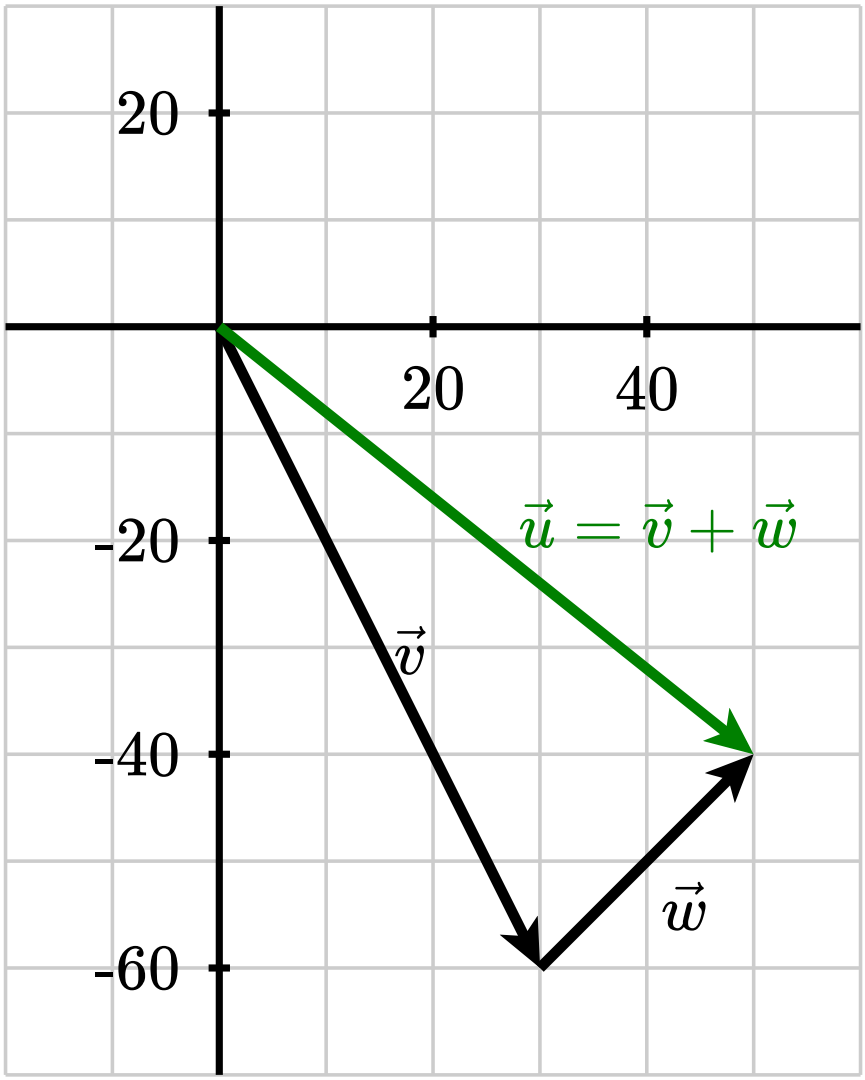


Figure 5 Graphical Addition of Vectors

To add $\vec{v}$ and $\vec{w}$ algebraically, we add *element-wise*, which means we add according to position. We will show this using both angle brackets and square brackets.

$$\begin{aligned}\vec{u} &= \vec{v} + \vec{w} \\ &= \langle 30, -60 \rangle + \langle 20, 20 \rangle \\ &= \langle 30 + 20, -60 + 20 \rangle \\ &= \langle 50, -40 \rangle\end{aligned}$$

$$\begin{aligned}\vec{u} &= \vec{v} + \vec{w} \\ &= \begin{bmatrix} 30 \\ -60 \end{bmatrix} + \begin{bmatrix} 20 \\ 20 \end{bmatrix} \\ &= \begin{bmatrix} 30 + 20 \\ -60 + 20 \end{bmatrix} \\ &= \begin{bmatrix} 50 \\ -40 \end{bmatrix}\end{aligned}$$

▲

Example 6 Graphical Addition of Vectors Without Coordinates. In Figure 5, we show how to add two vectors graphically without knowing the algebraic representations.

To add $\vec{v}$ and $\vec{w}$ graphically, we draw $\vec{w}$ with its tail at the head of $\vec{v}$. Then, the vector $\vec{u} = \vec{v} + \vec{w}$ will have its tail at the tail of $\vec{v}$ and its head at the head of $\vec{w}$.

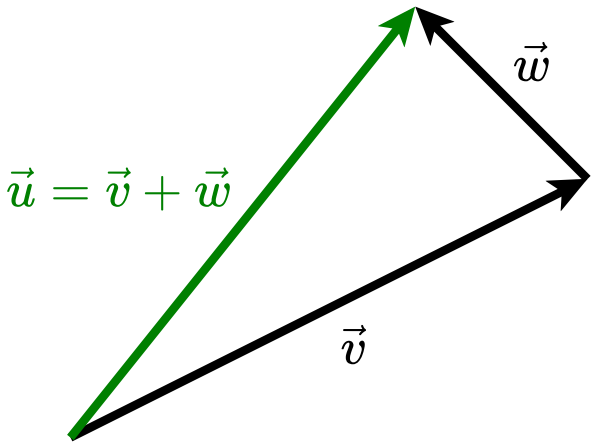


Figure 7 Graphical Addition of Vectors Without Coordinates

To add $\vec{v}$ and $\vec{w}$ algebraically, we add *element-wise*, which means we add according to position. We will show this using both angle brackets and square brackets.

$$\begin{aligned}\vec{u} &= \vec{v} + \vec{w} \\ &= \langle 30, -60 \rangle + \langle 20, 20 \rangle \\ &= \langle 30 + 20, -60 + 20 \rangle \\ &= \langle 50, -40 \rangle\end{aligned}$$

$$\begin{aligned}\vec{u} &= \vec{v} + \vec{w} \\ &= \begin{bmatrix} 30 \\ -60 \end{bmatrix} + \begin{bmatrix} 20 \\ 20 \end{bmatrix} \\ &= \begin{bmatrix} 30 + 20 \\ -60 + 20 \end{bmatrix} \\ &= \begin{bmatrix} 50 \\ -40 \end{bmatrix}\end{aligned}$$

▲

3.1.1.2 Extensions

While we have looked at vectors in two dimensions, in the xy-plane, there are also three dimensional vectors that behave in pretty much the exact same way! Depending on how the semester goes, we may briefly take a look at three-dimensional vectors.

There are lots of things that can be treated as vectors. Mathematicians define what it means to be a vector not so much in terms of what something *is*, but more in terms of how it *behaves*, but we won't get that in this class.

Another thing to be aware of is that sometimes we want to graphically add vectors that are not drawn tail-to-head. For example, we might wish to add two vectors that are currently drawn with their tails in the same place. (This is common for

forces acting on a body in physics applications.)

Example 8 Two Forces Acting on a Body.. Figure 9 shows two forces, represented by $\vec{F}_1$ and $\vec{F}_2$ acting on a body (represented as a rectangle).

To add the vectors, we duplicate $\vec{F}_2$ with its tail at the head of $\vec{F}_1$ and we duplicate $\vec{F}_1$ with its tail at the head of $\vec{F}_2$, as shown by the dashed vectors. While we really only need to duplicate one of these, duplicating both allows us to see that the resultant vector, $\vec{F}_r = \vec{F}_1 + \vec{F}_2$, lies along the diagonal of a parallelogram.

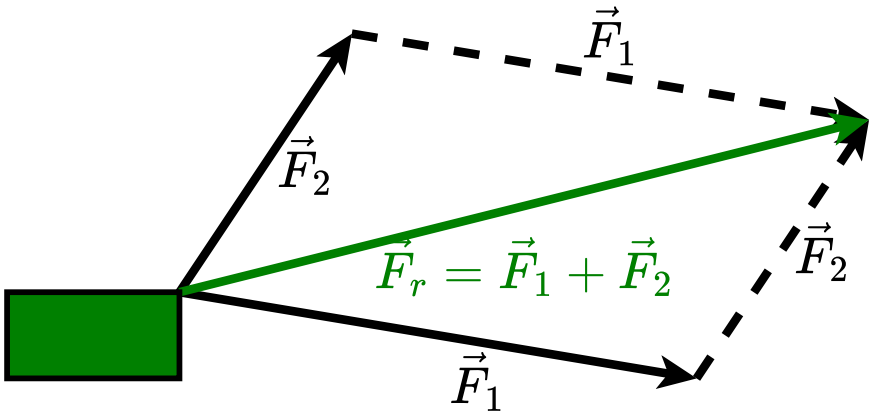


Figure 9 Showing $\vec{F}_r = \vec{F}_1 + \vec{F}_2$ where the tails of $\vec{F}_1$ and $\vec{F}_2$ coincide.

▲

3.1.2 Thursday, Sept. 3

Discussion of measures of success, very brief launch of ideas that lead to circular functions.

We will re-launch "Ferris Wheels and Circular Functions" on Tuesday.

Chapter 4

In-Class Activities

4.1 Representing Points and Motion

4.1.1 A Knight's Moves

In chess, the knight is one of the most misunderstood, under-utilized, and powerful pieces on the board. All of these traits are due to the way the knight moves. Two possible moves for a knight are shown in Figure 10. If we place "Start" at $(0, 0)$, the origin of a coordinate system, then *Move* 1 is associated with the point $(1, 2)$ and *Move* 2 is associated with the point $(2, 1)$.

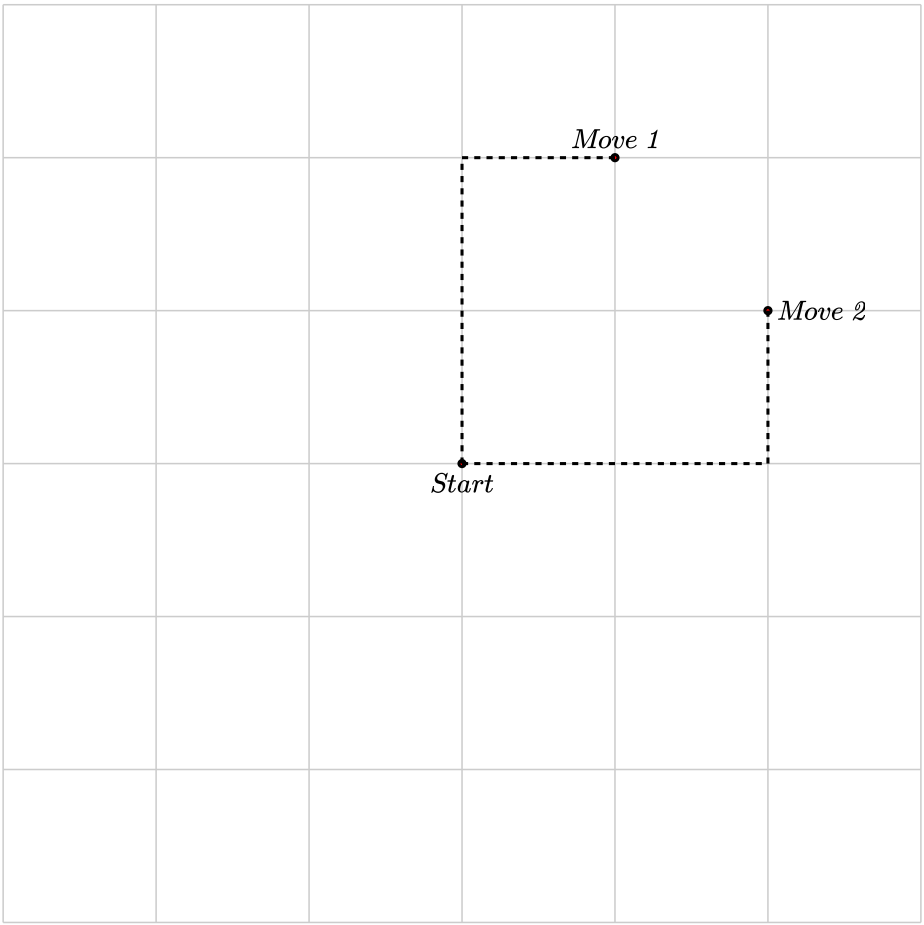


Figure 10 Two of a knight's possible moves in chess.

4.1.2 Vectors

Because *Move* 1 and *Move* 2 are motions, it is probably more correct to say that they are associated with moving *from* $(0,0)$ *to* $(1,2)$ and *from* $(0,0)$ *to* $(2,1)$, respectively.

Using *vectors*, we can represent *Move* 1 by the vector $\vec{k}_1 = \begin{bmatrix} +1 \\ +2 \end{bmatrix}$ and *Move* 2 by $\vec{k}_2 = \begin{bmatrix} +2 \\ +1 \end{bmatrix}$. (We generally don't bother writing "+", though. Instead, a positive value is a motion right or up, and a negative value is left or down.)

However, we don't really have an easy way to represent performing *Move* 1 twice, or *Move* 1 followed by *Move* 2, and so forth. What we can do, though is think of vectors as *relative* position or *motion*. Thus, we can think of the vector $\vec{k}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ representing a move 1 unit to the right and 2 units up *from any point*, and similarly, the vector $\vec{k}_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ representing a move 2 units to the right and 1 unit up from any point.

Now, we can think about combining the vectors. Starting from *Start*, $(0,0)$, we can first perform *Move* 1, $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, arriving at $(1,2)$, and then perform *Move* 2, $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, arriving at $(3,3)$. We write this combination of moves as

$$\begin{aligned} \vec{k}_1 + \vec{k}_2 &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1+2 \\ 2+1 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 3 \end{bmatrix} \end{aligned}$$

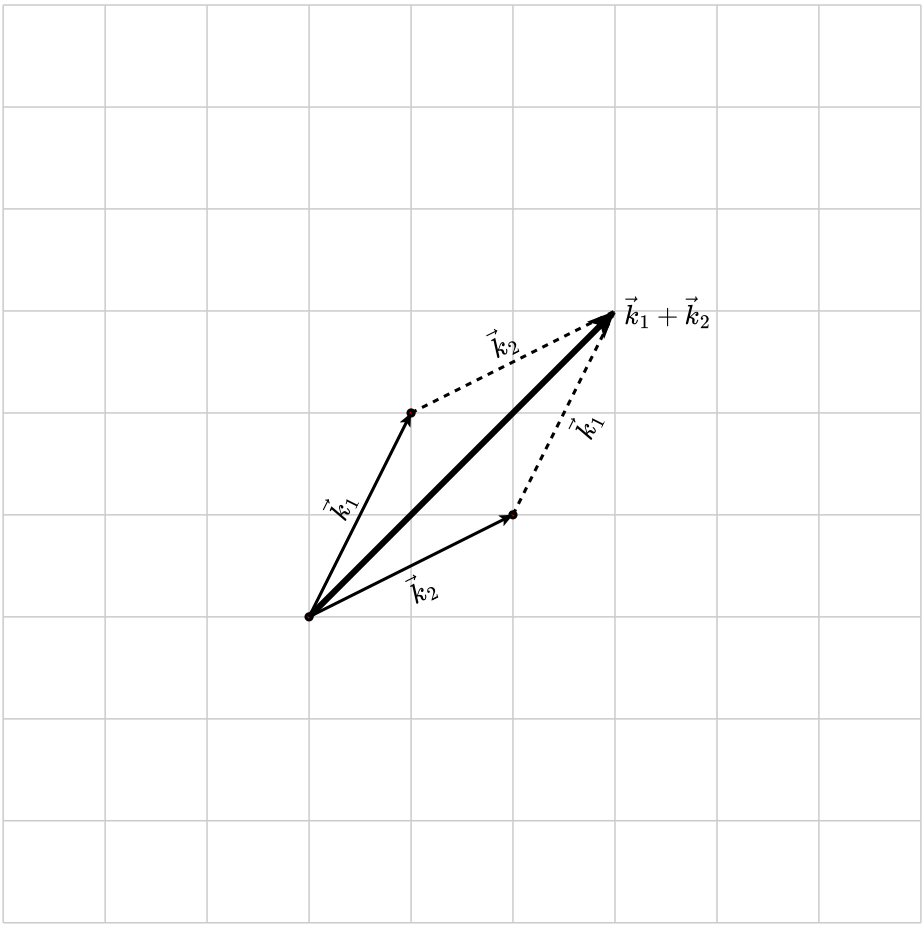


Figure 11 Adding Vectors Graphically.

Graphically, we generally represent vectors as arrows from their starting point, called the *tail*, to their ending point, called the *head*. Since vectors represent *relative* motion, the vector from $(0, 0)$ to $(1, 2)$ and the vector from $(2, 1)$ to $(3, 3)$ are *exactly the same vector*. Because of this, we can represent adding vectors by placing them head-to-tail, as shown in Figure 11 where the *resultant* vector of the addition $\vec{k}_1 + \vec{k}_2$ is shown with a thick arrow.

1. How do you think you would represent a motion two units right and one unit down, graphically and symbolically?
2. How do you think you would represent a motion two units left and one unit up, graphically and symbolically?
3. What vector is the result of performing *Move* 1 twice?
4. What do you think the notation $2\vec{k}_1$ means?

4.1.3 Combining Moves

An interesting (and rather difficult) set of mathematical problems arise from considering which squares on different sized chess boards can be reached by a cohort of several knights. In this activity we will consider a slightly different but related idea: how can we reach various squares using linear combinations of $\vec{k}_1$ and $\vec{k}_2$?

1. We know that $\vec{k}_1 + \vec{k}_2$ allows you to reach $(3, 3)$. Will any other combination of $\vec{k}_1$ and $\vec{k}_2$ allow you to reach the same point? Explain.
2. Is it possible to reach $(-1, 4)$ using some combination of $\vec{k}_1$ and $\vec{k}_2$? How (or why not)?
3. Is it possible to reach $(-1, -1)$? How (or why not)?
4. What do you think the notation $-1\vec{k}_1$ means?

5. What do you think the notation $-2\vec{k}_2$ means?
6. What do you think the notation $\vec{k}_1 - 2\vec{k}_2$ means?
7. Would your responses to any of the above questions change in light of your answers to Item 4, Item 5, and Item 6?
8. Would your responses to any of the above questions change if you could Move a fractional part of $\vec{k}_1$ and $\vec{k}_2$? (Say $\frac{1}{3}\vec{k}_1$, for example.)
9. If any real number multiples of $\vec{k}_1$ and $\vec{k}_2$ are allowed, can you reach the point $(1, 0)$? Explain.
10. If any real number multiples of $\vec{k}_1$ and $\vec{k}_2$ are allowed, can you reach the point $(0, 1)$? Explain.
11. If any real number multiples of $\vec{k}_1$ and $\vec{k}_2$ are allowed, which squares can you reach? Explain.
12. Now let's do away with the idea that we have to land on a corner of a square and think about points in the plane more generally. How could you reach the point $\left(\frac{1}{2}, \frac{3}{4}\right)$ using a combination of $\vec{k}_1$ and $\vec{k}_2$?
13. More generally, what is the process to reach a given a point (a, b) in the plane using combinations of $\vec{k}_1$ and $\vec{k}_2$?

4.2 Ferris Wheels, Angles, and Distances

4.2.1 Height of a Ferris Wheel Car

Figure 12 shows the height of a Ferris wheel car as a function of the distance the car has travelled from the bottom. A full animation is available from the National Council of Teachers of Mathematics [NCTM] *Illuminations* classroom resources, Investigating Functions with a Ferris Wheel: Distance vs. Height (gvsu.edu/s/3MV).

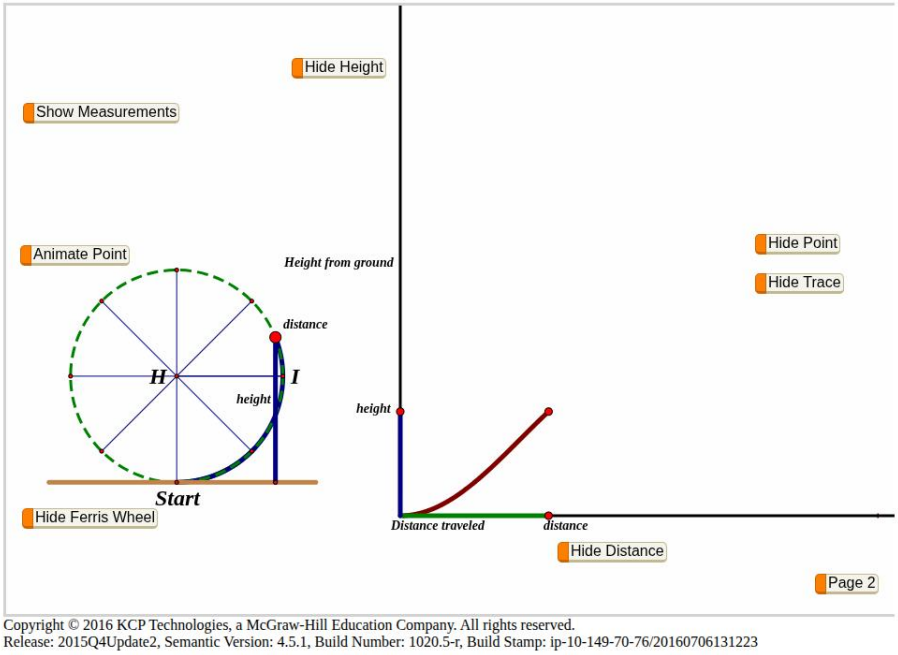


Figure 12 Height of a Ferris Wheel Car

1. For the calculations in this exercise, assume that the circumference of the Ferris wheel is 128m, and recall that the circumference, C, of a circle is given by $C = 2\pi r$, where r is the radius of the circle.

We will refer to the height as h and the distance travelled as d.

(a) Find the total height of the Ferris wheel (in meters).

Answer. 40.74367m

Solution. The total height is twice the radius. Since the radius is $\frac{128\text{m}}{2\pi}$, the total height is $\frac{128\text{m}}{\pi} \approx 40.74367\text{m}$.

(b) When $d = 32\text{m}$, find the value of h. That is, when the car when has travelled 32m, how high is it?

Answer. 20.37183m

Solution. When the car has travelled 32m, it has travelled one quarter of the way around, or half-way to the top. Thus its height is half the total height, approximately 20.37183m.

(c) Find h when $d = 64\text{m}$.

Answer. 40.74367m

Solution. When the car has travelled 64m, it has travelled half of the way around, so it is at the top. Thus its height is the same as the total height, approximately 40.7467m.

(d) How much distance along the circle has the cab traversed at the moment it first reaches a height of $\frac{128\text{m}}{2\pi} \approx 20.37183\text{m}$?

Answer. 32m.

Solution. We know that $\frac{128\text{m}}{2\pi}$ is the radius, which is also half the diameter, and so the height is half of the total height. The first time the car reaches this height is when it has travelled $\frac{1}{4}$ of the way around, which is $\frac{128\text{m}}{4} = 32\text{m}$.

(e) How much distance along the circle has the cab traversed at the moment it reaches a height of $\frac{128\text{m}}{2\pi} \approx 20.37183\text{m}$ for the *second* time?

Answer. 96m.

Solution. The second time the car reaches a height of is when it has travelled $\frac{3}{4}$ of the way around, which is $\frac{3}{4}128\text{m} = 96\text{m}$.

(f) Can we express h as a function of d?

Answer. Yes.

Solution. Every distance travelled corresponds to exactly one position on the circular path of the Ferris wheel car, and each position has a particular height. Since there is always exactly one value of h for each value of d, we can express h as a function of d.

(g) Can we express d as a function of h?

Answer. No.

Solution. As the Ferris wheel car travels around its circular path, it passes through each height multiple times. Since there is *not* always exactly one value of d for each value of h, we cannot express d as a function of h.

An important idea in setting up mathematical models that is often assumed or overlooked is the fact that ***you get to choose where you measure from***. Sometimes the choice of your "referece point" is the difference between straight-forward calculations and much more complicated ones. With that in mind, we will revisit the Ferris wheel problem, but with a couple of slight twists.

First, instead of measuring height above the ground, we will measure the height, h, as the vertical distance from a horizontal line ***through the center*** of the Ferris wheel. Of course, we will need to indicate whether a given height is *above* the center or *below* the center, so we will use *positive* values of h to indicate distance *above* the center line, and *negative* values to indicate distance *below* the center line.

Second, we will measure the distance the Ferris wheel car has travelled relative to the point where it is farthest to the right. In other words, when it is on the center line.

Here, distances that are *before* reaching the line will be negative, and distances after will be positive.

2. For this problem, we will consider a Ferris wheel whose total height is 40m.

Recall that we are measuring distance travelled, d, relative to the point on the circle to the extreme right, and we are measuring the height, h, relative to the center of the Ferris wheel, so, for example, when $d = 0\text{m}$, $h = 0\text{m}$.

(a) How high is the center of the Ferris wheel above the ground?

Answer. 20m.

Solution. The total height is 40m, so the center is half that distance above the ground, which is 20m.

(b) What is the circumference of the Ferris wheel?

Answer. $\approx 125.6637\text{m}$.

Solution. We will call the radius of the Ferris wheel r and the circumference C. Since the total height is twice the radius, we have $r = 20\text{m}$, and thus $C = 2\pi r = 2\pi \cdot 20\text{m} \approx 125.6637\text{m}$.

(c) What is the value of h when 31.4159m?

Answer. 20m.

Solution. When the car has travelled $31.4159\text{m} = \frac{1}{4} \cdot 125.6637\text{m}$, it has travelled $\frac{1}{4}$ of the circumference. Because we are measuring d from the point on the extreme right, this means that the Ferris wheel car is at its highest point. Since we are measuring h from the center of the Ferris wheel, that means that $h = 20\text{m}$.

(d) What is the value of h when $d = 62.8315\text{m}$?

Answer. 0m.

Solution. When the car has travelled $62.8315\text{m} = \frac{1}{2} \cdot 125.6637\text{m}$, it has travelled $\frac{1}{2}$ of the circumference. Because we are measuring d from the point on the extreme right, this means that the Ferris wheel car is

at the extreme left. Since we are measuring h from the center of the Ferris wheel and we are back at the original height, that means that $h = 0$m.

(e) What is the value of h when $d = 94.2478$m?

Answer. -20m.

Solution. When the car has travelled $94.2478\text{m} = \frac{3}{4} \cdot 125.6637\text{m}$, it has travelled $\frac{3}{4}$ of the circumference. Because we are measuring d from the point on the extreme right, this means that the Ferris wheel car is at the bottom of its path, so it is 20m below the center, and therefore $h = -20$m.

4.2.2 Distance and Angles

In Worksheet Exercise 4.2.1.2, we know that the Ferris wheel is 40m tall, so it has a radius of 20m. We also know that the circumference is $C = 2\pi r \approx 125.6637$m. Said another way, the radius would fit $2\pi \approx 6.28$ times around the circle. Given two points on a circle, the part of the circle between them is called an **arc**, and its length is, appropriately, called the **arc length**. So, there are 2π radii of arc length in a circle.

It's not too much of a stretch to wonder how many radii of arc length lie between two points on a circle, and in fact, it gives a very natural way to measure angles, because given any angle, the arc length determined by the angle depends on the radius of the circle, and if an angle has an arc length of one radius on one circle, that same angle has one radius of arc length on any circle!

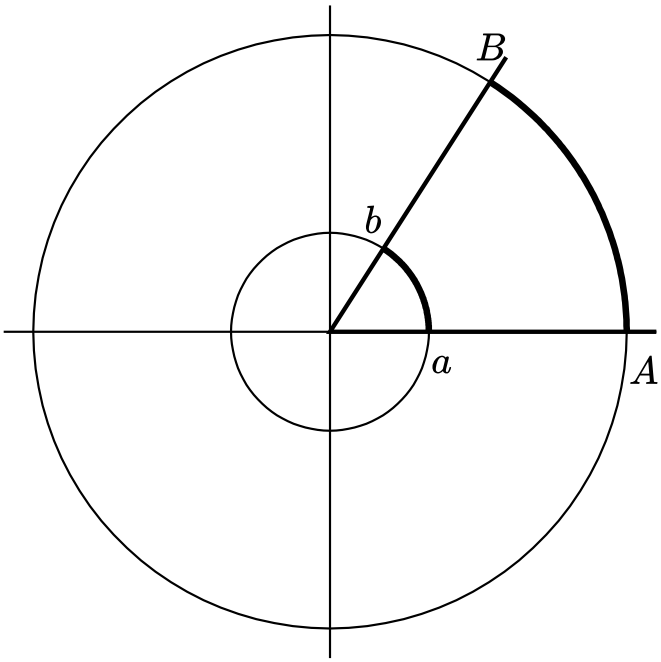


Figure 13 One radius of arc length on two different circles

An angle that has one radius of arc length has a measure of 1 **radian**. Both of the angles shown in Figure 13, $\angle AOB$ and $\angle aOb$, have a measure of 1 radian, which we will often abbreviate as 1 rad.

1. In this problem, we again consider a circle (a Ferris wheel) with a radius of 20m.

(a) When the car has travelled through 1 rad of angle, how much distance (arc length) has it travelled?

Answer. 20m

Solution. When the car has travelled one radian, it has travelled a distance (arc length) equal to its radius. Since the radius is 20m, the car has travelled 20m.

(b) When the car has travelled through an angle of 2.5 rad, how much arc length has it travelled?

Answer. 50m.

Solution. When the car has travelled 2.5 rad, it has travelled an arc length that is 2.5 times its radius, so it has travelled an arc length of $2.5 \cdot 20\text{m} = 50\text{m}$.

(c) When the car has travelled through 30m of arc length, what angle (in radians) has it travelled through?

Answer. 1.5 rad.

Solution. When when the car has travelled through θ rad, it has travelled a distance of $\theta \cdot 20$m, so we can solve $20\text{m} \cdot \theta = 30\text{m}$ to get $\theta = \frac{30\text{m}}{20\text{m}} = 1.5$ rad.[2]

(d) What is the arc length of one quarter of the circle?

Answer. 31.4159m

Solution. The total circumference is $2\pi \cdot 20\text{m} = 40\pi\text{m} \approx 125.6637\text{m}$, so one quarter of the circumference is $40\pi\text{m}/4 = 10\pi \approx 31.4159\text{m}$.

(e) What is the radian measure of one quarter of a revolution (the angle for one fourth of a circle)?

Answer. $\pi/2 \approx 1.57079$ rad.

Solution. There are two ways we could approach this. We could use our previous result for the arc length of a quarter circle is approximately 31.4159m and divide by the radius to find the angle,

$$\frac{31.4159\text{m}}{20\text{m}} \approx 1.57079$$

or we could realize that there are 2π radii in the circumference, and thus a total revolution is 2π radians. Then, we get that one quarter of that is $\frac{2\pi}{4} = \frac{\pi}{2} \approx 1.57079$ radians.

(f) What is the radian measure of one eighth of a revolution?

Answer. $\pi/4 \approx 0.78534$ rad.

Solution. Since there are 2π radii in the circumference, and thus a total revolution is 2π rad, we get that one eighth of that is $\frac{2\pi}{8} = \frac{\pi}{4} \approx 0.78534$ rad.

(g) What fraction of the distance around the circle is $\frac{\pi}{4}$ rad?

Answer. One eighth.

A couple of important things about radians before we move on.

- As we noted in the solution to Task 4.2.2.1.c, radians are **unitless** from a dimensional/unit analysis point of view.
- One full revolution, what we are used to thinking of as 360°, is 2π rad. This means that $180° = \pi$ rad, and a right angle is $90° = \frac{\pi}{2}$ rad. In general, the conversions between degrees and radians are

$$d \text{ deg} = \frac{\pi}{180} d \text{ rad}, \qquad \theta \text{ rad} = \theta\left(\frac{180}{\pi}\right) \text{ deg}.$$

- Some important angles to become familiar with are:

$$\frac{\pi}{6} = 30°, \quad \frac{\pi}{4} = 45°, \quad \frac{\pi}{3} = 60°, \quad \frac{\pi}{2} = 90°,$$

$$\frac{2\pi}{3} = 120°, \quad \frac{3\pi}{4} = 135°, \quad \frac{5\pi}{6} = 150°, \quad \pi = 180°,$$

$$\frac{7\pi}{6} = 210°, \quad \frac{5\pi}{4} = 225°, \quad \frac{4\pi}{3} = 240°, \quad \frac{3\pi}{2} = 270°,$$

$$\frac{5\pi}{3} = 300°, \quad \frac{7\pi}{4} = 315°, \quad \frac{11\pi}{6} = 330°, \quad 2\pi = 360°.$$

[2] It is worth noting here that if we do unit analysis of 20m/30m, we see that the units cancel out. Therefore, technically radians are ***unitless***.

4.2.3 The Sine and Cosine Functions

Probably the two most important functions in trigonometry are defined based on angles and the coordinates of points on a circle of radius 1, which we refer to as **the unit circle**.

For any point, P, on the unit circle, we measure the angle for P ***in radians*** from the positive x-axis counter-clockwise to P. Negative angles and angles greater than 2π are allowed: negative angles are measured counter-clockwise from the positive x-axis, and angles greater than 2π (or less than -2π) start around the circle again.

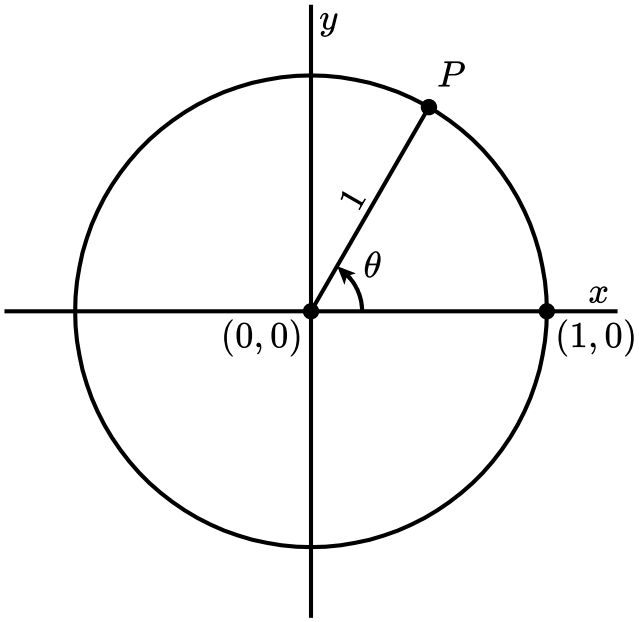


Figure 14 The Unit Circle with an Angle Shown

Figure 14 shows the unit circle with a point $P = (a, b)$ and its angle, θ.

Now that we have the unit circle and have explained how we measure angles on the unit circle, we are ready to define the circular functions **sine** and **cosine**.

Definition 15 The Sine and Cosine Functions. For any real number θ, define the point $P = (a, b)$ on the unit circle whose angle (measured from the positive x-axis) is θ rad. Then the cosine of θ is the x-coordinate of P and the sine of θ is the y-coordinate of P. That is,

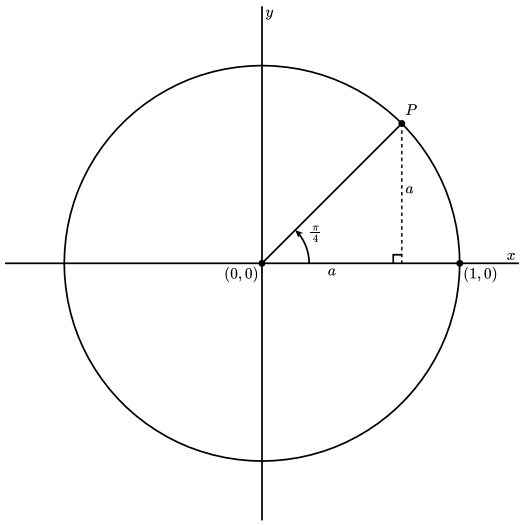
$$\cos(\theta) = a, \qquad \sin(\theta) = b.$$

◆

Example 16 We will find $\sin\left(\frac{\pi}{4}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.

We know that $\frac{\pi}{4}$ is $45°$, so we start by drawing a point on the unit circle that makes an angle of $\frac{\pi}{4}$ with the positive x-axis and drop a perpendicular from the point to the x-axis.

Because we now have a $45, 45, 90$ triangle, we know the x- and y-coordinates of P will be the same size, and because we are in the first quadrant, they will both be positive. So, let $P = (a, a)$, where $a > 0$, and we will find the value of a, and therefore the sine and cosine of $\frac{\pi}{4}$.



From the Pythagorean theorem, since the unit circle has radius 1 and the radius of the unit circle is the hypotenuse of the right triangle, we have

$$a^2 + a^2 = 1^2$$

so

$$2a^2 = 1.$$

Dividing by 2 gives us

$$a^2 = \frac{1}{2},$$

and then taking the (positive) square root give us

$$\begin{aligned} a &= \sqrt{\frac{1}{2}} \\ &= \frac{\sqrt{1}}{\sqrt{2}} \\ &= \frac{1}{\sqrt{2}}. \end{aligned}$$

If we want to rationalize the denominator, we can multiply both the numerator and denominator by $\sqrt{2}$ to get

$$\begin{aligned} a &= \frac{1 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} \\ &= \frac{\sqrt{2}}{2}. \end{aligned}$$

Therefore, we now know that $\cos\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ and $\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$. ▲

1. In the following explorations, use symmetry, the Pythagorean theorem, and geometry to relate angles and the values of sine and cosine.

One geometry fact you may not know is the following: in a triangle with angles $\frac{\pi}{6} = 30°$, $\frac{\pi}{3} = 60°$, and $\frac{\pi}{2} = 90°$ (a 30-60-90 triangle) , the shortest side (opposite the $30° = \frac{\pi}{6}$ angle) is half the length of the hypotenuse (longest side).

(a) Find $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{6}\right)$.

Answer.

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

Solution. Dropping a perpendicular from the point at $\frac{\pi}{6}$ on the unit circle creates a 30-60-90 triangle with the perpendicular being the shortest side. Since the hypotenuse has length 1, that means that the y-coordinate of the point is $\frac{1}{2}$, and so $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$. Then, from the Pythagorean theorem, if we let the x-coordinate of the point be a, we have

$$a^2 + \left(\frac{1}{2}\right)^2 = 1^2$$

so

$$a^2 + \frac{1}{4} = 1$$

and therefore $a^2 = \frac{3}{4}$. Since we know that $a > 0$, we know that $a = \frac{\sqrt{3}}{2}$. Thus, $\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$.

(b) Find $\sin\left(\frac{\pi}{3}\right)$ and $\cos\left(\frac{\pi}{3}\right)$.

Hint. Use symmetry and the previous problem.

Answer.

$$\sin\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$
$$\cos\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

(c) Find $\sin\left(\frac{5\pi}{6}\right)$ and $\cos\left(\frac{5\pi}{6}\right)$.

Answer.

$$\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$$
$$\cos\left(\frac{5\pi}{6}\right) = \frac{-\sqrt{3}}{2}$$

Solution. Dropping a perpendicular from the point at $\frac{5\pi}{6}$ on the unit circle creates a 30-60-90 triangle with the perpendicular being the shortest side. Since the hypotenuse has length 1, that means that the y-coordinate of the point is $\frac{1}{2}$, and so $\sin\left(\frac{5\pi}{6}\right) = \frac{1}{2}$. Then, from the Pythagorean theorem, if we let the x-coordinate of the point be a, we have

$$a^2 + \left(\frac{1}{2}\right)^2 = 1^2$$

so

$$a^2 + \frac{1}{4} = 1$$

and therefore $a^2 = \frac{3}{4}$. Since the point is in the second quadrant, we know that $a < 0$, so we know that $a = \frac{-\sqrt{3}}{2}$. Thus, $\cos\left(\frac{5\pi}{6}\right) = \frac{-\sqrt{3}}{2}$.

(d) Use symmetry to find the coordinates of each of the points at the important angles shown in Figure 17.

Answer.

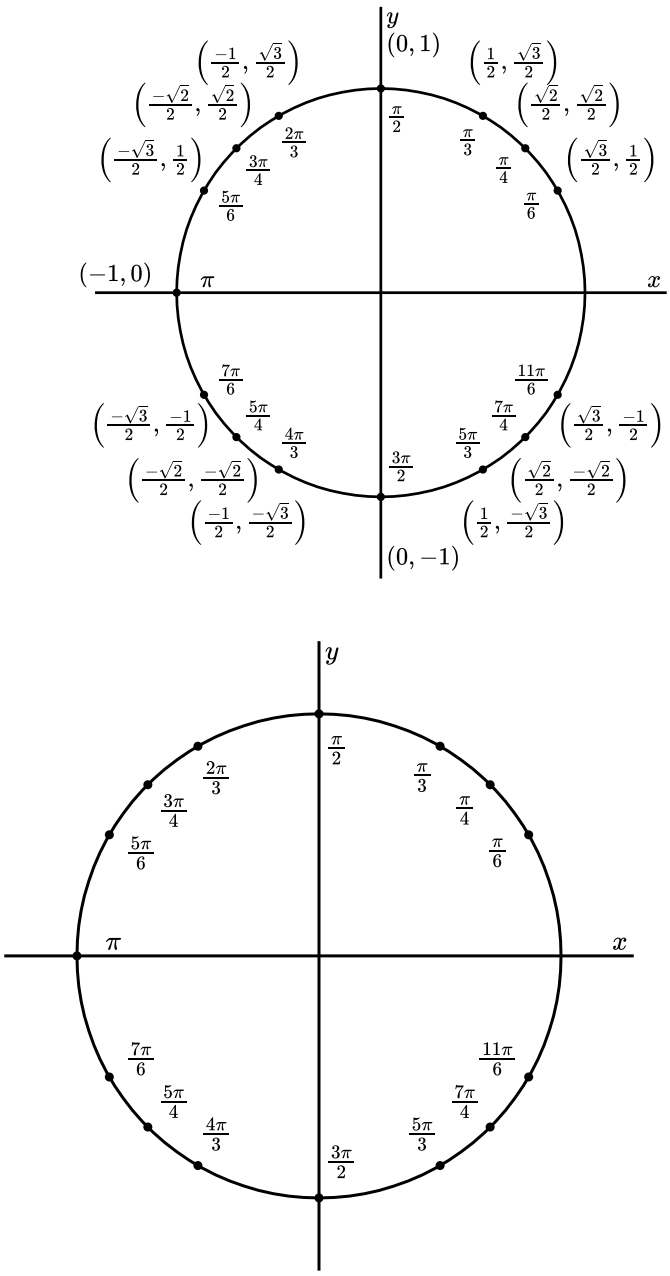


Figure 17 The Unit Circle with Important Angles

4.2.4 Similar Triangles and the Pythagorean Theorem

An extremely important concept in physics, trigonometry, calculus, and even carpentry is the notion of *similar triangles*. We will briefly discuss how to recognize similar triangles and what properties they have before returning to the ideas we explored in Worksheet 4.2.1.

Combining the definitions of sine and cosine with similar triangles and the Pythagorean theorem gives us tools that make these functions extremely useful in a variety of situations.

Definition 18 Similar Triangles. When two triangles have exactly the same three angles, we say that they are *similar triangles* or that the triangles are *similar*. ◆

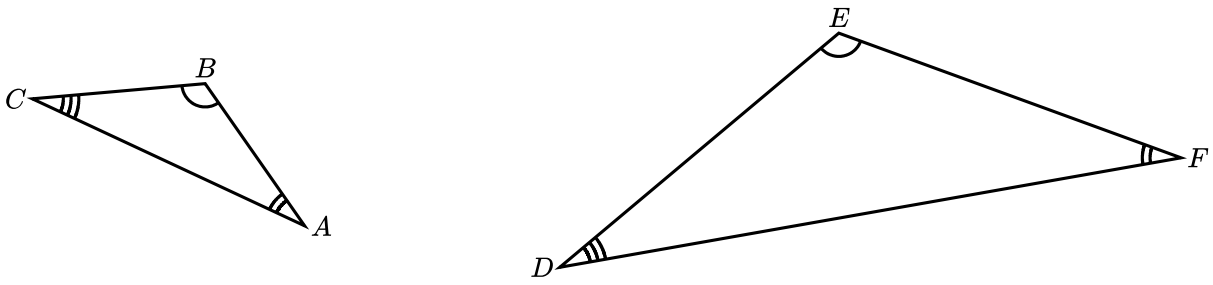


Figure 19 An Example of Similar Triangles

The triangles shown in Figure 19 are similar because angle C is the same as angle D ($\angle C \cong \angle D$), angle B is the same as angle E ($\angle B \cong \angle E$), and angle A is the same as angle F ($\angle A \cong \angle F$).

We say that $\angle C$ and $\angle D$, $\angle B$ and $\angle E$, $\angle A$ and $\angle F$ are *corresponding angles*, and similarly, we say that sides $\overline{CB}$ and $\overline{DE}$, sides $\overline{BA}$ and $\overline{EF}$, sides $\overline{AC}$ and $\overline{FD}$ are *corresponding sides*.

Furthermore, we say that $\triangle CBA \sim \triangle DEF$, and the order matters!

Theorem 20 Corresponding Sides of Similar Triangles. *Given two triangles, $\triangle MNE$ and $\triangle PQR$, when the triangles are similar ($\triangle MNE \sim \triangle PQR$), the lengths of the corresponding sides are proportional.*

Applying Theorem 20 to Figure 19, for example, we have that

$$\frac{f}{a} = \frac{e}{b} = \frac{d}{c},$$

and, equivalently,

$$\frac{a}{c} = \frac{f}{d}, \qquad \frac{a}{b} = \frac{f}{e}, \qquad \frac{b}{c} = \frac{d}{e}.$$

Theorem 21 The Pythagorean Theorem. *For any positive numbers, a, b, and c, if a and b are the lengths of two legs of a right triangle and c is the length of the hypoteneuse, then $a^2 + b^2 = c^2$.*

Furthermore, for any triangle with side lengths a, b, and c, if $a^2 + b^2 = c^2$, then the triangle is a right triangle, the sides with lengths a and b are the legs, and the side with length c is the hypotenuse.

1. In Figure 22, $\overline{OA}$ has length 0.5m, $\overline{OB}$ has length 1m, and $\overline{OD}$ has length 3m.

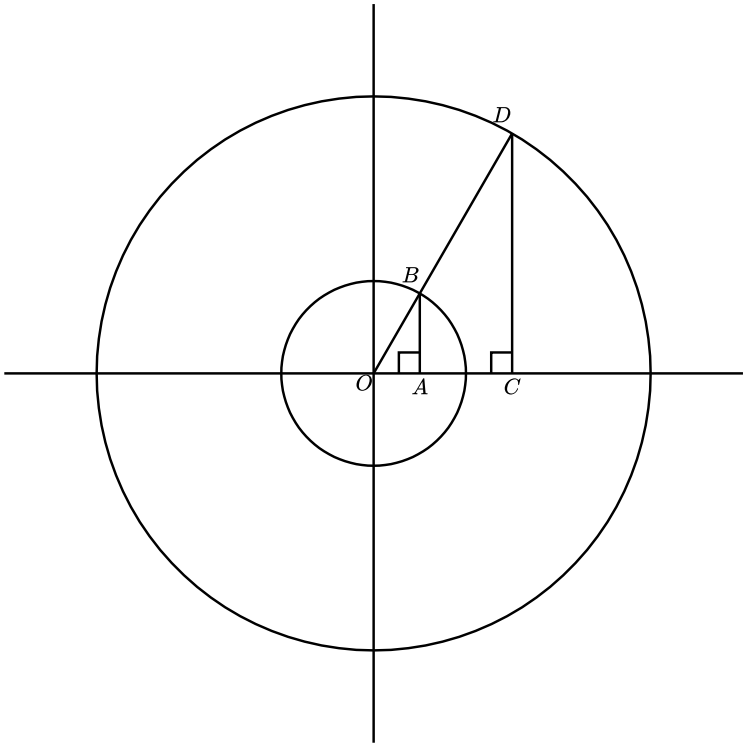


Figure 22 Circles with Similar Triangles

(a) Explain why $\triangle OBA$ is similar to $\triangle ODC$.

Solution. Both triangles share $\angle AOB \cong \angle COD$, and $\angle DCO$ and $\angle BAO$ are both right angles. Because the measures of the three angles of any triangle add to $180° = \pi$ radians, we must have $\angle OBA \cong \angle ODC$.

Since all three angles of $\triangle OBA$ are congruent to the respective angles of $\triangle ODC$, the two triangles are similar, $\triangle OBA \sim \triangle ODC$.

(b) Use the fact that $\triangle OBA \sim \triangle ODC$ to find the length of $\overline{OC}$.

Answer. 1.5m.

Solution. We will represent the length of a segment as the two endpoints, so, for example, we will say that $OA = 0.5\text{m}$.

We know that $OA = 0.5\text{m}$, $OB = 1\text{m}$, and $OD = 3\text{m}$.

Because $\triangle OBA \sim \triangle ODC$, we know that $\frac{OA}{OB} = \frac{OC}{OD}$, so substituting the known lengths, we have

$$\begin{aligned} \frac{OC}{OD} = \frac{OA}{OB} &\rightarrow \frac{OC}{3\text{m}} = \frac{0.5\text{m}}{1\text{m}} \\ &\rightarrow OC = 3\text{m}\frac{0.5\text{m}}{1\text{m}} \\ &\rightarrow OC = 1.5\text{m} \end{aligned}$$

(c) Use the Pythagorean theorem to find AB, the length of $\overline{AB}$.

Answer. $\frac{\sqrt{3}}{2}\text{m} \approx 0.866025\text{m}$.

Solution. From the Pythagorean theorem, since $\triangle BAO$ is a right triangle, we have $(BA)^2 + (OA)^2 = (OB)^2$, so $(BA)^2 + (0.5)^2 = (1)^2$, so

$$\begin{aligned} (BA)^2 &= (1)^2 - (0.5)^2 \\ &= 1 - 0.25 \end{aligned}$$

$$= 0.75$$
$$= \frac{3}{4}$$

so

$$BA = \sqrt{\frac{3}{4}}$$
$$= \frac{\sqrt{3}}{\sqrt{4}}$$
$$= \frac{\sqrt{3}}{2}$$
$$\approx 0.866025.$$

(d) Find the length of CD.

Answer. $\frac{3\sqrt{3}}{2}\text{m} \approx 2.5981\text{m}$

4.3 Practical Applications of Sine and Cosine

Example 23 We will show how to use sine, cosine, and similar triangles to solve a practical problem.

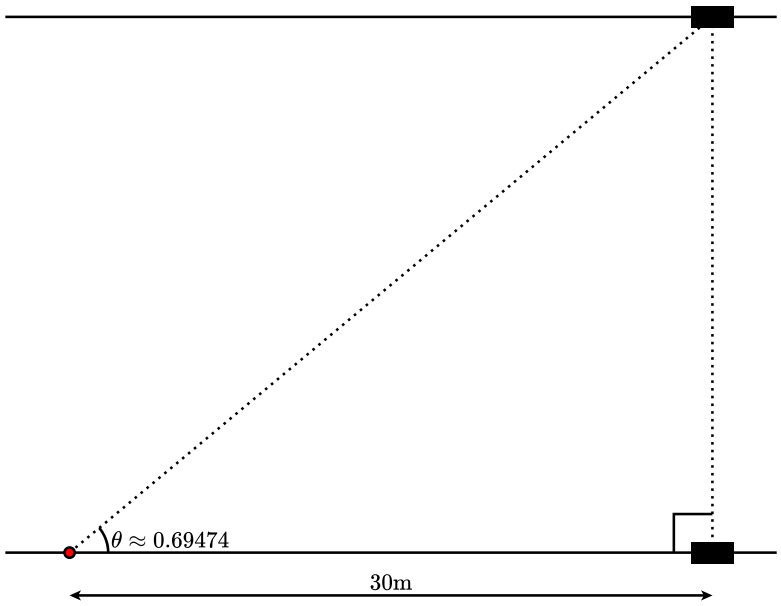


Figure 24 Two Gatehouses on a Canal Lock

Amir is standing at the edge of a canal lock. There are gatehouses on each side of the lock. The gatehouse on the same side as Amir is located 30m from them. Amir wants to find out how wide the lock is by finding how far one gatehouse is from the other. Amir measures the angle from the near gatehouse to the far gatehouse and finds that it is 0.69474 rad.

How wide is the lock?

We start by drawing a copy of the unit circle centered on Amir's position, and we will assign the distance between the two gatehouses to the variable d. Then, by dropping a perpendicular from the point, P, where the line from Amir to the far gatehouse crosses the circle, we have similar triangles.

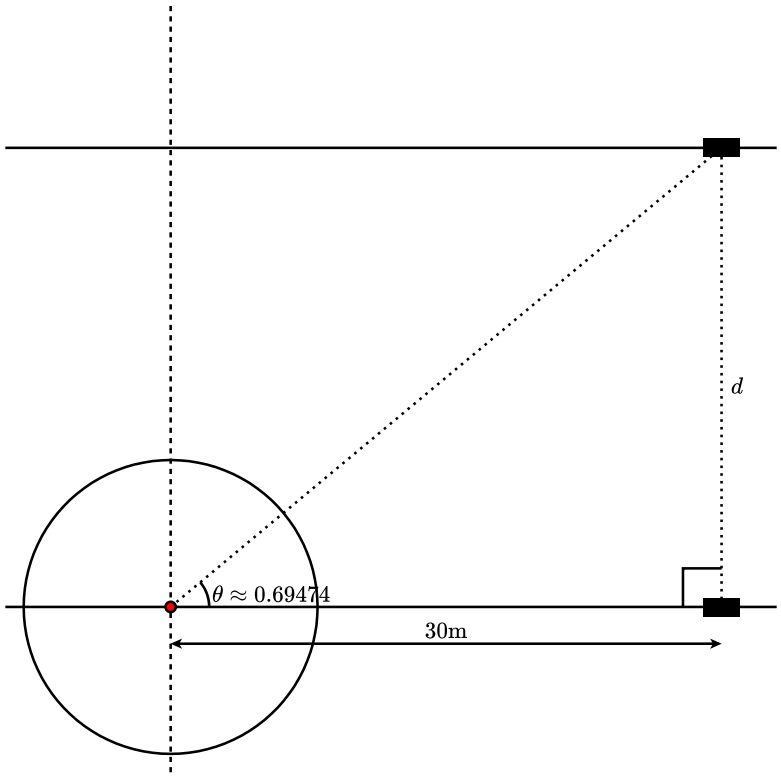


Figure 25 Canal Lock with Unit Circle

We know that P has coordinates $(\cos(0.69474), \sin(0.69474))$ (which we will leave unevaluated until the end). Furthermore, we know from similar triangles that corresponding parts are proportional, so we have

$$\frac{d}{\sin(0.69474)} = \frac{30\text{m}}{\cos(0.69474)}.$$

Multiplying both sides by $\sin(0.69474)$, we have

$$\begin{aligned} d &= \frac{30\text{m}\sin(0.69474)}{\cos(0.69474)} \\ &\approx 25\text{m} \end{aligned}$$

▲

1. Inaya wants to use trigonometry to measure the height of a tree. She walks 10m away from the base of the tree and measures the angle to the top of the tree as shown in Figure 26.

To the nearest tenth of a meter, how tall is the tree? That is, what is the value of d in meters?

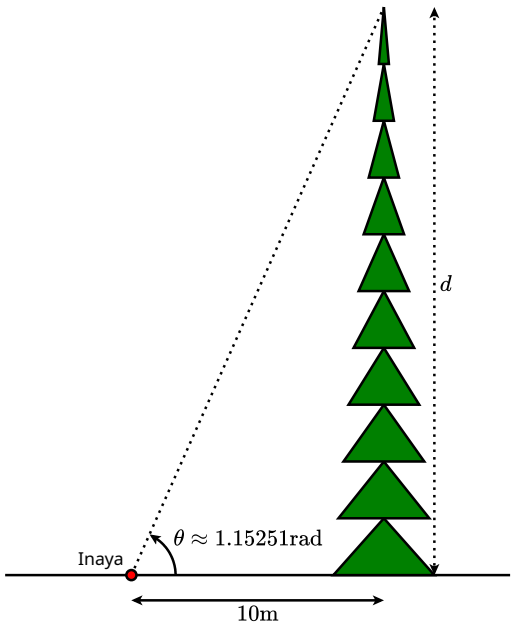


Figure 26 An Observer and a Tree

Answer. The tree is approximately 22.5m tall.

Solution. Draw a circle that represents the unit circle around the point representing Inaya and set your coordinate axes so that this is the origin.

Drop a perpendicular from where the circle intersects the line from Inaya to the top of the tree. We then have a small triangle with corners at the origin, on the unit circle, and on the x-axis. This triangle is similar to the triangle from the origin to the top of the tree to the x-axis because the share the angle θ and they both have right angles on the x-axis.

We know the coordinates of the point on the unit circle are $(\cos(\theta), \sin(\theta))$, and we know that the sides of the small triangle are proportional to the sides of the big triangle. Thus, $\frac{d}{10\text{m}} = \frac{\sin(\theta)}{\cos(\theta)}$, so we have

$$\begin{aligned} d &= 10\text{m}\frac{\sin(\theta)}{\cos(\theta)} \\ &= 10\text{m}\frac{\sin(1.15251)}{\cos(1.15251)} \\ &= 22.5\text{m} \end{aligned}$$

2. Micah is watching a train as it approaches. He is 10m from the closest point on the track.

He observes that the train makes an angle of approximately 0.98273 radians, and then one second later an angle of approximately 0.89594 radians. How fast is the train travelling?

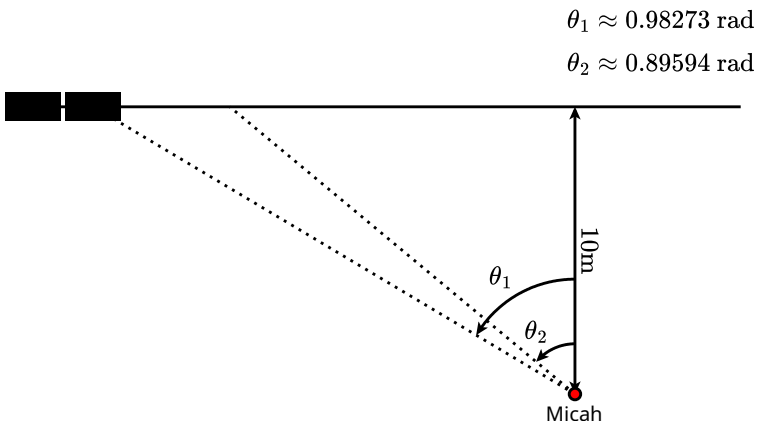


Figure 27 A Train on a Track

4.4 Transformations of Sine and Cosine

In this activity we will first review transformations of functions using Desmos, and then we will apply what we notice to graphs of the sine and cosine.

1. **Review of Transformations of a Graph.** Open the Desmos activity Review of Transformations, https://www.desmos.com/calculator/m12t66m4ok.

   This graph shows a function, $f(x)$, in blue, and the function

   $$g(x) = a \cdot f\left(b(x - c)\right) + d$$

   in red, and has a set of sliders, one for each of the parameters a, b, c, and d.

   (a) *The Effect of a.*

   (i) Move the slider for a to 3. How does the transformed graph (red) compare to the original (blue).

   (ii) Now move the slider for a to 0.5. How does the transformed graph (red) compare to the original (blue).

   (iii) Now move the slider for a to -2. How does the transformed graph (red) compare to the original (blue).

   (iv) How would you describe the effect of a on the graph?

   (b) *The Effect of b.*

   (i) Move the slider for b to 3. How does the transformed graph (red) compare to the original (blue).

   (ii) Now move the slider for b to 0.5. How does the transformed graph (red) compare to the original (blue).

   (iii) Now move the slider for b to -2. How does the transformed graph (red) compare to the original (blue).

   (iv) How would you describe the effect of b on the graph?

(c) *The Effect of c.*

(i) Move the slider for c to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for c to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for c to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of c on the graph?

(d) *The Effect of d.*

(i) Move the slider for d to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for d to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for d to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of d on the graph?

2. Transformations Sine. Open the Desmos activity Transformations of Sine, https://www.desmos.com/calculator/6dk4cjfgsu.

This graph shows the function, $f(x) = \sin(x)$ in blue, and the function

$$g(x) = a \cdot f\left(b(x - c)\right) + d$$

in red, and has a set of sliders, one for each of the parameters a, b, c, and d.

(a) *The Effect of a.*

(i) Move the slider for a to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for a to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for a to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of a on the graph?

(b) *The Effect of b.*

(i) Move the slider for b to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for b to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for b to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of b on the graph?

(c) *The Effect of c.*

(i) Move the slider for c to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for c to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for c to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of c on the graph?

(d) *The Effect of d.*

(i) Move the slider for d to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for d to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for d to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of d on the graph?

3. Transformations Cosine. Open the Desmos activity Transformations of Cosine, https://www.desmos.com/calculator/7dwhncpsfz.

This graph shows the function, $f(x) = \cos(x)$ in blue, and the function

$$g(x) = a \cdot f\left(b(x - c)\right) + d$$

in red, and has a set of sliders, one for each of the parameters a, b, c, and d.

(a) *The Effect of a.*

(i) Move the slider for a to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for a to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for a to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of a on the graph?

(b) *The Effect of b.*

(i) Move the slider for b to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for b to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for b to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of b on the graph?

(c) *The Effect of c.*

(i) Move the slider for c to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for c to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for c to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of c on the graph?

(d) *The Effect of d.*

(i) Move the slider for d to 3. How does the transformed graph (red) compare to the original (blue).

(ii) Now move the slider for d to 0.5. How does the transformed graph (red) compare to the original (blue).

(iii) Now move the slider for d to -2. How does the transformed graph (red) compare to the original (blue).

(iv) How would you describe the effect of d on the graph?

Functions of the form $f(x) = a\sin\big(b(x-c)\big) + d$ and $g(x) = a\cos\big(b(x-c)\big) + d$ are called **sinusoidal functions** or simply **sinusoids**.

For the graphs of sinusoidal functions, the various sliders affect aspects of the graphs that we define below with specific names.

Definition 28 Midline, Amplitude, Period, and Phase Shift of Sinusoids.

- The **midline** of a sinusoidal graph (that is, sine or cosine) is the vertical position that is halfway between the maxima and minima of the graph. For an untransformed sine or cosine the midline is $y = 0$. For the functions $f(x) = a\sin\big(b(x-c)\big) + d$ and $g(x) = a\cos\big(b(x-c)\big) + d$, the midline is $y = d$.
- The **amplitude** of a sinusoidal graph is the vertical distance between the midline and the maxima, or half the distance between the minima and maxima. For an untransformed sine or cosine the amplitude is 1. For the functions $f(x) = a\sin\big(b(x-c)\big) + d$ and $g(x) = a\cos\big(b(x-c)\big) + d$, the amplitude is $|a|$.
- The **period** of a sinusoidal graph is the minimum horizontal distance between repeats (also called cycles or periods). For an untransformed sine or cosine the period is 2π. For the functions $f(x) = a\sin\big(b(x-c)\big) + d$ and $g(x) = a\cos\big(b(x-c)\big) + d$, the period is $\frac{2\pi}{|b|}$.
- The **phase shift** of a sinusoidal graph the horizontal shift of the sinusoid (relative to the untransformed function). For an untransformed sine or cosine the phase shift is 0. Note that because sinusoids are periodic, the phase shift can always be described with either a positive (rightward) shift that is between 0 and the period or with a shift that is between $\frac{-1}{2}$ times the period and $\frac{1}{2}$ times the period. For the functions $f(x) = a\sin\big(b(x-c)\big) + d$ and $g(x) = a\cos\big(b(x-c)\big) + d$, the phase shift is c.

◆

In some cases, rather than specifying the *period*, we may specify the **frequency** of the sinusoid. The frequency of a sinusoid is the number of periods per unit time. Thus, if F is the frequency and P is the period, we have $F = \frac{1}{P}$ or $P = \frac{1}{F}$. In relation to the sliders in this activity, we have $F = \frac{|b|}{2\pi}$.

4.5 Oscillatory Motion

In this activity, we will use transformations of sine and cosine graphs to model the motion of a pendulum.

1. **Modeling the Motion of a Pendulum.** We have a pendulum suspended from a point (on a stand) 20cm away from a wall and 15cm above the table top. At its farthest from the wall, it is 25cm away from the wall and at its closest it is 15cm away from the wall (see Figure 29).

   Every two seconds, it completes one swing, from farthest point back to farthest point.

   We want to find a model sinusoidal that will give us the pendulum bob's distance away from the wall as a function of time.

   As a stretch goal, we want to model the pendulum bob's distance away from the table top as a function of time using a sinusoidal function.

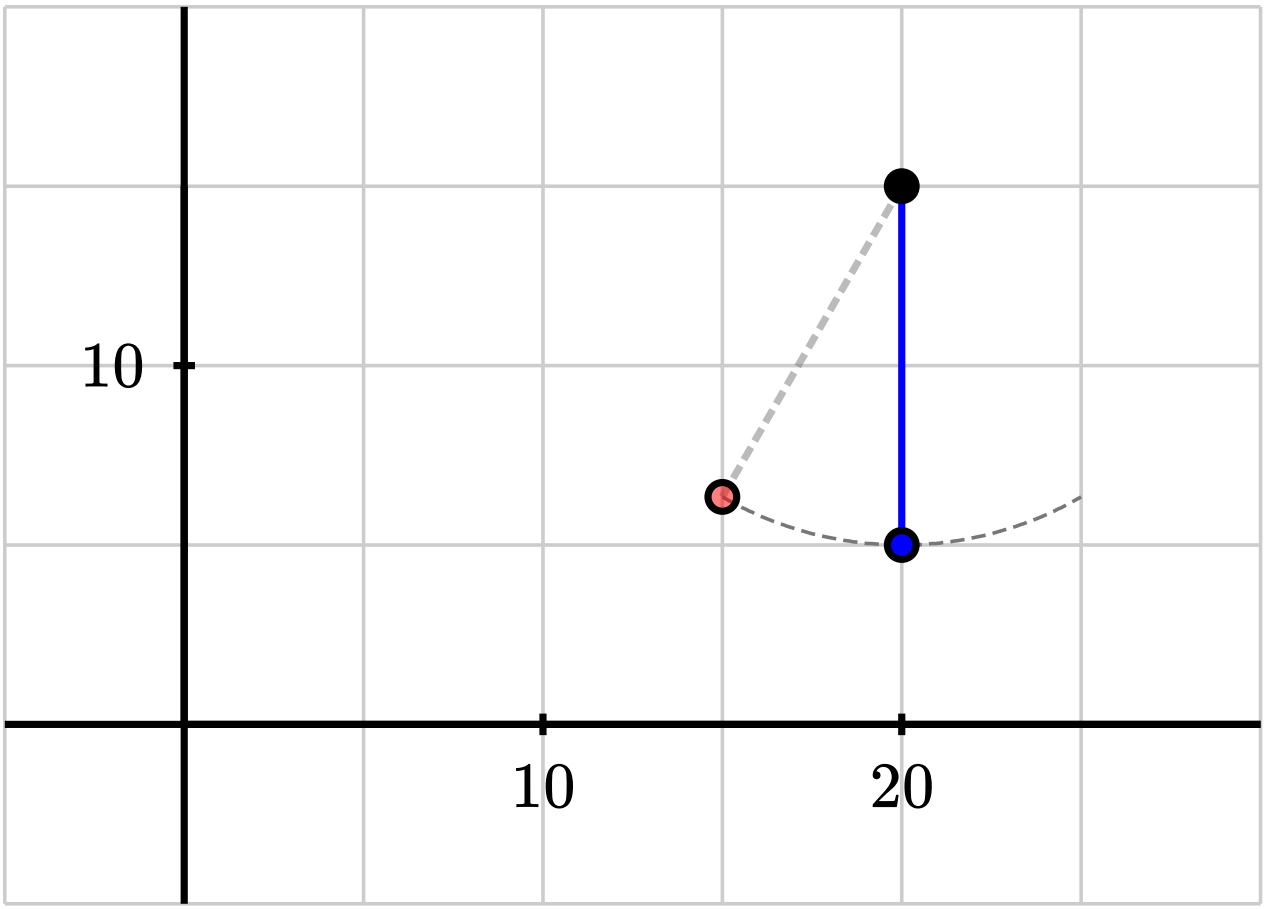


Figure 29 A Pendulum

(a) *Modeling the Horizontal Motion.*

For this exercise, let $f_x(t)$ give the horizontal distance of the pendulum from the wall, in centimeters, at time t seconds. (We will decide on what time $t = 0$ means later.)

(i) What is the maximum value of $f_x(t)$? That is, what is the maximum horizontal distance of the pendulum from the wall?

Answer. 25cm.

(ii) What is the minimum value of $f_x(t)$? That is, what is the minimum horizontal distance of the pendulum from the wall?

Answer. 15cm.

(iii) Use the maximum and minimum values to calculate the midline for $f_x(t)$.

Answer. 20cm.

(iv) Use the maximum and minimum values to calculate the amplitude for $f_x(t)$.

Answer. 20cm.

(v) Sketch a graph of a sinusoidal function that fits your previous answers.
In your sketch, what is the pendulum doing at time $t = 0$, specifically, how far from the wall is the pendulum and is it moving toward or away from the wall?

(vi) How long does it take for the distance from the wall of the pendulum to return to the same position and direction?

Answer. 2s.

(vii) Assume that at time $t = 0$, the pendulum is at its farthest point from the wall.
Find parameters a, b, c, and d so that the function $f_x(t) = a\cos(b(t-c)) + d$ fits all our information.

Answer. $f_x(t) = 5\cos(\pi(t-0)) + 20 = 5\cos(\pi t) + 20$.

(viii) Assume that at time $t = 0$, the pendulum is at its nearest point to the wall.
Find parameters a, b, c, and d so that the function $f_x(t) = a\sin(b(t-c)) + d$ fits all our information.

Answer. $f_x(t) = 5\sin(\pi(t-0.5)) + 20$.

(b) *Modeling the Vertical Motion.*

For this exercise, let $f_y(t)$ give the horizontal distance of the pendulum from the wall, in centimeters, at time t seconds. (We will decide on what time $t = 0$ means later.)

(i) What is the minimum value of $f_y(t)$? That is, what is the minimum vertical distance of the pendulum from the floor?

Answer. 5cm.

(ii) Note that when the pendulum is closest to the wall, this corresponds to a point of maximum height. In Figure 29, if we draw a horizontal line through the position of the pendulum when it is closest to the wall, we form a right triangle with the vertical line and the string. The string (which has length 10cm) is the hypotenuse of this right triangle.

(A) What is the length of the horizontal leg of the right triangle?

Answer. 5cm

(B) Use the Pythagorean theorem to find the length of the other (vertical) leg of the right triangle.

Answer. $5\sqrt{3}$cm

Solution. Since the hypotenuse is 10cm and one leg is 5cm, if we let the length of the other leg be z, we have $5^2 + z^2 = 10^2$, so $z^2 = 100 - 25 = 75$. Since distance is positive, we have $z = \sqrt{75} = \sqrt{3 \cdot 25} = 5\sqrt{3}$. Thus, the length of the leg is $5\sqrt{3}$cm.

(C) Calculate the maximum value of $f_y(t)$. That is, find the minimum vertical distance of the pendulum from the floor.

Hint. The distance from the floor to the pivot point is 15cm and you found the vertical distance from the pivot point to highest point of the pendulum's path in Task 4.5.1.b.b.ii.C.

Answer. $(15 - 5\sqrt{3})$cm

(iii) Use the maximum and minimum values to calculate the midline for $f_y(t)$.

Answer. $\left(10 + \frac{5\sqrt{3}}{2}\right)$ cm.

(iv) Use the maximum and minimum values to calculate the amplitude for $f_y(t)$.

Answer. $\left(5 - \frac{5\sqrt{3}}{2}\right)$ cm.

(v) Sketch a graph of a sinusoidal function that fits your previous answers.
In your sketch, what is the pendulum doing at time $t = 0$, specifically, how far from the floor is the pendulum and is it moving toward or away from the floor?

(vi) How long does it take for the distance from the floor of the pendulum to return to the same position and direction?

Answer. 1s.

(vii) Assume that at time $t = 0$, the pendulum is at its farthest point from the wall.
Find parameters a, b, c, and d so that the function $f_y(t) = a\cos(b(t-c)) + d$ fits all our information.

Answer. $f_y(t) = \left(5 - \frac{5\sqrt{3}}{2}\right)\cos(2\pi(t-0)) + \left(10 + \frac{5\sqrt{3}}{2}\right)$.

4.6 Other Trigonometric Functions

In addition to sine ($\sin(\theta)$) and cosine ($\cos(\theta)$) there are four other trigonometric functions, each based on one or both or sine and cosine. They are:

Tangent $\tan(\theta) = \dfrac{\sin(\theta)}{\cos(\theta)}$

Cotangent $\cot(\theta) = \dfrac{\cos(\theta)}{\sin(\theta)}$

Secant $\sec(\theta) = \dfrac{1}{\cos(\theta)}$

Cosecant $\csc(\theta) = \dfrac{1}{\sin(\theta)}$

In terms of the unit circle, recall that $\cos(\theta)$ is the x-coordinate of the point, let's call it P, on the unit circle at an angle θ counter-clockwise from the point $(1, 0)$ and $\sin(\theta)$ is the y-coordinate of P. That is, $P = (\cos(\theta), \sin(\theta))$.

With this in mind, we see that

$$\begin{aligned}\tan(\theta) &= \frac{\sin(\theta)}{\cos(\theta)} \\ &= \frac{\sin(\theta) - 0}{\cos(\theta) - 0} \\ &= \frac{\Delta y}{\Delta x},\end{aligned}$$

so $\tan(\theta)$ can be thought of as the slope of the line through $(0, 0)$ and P.

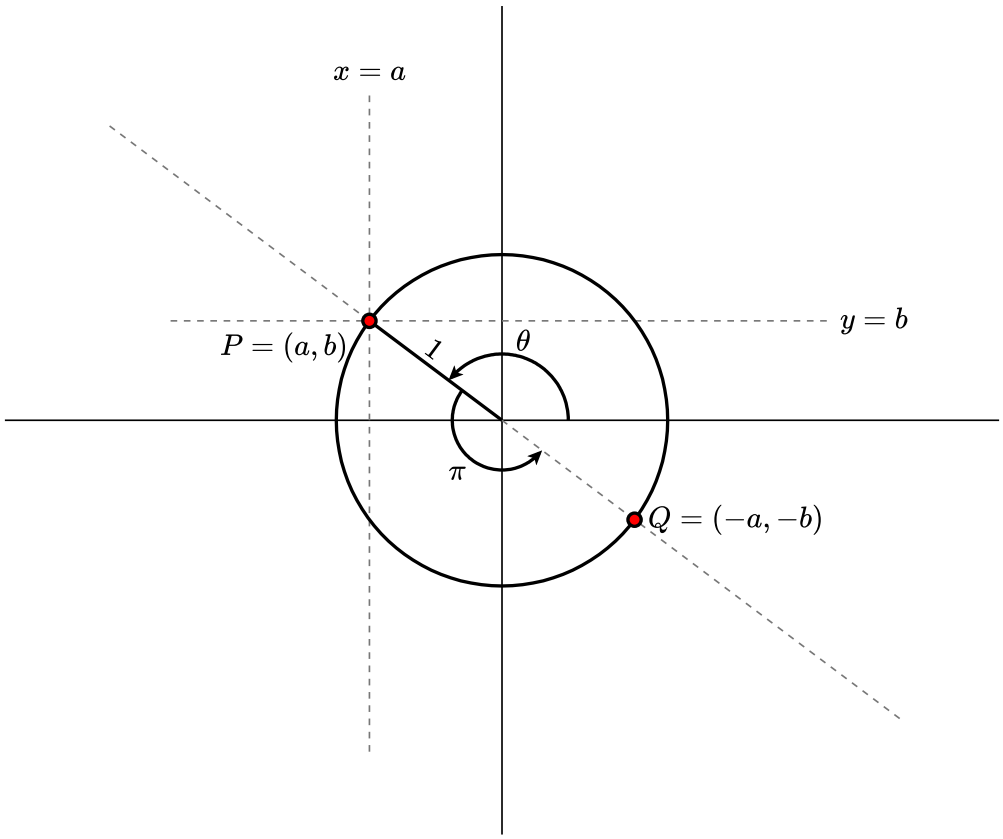


Figure 30 The unit circle showing $\cos(\theta)$, $\sin(\theta)$, and $\tan(\theta)$.

1. In this problem, we will assume that the point on the unit circle at angle θ is $P = (a, b)$ and that the point $Q = (-a, -b)$ is obtained by rotating P through π radians (180°), as shown in Figure 30.

(a) In Figure 30, what is the value of $\cos(\theta)$?

(b) In Figure 30, what is the value of $\sin(\theta)$?

(c) In Figure 30, what is the value of $\tan(\theta)$?

(d) In Figure 30, what is the value of $\cos(\theta + \pi)$?

(e) In Figure 30, what is the value of $\sin(\theta + \pi)$?

(f) In Figure 30, what is the value of $\tan(\theta + \pi)$?

We have just discovered three **trigonometric identities**:for any real number θ,

- $\cos(\theta + \pi) = -1 \cdot \cos(\theta)$;
- $\sin(\theta + \pi) = -1 \cdot \sin(\theta)$; and
- $\tan(\theta + \pi) = \tan(\theta)$.

We actually know a couple more identities, we just don't realize it yet! Because of the periodicity of sine and cosine, we have: for any real number θ, $\cos(\theta + 2\pi) = \cos(\theta)$ and $\sin(\theta + 2\pi) = \sin(\theta)$.

2. Although they are not used nearly as often as sine, cosine, and tangent, the functions cotangent, secant, and cosecant do occasionally show up, so it's important to know them. We will again use Figure 30 in this activity.

(a) In Figure 30, what is the value of $\sec(\theta)$?

(b) In Figure 30, what is the value of $\csc(\theta)$?

(c) In Figure 30, what is the value of $\cot(\theta)$?

(d) In Figure 30, what is the value of $\sec(\theta + \pi)$?

(e) In Figure 30, what is the value of $\csc(\theta + \pi)$?

(f) In Figure 30, what is the value of $\cot(\theta + \pi)$?

4.7 Sine, Cosine, and Triangles

In this activity, we will use the unit circle and similar triangle to investigate how sine and cosine can be used with triangles.

1. Investigate the following questions using Figure 31.

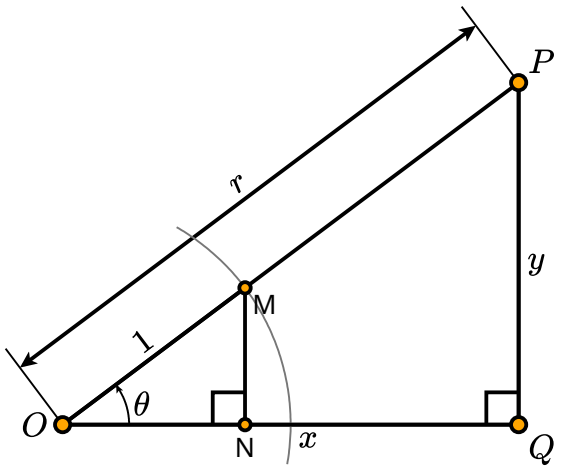


Figure 31 Two Triangles

(a) Explain why $\triangle OPQ$ and $\triangle OMN$ are similar triangles.

Solution. Both $\triangle OPQ$ and $\triangle OMN$ are right triangles with right angles $\angle OQP$ and $\angle ONM$, respectively, so $\angle OQP \cong \angle ONM$.

Next, $\angle POQ$ and $\angle MON$ are the same angle, so $\angle POQ \cong \angle MON$.

Finally, because the sum of the measures of the three angles in a triangle is always π radians (or $180°$), we must have $\angle OMN \cong \angle OPQ$.

Because $\angle OQP \cong \angle ONM$, $\angle POQ \cong \angle MON$, and $\angle OMN \cong \angle OPQ$, we have $\triangle OPQ \sim \triangle OMN$.

(b) What is the value of the ratio $\frac{OP}{OM}$? What does this (and the fact that the triangles are similar) tell you about the ratios $\frac{OQ}{ON}$ and $\frac{PQ}{MN}$?

Solution.

$$\begin{aligned}\frac{OP}{OM} &= \frac{r}{1} \\ &= r.\end{aligned}$$

Because corresponding sides of similar triangles have proportional lengths,

$$\begin{aligned}\frac{OQ}{ON} &= \frac{PQ}{MN} \\ &= r.\end{aligned}$$

(c) **(i)** What is the value of ON in terms of θ?

Hint. Use the fact that M is a point on the unit circle.

Answer. $ON = \cos(\theta)$

(ii) What is the value of MN in terms of θ?

Answer. $MN = \sin(\theta)$

(d) Use your conclusions in Task 4.7.1.b and Task 4.7.1.c to express x and y in terms of r and θ.

Hint. Notice, for example, that $ON = \frac{ON}{1} = \frac{ON}{OM}$.

Solution. We know that $x = OQ$ and $ON = \cos(\theta)$. By similar triangles,

$$\begin{aligned} r &= \frac{PO}{MO} \\ &= \frac{OQ}{ON} \\ &= \frac{x}{\cos(\theta)}, \end{aligned}$$

so, $r = \frac{x}{\cos(\theta)}$.

Multiplying both sides by $\cos(\theta)$ gives us $r\cos(\theta) = x$.

Similarly, $y = PQ$, and $r = \frac{PQ}{\sin(\theta)} = \frac{y}{\sin(\theta)}$, so $y = r\sin(\theta)$.

2. Investigate the following questions using Figure 32.

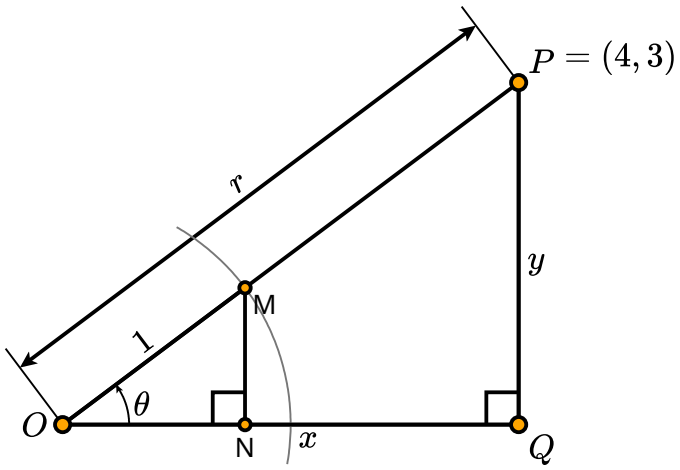


Figure 32 Two Triangles

(a) Explain why $\triangle OPQ$ and $\triangle OMN$ are similar triangles.

Answer. This is essentially the same argument as in Solution 4.7.1.a.1.

(b) Use the Pythagorean theorem to calculate r, the length of OP.

Answer. $r = \sqrt{(4)^2 + (3)^2} = 5$.

(c) Find the coordinates for Q.

Answer. $(4, 0)$.

(d) Use your results from Task 4.7.2.b and Task 4.7.2.c to answer the following questions.

(i) What is the value of $\frac{y}{r}$?

Answer. $\frac{3}{5}$

(ii) What is the value of $\frac{x}{r}$?

Answer. $\frac{4}{5}$

(e) Use the your result from Task 4.7.2.b to find the coordinates of M and N.

Solution 1. Using similar triangles, we know that $\frac{OQ}{ON} = 5$ and $OQ = 4$, so $ON = \frac{4}{5}$. We also know that $QP = 3$ and $\frac{QP}{NM} = 5$, so $QP = \frac{3}{5}$. This means that $N = \left(\frac{4}{5}, 0\right)$ and $M = \left(\frac{4}{5}, \frac{3}{5}\right)$.

Solution 2. Using vectors, we know that $P = (4, 3)$ is associated with the vector $\vec{p} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, and M is associated with a unit vector, call it $\vec{u}$, in the same direction. Since $|\vec{p}| = 5$, we know that $\vec{p} = 5\vec{u}$, so

$$\begin{aligned} \vec{u} &= \frac{1}{|\vec{p}|}\vec{p} \\ &= \frac{1}{5}\vec{p} \\ &= \frac{1}{5}\begin{bmatrix} 4 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}. \end{aligned}$$

Thus, $M = \left(\frac{4}{5}, \frac{3}{5}\right)$, and N is directly below it on the x-axis, so $N = \left(\frac{4}{5}, 0\right)$.

(f) Use the fact that M is on the unit circle to answer the following questions.

(i) What is the value of $\sin(\theta)$?

Answer. $\frac{3}{5}$

(ii) What is the value of $\cos(\theta)$?

Answer. $\frac{4}{5}$

(g) What do you notice about your results from Task 4.7.2.d and Task 4.7.2.f?

4.8 Trigonometric Identities

In this activity, we will investigate some important trigonometric **identities**. These are relationships among trigonometric functions that are universally true.

1. **The Pythagorean Identity for Sine and Cosine.** Recall that sine and cosine are defined in terms of points on the unit circle. Explain why this means that for any angle θ, $(\cos(\theta))^2 + (\sin(\theta))^2 = 1$.

2. **Symmetry Identities for Sine and Cosine.**

   (a) Examine the graph of $y = \sin(x)$. Explain why, in terms of transformations of graphs and symmetry, for any angle x, $\sin(-1 \cdot x) = -1 \cdot \sin(x)$

   (b) Examine the graph of $y = \cos(x)$. Explain why, in terms of transformations of graphs and symmetry, for any angle x, $\cos(-1 \cdot x) = \cos(x)$.

3. **Co-Function Identities for Sine and Cosine.** Sine and Cosine, Tangent and Cotangent, and Secant and Cosecant are called **co-function** pairs, or simply co-functions. The "co-" in co-function (and in cosine, cotangent, and cosecant) comes from complementary angles. Recall that complementary angles are angles that add up to a right angle. Thus, if θ and ϕ are complementary angles, $\theta + \phi = \frac{\pi}{2}$, or equivalently $\phi = \frac{\pi}{2} - \theta$ or $\theta = \frac{\pi}{2} - \phi$.

   (a) Use Desmos or a graphing calculator to graph $\sin\left(x - \frac{\pi}{2}\right)$ and $\cos(x)$ and describe their relationship.

   (b) Explain, based on previous exercises in this worksheet, why $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$.

   Hint. $\frac{\pi}{2} - x = -1 \cdot \left(x - \frac{\pi}{2}\right)$.

   (c) Substitute $x = \frac{\pi}{2} - \theta$ in the identity $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$. Simplify $\frac{\pi}{2} - (\frac{\pi}{2} - \theta)$. What identity do you now have for $\cos\left(\frac{\pi}{2} - \theta\right)$?

4. **Angle Sum Identities for Sine and Cosine.** For this exercise, we will use Figure 33. This figure begins with the unit circle with center $O = (0, 0)$ and the x- and y-axes, with point B at angle θ and point A at angle $\theta + \phi$ on the unit circle. We then draw the following constructions.

   - Draw segment $\overline{OA}$;
   - draw segment $\overline{OB}$;
   - draw a perpendicular line from A to $\overline{OB}$, intersecting at C;
   - draw a vertical line from A to the x-axis, intersecting at F;

- draw a vertical line from C to the x-axis, intersecting at D;
- draw a horizontal segment from C to $\overline{AF}$, intersecting at E;
- draw a vertical line from B to the x-axis, intersecting at G.

We have highlighted $\triangle AEC$ in blue and $\triangle ODC$ in green.

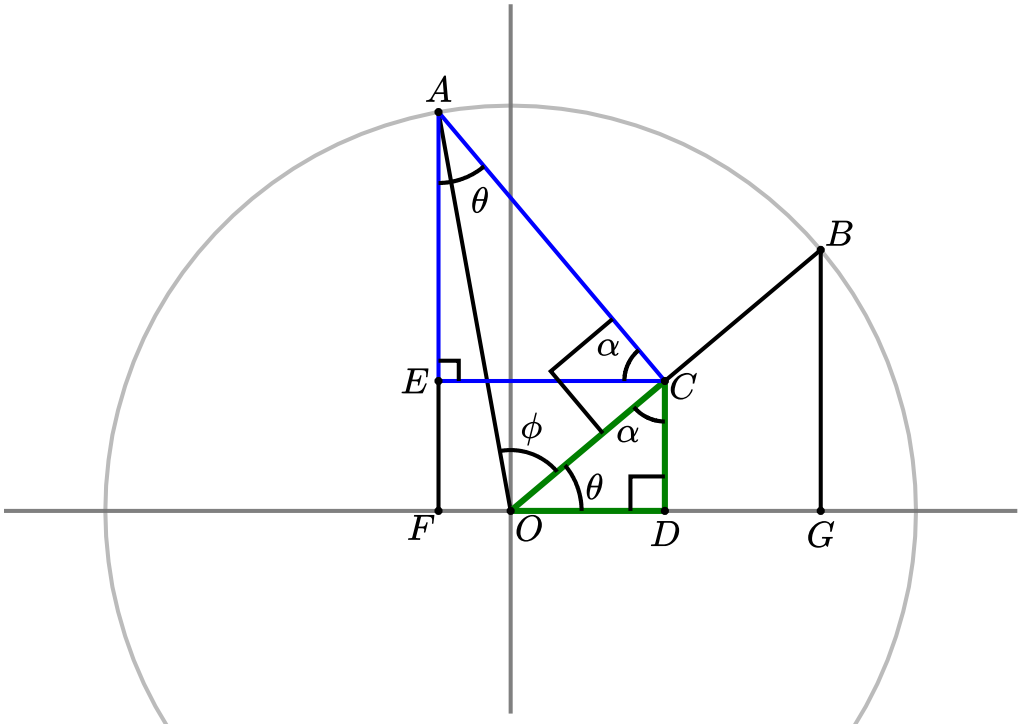


Figure 33 The Unit Circle with Points at Angles θ and $\theta + \phi$

(a) Explain why $\angle ACE \cong \angle OCD$.

(b) Explain why $\angle EAC \cong \angle DOC$.

(c) Explain why $AC = \sin(\phi)$ and $OC = \cos(\phi)$.

Hint. Rotate the unit circle and all points so that B lies on the x-axis.

(d) Explain why $\frac{DC}{OC} = \sin(\theta)$ and $\frac{OD}{OC} = \cos(\theta)$

(e) Explain why $\frac{EC}{AC} = \sin(\theta)$ and $\frac{EA}{AC} = \cos(\theta)$

(f) Note that $\sin(\theta+\phi) = FA = FE + EA$ and $FE = DC$. Use Task c, Task d, Task e to express $\sin(\theta+\phi)$ in terms of $\sin(\theta)$, $\cos(\theta)$, $\sin(\phi)$, and $\cos(\phi)$.

(g) Note that $\cos(\theta+\phi) = OD - DF$ and $DF = CE$. Use Task c, Task d, Task e to express $\cos(\theta+\phi)$ in terms of $\sin(\theta)$, $\cos(\theta)$, $\sin(\phi)$, and $\cos(\phi)$.

(Note that in this figure, $\theta+\phi$ is in the second quadrant, so $\cos(\theta+\phi) < 0$.)

The specific geometric arguments in this activity are only valid when $\theta > 0$, $\phi > 0$, and $\theta + \phi < \pi$, but similar arguments can be made, making note of when values of sine and cosine are negative, for other cases, and the angle sum identities are valid for all values of θ and ϕ.

5. Angle Difference Identities for Sine and Cosine.

(a) Rewriting $\theta - \phi$ as $\theta + (-1 \cdot \phi)$ and using Worksheet Exercise 2, Worksheet Exercise 4, express $\sin(\theta-\phi)$ in terms of $\sin(\theta)$, $\cos(\theta)$, $\sin(\phi)$, and $\cos(\phi)$.

(b) Rewriting $\theta - \phi$ as $\theta + (-1 \cdot \phi)$ and using Worksheet Exercise 2, Worksheet Exercise 4, express $\cos(\theta-\phi)$ in terms of $\sin(\theta)$, $\cos(\theta)$, $\sin(\phi)$, and $\cos(\phi)$.

4.9 The Dot Product, Angles, and Projection

In this activity, we will extend our definition of vector, investigate an operation on vectors called the **dot product**, and use the dot product to determine angles and **projection** (technically **orthogonal projection**) of a vector onto a vector.

4.9.1 More Vectors

So far, we have limited our study of vectors to vectors such as $\vec{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$. We have discussed them in relation to points in the xy-plane. However, we live in a three dimensional world[3]. We can describe points using x, y, and z coordinates, like $P = (3, 5, 2)$. Typically we visualize this by thinking of rotating the xy-plane so that it is horizontal (like a sheet of paper on a desk) and using the z-coordinate to describe how far above or below the plane a point is. Thus, $(3, 5, 2)$ is the point that is exactly 2 units directly above the point $(3, 5)$ in the xy-plane.

Since we can have points in three dimensions, it makes sense that we could have vectors with three entries, too, such as $\begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$. However, there are situations where it makes sense to have vectors that have 4, 5, and even more entries. This leads us to the following definition.

Definition 34 For any positive integer, n, the set of all vectors consisting of an array of n real numbers is referred to as $\mathbb{R}^n$. Any vector in this set is called an **n-vector**. That is,

$$\mathbb{R}^n = \left\{ \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} : a_1, a_2, \ldots, a_n \in \mathbb{R} \right\}$$

◆

So, for example, we would say that $\begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix}$ is a 3-vector, the set of all 3-vectors is $\mathbb{R}^3$, $\begin{bmatrix} 3 \\ -5 \\ 2 \\ -2 \end{bmatrix}$ is a 4-vector, and the set of all 4-vectors is $\mathbb{R}^4$.

Definition 35 Vector Arithmetic. For any n-vectors $\vec{a}$ and $\vec{b}$, the **vector sum** $\vec{a} + \vec{b}$ is the n-vector obtained by adding the entries of $\vec{a}$ and $\vec{b}$ in corresponding positions.

For any n-vector $\vec{a}$ and and real number b, the **scalar product** $b\vec{a}$ is the n-vector obtained by multiplying all the entries of $\vec{a}$ by b, and $b\vec{a}$ is said to be a **scalar multiple** of $\vec{a}$. ◆

4.9.2 The Dot Product

We will now define the **dot product** of two vectors.

Definition 36 The Dot Product. Given a positive integer n, the **dot product** of two n-vectors, $\vec{a}$ and $\vec{b}$ obtained by multiplying corresponding entries of the vectors and adding the results. That is,

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 + \cdots + a_n b_n.$$

The dot product is also sometimes called the **inner product** and written as $\left\langle \vec{b}, \vec{a} \right\rangle$ or $\left\langle \vec{b} \middle| \vec{a} \right\rangle$. ◆

Theorem 37 *For any n-vector, $\vec{v}$, the dot product of the vector with itself is the square of its magnitude. That is,*

$$\vec{v} \cdot \vec{v} = |\vec{v}|^2.$$

Proof. We will prove this only for the case of 2-vectors.

[3] Physicists actually use models that describe the number of physical dimensions as more than three, but that's beyond the scope of this course.

Let $\vec{v} = \begin{bmatrix} a \\ b \end{bmatrix}$ be any 2-vector. That is, let a and b in the definition of $\vec{v}$ be any real numbers. We will show that $\vec{v} \cdot \vec{v} = |\vec{v}|^2$.
First, we note from the definition of magnitude of a vector that $|\vec{v}| = \sqrt{a^2 + b^2}$, and so $|\vec{v}|^2 = a^2 + b^2$.
Using the definition of dot product, we have

$$\begin{aligned} \vec{v} \cdot \vec{v} &= a \cdot a + b \cdot b \\ &= a^2 + b^2 \\ &= |\vec{v}|^2 \end{aligned}$$

as was to be shown. ■

Checkpoint 38 Calculate the dot product for each given pair of vectors.

(a) $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

Answer. $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2.$

(b) $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Answer. $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 0.$

(c) $\begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

Answer. $\begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = 16.$

(d) $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Answer. $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0.$

(e) $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -3 \end{bmatrix}$

Answer. $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -3 \end{bmatrix} = 0.$

Checkpoint 39 For each of the given pairs of vectors, find the direction of each vector in terms of slope. Then, calculate the dot product of the vectors. What pattern do you notice about the slopes? What pattern do you notice about the dot products?

(a) $\begin{bmatrix} 3 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(b) $\begin{bmatrix} 5 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} 7 \\ -5 \end{bmatrix}$

(c) $\begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(d) $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(e) $\begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -3 \end{bmatrix}$

Checkpoint 40 Recall that any vector can be expressed in terms of its length and a unit vector, and that every unit vector has the form $\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}$. So, let $\vec{v}_2$ and $\vec{v}_2$ be any vectors, and we can write

$$\vec{v}_1 = \ell_1 \begin{bmatrix} \cos(\theta_1) \\ \sin(\theta_1) \end{bmatrix} \qquad \vec{v}_2 = \ell_2 \begin{bmatrix} \cos(\theta_2) \\ \sin(\theta_2) \end{bmatrix}$$
$$= \begin{bmatrix} \ell_1 \cos(\theta_1) \\ \ell_1 \sin(\theta_1) \end{bmatrix} \qquad = \begin{bmatrix} \ell_2 \cos(\theta_2) \\ \ell_2 \sin(\theta_2) \end{bmatrix}$$

where $\ell_1 = |\vec{v}_1|$ and $\ell_2 = |\vec{v}_2|$.

Calculate $\vec{v}_1 \cdot \vec{v}_2$ and use algebra and trigonometric identities to verify that

$$\vec{v}_1 \cdot \vec{v}_2 = |\vec{v}_1|\,|\vec{v}_2| \cos(\theta_1 - \theta_2).$$

Checkpoint 40 gives us a *geometric* interpretation of the dot product: the dot product of two vectors is the product of the lengths of the vectors multiplied by the cosine of the angle between them. We also saw, in Checkpoint 39, that when two vectors are perpendicular (or more generally, **orthogonal**) to each other, their dot product is zero. This is consistent, since $\cos(90°) = \cos\left(\frac{\pi}{2}\right) = 0$.

If the angle between two vectors is greater than $\frac{\pi}{2}$, then the cosine of the angle is negative. Thus, the dot product gives us an indication of how much two vectors point in the same general direction versus or in generally opposite directions.

4.9.3 Orthogonal Projection

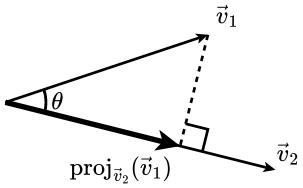


Figure 41 Orthogonal Projection of $\vec{v}_1$ onto $\vec{v}_2$

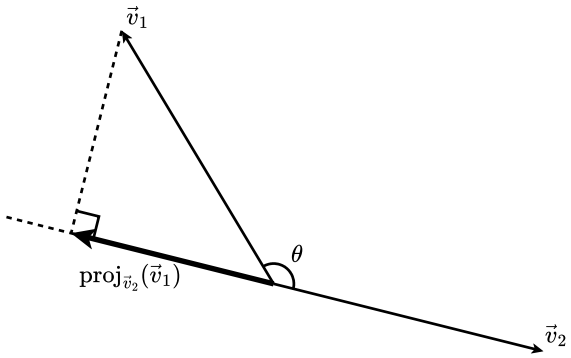


Figure 42 Orthogonal Projection of $\vec{v}_1$ onto $\vec{v}_2$

An important use of the dot product is in **orthogonal projection** of one vector onto another. This allows us to easily decompose a vector into a part that points in the direction of (or in the direction opposite to) another vector. We denote the projection of $\vec{v}_1$ onto $\vec{v}_2$ by $\text{proj}_{\vec{v}_2}(\vec{v}_1)$.

For vectors that point more or less in the same direction, as shown in Figure 41, the projection of $\vec{v}_1$ onto $\vec{v}_2$ points in the direction of $\vec{v}_2$, but its length is determined by the angle between the vectors as well as the length of the vector $\vec{v}_1$.

Referring to Figure 41, let $\vec{p} = \text{proj}_{\vec{v}_2}(\vec{v}_1)$. We can see that $\frac{|\vec{p}|}{|\vec{v}_1|} = \cos(\theta)$, so $|\vec{p}| = |\vec{v}_1|\cos(\theta)$. The direction of $\vec{p}$ is the direction of $\vec{v}_2$, which can be expressed as the unit vector $\vec{u}_2 = \frac{1}{|\vec{v}_2|}\vec{v}_2$.

Since a vector is determined by its magnitude and its direction, we see that

$$\begin{aligned} \text{proj}_{\vec{v}_2}(\vec{v}_1) &= |\vec{v}_1| \cos(\theta) \frac{1}{|\vec{v}_2|} \vec{v}_2 \\ &= \frac{|\vec{v}_1|}{|\vec{v}_2|} \cos(\theta) \vec{v}_2 \\ &= \frac{|\vec{v}_1|\,|\vec{v}_2| \cos(\theta)}{|\vec{v}_2|^2} \vec{v}_2 \end{aligned}$$

$$= \left(\frac{\vec{v}_1 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \right) \vec{v}_2$$

In the case that the angle between the vectors is greater than $\pi/2$, the cosine will be negative, but the direction will be $-\vec{u}_2$, so that the same formula applies. Thus, we have the following definition.

Definition 43 Orthogonal Projection of One Vector Onto Another. Given any vector, $\vec{v}_1$ and any non-zero vector $\vec{v}_2$, the **orthogonal projection** of $\vec{v}_1$ onto $\vec{v}_2$ is given by

$$\text{proj}_{\vec{v}_2}(\vec{v}_1) = \left(\frac{\vec{v}_1 \cdot \vec{v}_2}{\vec{v}_2 \cdot \vec{v}_2} \right) \vec{v}_2.$$

If $\vec{p} = \text{proj}_{\vec{v}_2}(\vec{v}_1)$, then $\vec{p}$ is perpendicular to $\vec{v}_1 - \vec{p}$, and $|\vec{v}_1 - \vec{p}|$ is as small as possible. ◆

4.9.4 Practical Application of the Dot Product

In this subsection, our goal is to solve the following problem.

A box with a mass of 15 kilograms is being pushed up a ramp. The ramp covers 3m horizontally and rises 0.2m. Ignoring friction, determine the minimum amount of force that must be applied to the box in the horizontal direction in order to overcome gravity and move the cart up the ramp.

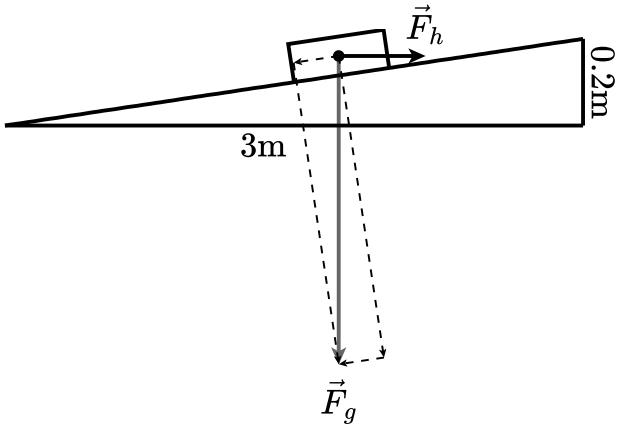


Figure 44 A 3 Meter Ramp with a 15kg Object

1. The object has a mass of 15kg, and from Newton's first law, force is equal to mass times acceleration. In this case, since we are using SI (metric) units, acceleration due to gravity is 9.8m/s^2, and gravity accelerates objects straight down. (The SI unit of force is a **Newton**, defined to be $1\text{N} = 1\text{kg} \cdot \text{m/s}^2$.)

Express the force from gravity as a vector, $\vec{F}_g$.

Solution. The magnitude is

$$9.8\text{m/s}^2 \cdot 15\text{kg} = 147\text{N}$$

and the direction is directly downward, which is indicated by the unit vector $\langle 0, -1 \rangle$. Therefore, we have

$$\begin{aligned} \vec{F}_g &= 147 \begin{bmatrix} 0 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ -147 \end{bmatrix}. \end{aligned}$$

2. Express the movement from the start of the ramp to the end of the ramp as a vector, $\vec{R}$.

Solution. The movement from the bottom of the ramp to the top is 3m to the right and 0.2m upward. Thus

$$\vec{R} = \begin{bmatrix} 3 \\ 0.2 \end{bmatrix}.$$

3. Calculate $\vec{F}_R = \text{proj}_{\vec{R}}(\vec{F}_g)$.

Solution. Using the formula for orthogonal projection, we have

$$\begin{aligned} \vec{F}_R &= \text{proj}_{\vec{R}}(\vec{F}_g) \\ &= \frac{\vec{F}_g \cdot \vec{R}}{\vec{R} \cdot \vec{R}} \vec{R} \\ &= \frac{\begin{bmatrix} 0 \\ -147 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0.2 \end{bmatrix}}{\begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0.2 \end{bmatrix}} \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \end{aligned}$$

$$
\begin{aligned}
&= \frac{0 \cdot 3 + 0.2 \cdot -147}{3 \cdot 3 + 0.2 \cdot 0.2} \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= \frac{-29.4}{9 + 0.04} \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= -3.25221239 \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= \begin{bmatrix} -9.7566 \\ -0.65044 \end{bmatrix}
\end{aligned}
$$

4. Calculate $\vec{F}_\perp = \vec{F}_g - \vec{F}_R$.

Solution. We have

$$
\begin{aligned}
\vec{F}_\perp &= \vec{F}_g - \vec{F}_R \\
&= \begin{bmatrix} 0 \\ -147 \end{bmatrix} - \begin{bmatrix} -9.7566 \\ -0.65044 \end{bmatrix} \\
&= \begin{bmatrix} 9.7566 \\ -146.34955 \end{bmatrix}
\end{aligned}
$$

5. Verify that $\vec{F}_\perp \cdot \vec{F}_R = 0$, and therefore they are perpendicular to each other.

Solution.

$$
\begin{aligned}
\vec{F}_\perp \cdot \vec{F}_R &= (9.7566) \cdot (-9.7566) + (-146.34955)(-0.65044) \\
&= (-95.19124) + (95.19124) \\
&= 0.
\end{aligned}
$$

6. Let $\vec{F}_h$ be the horizontal pulling force, given by

$$
\vec{F}_h = \begin{bmatrix} \alpha \\ 0 \end{bmatrix}
$$

where the parameter α is an unspecified quantity.

Find $\text{proj}_{\vec{R}}(\vec{F}_h)$.

Compare $\text{proj}_{\vec{R}}(\vec{F}_h)$ to $\text{proj}_{\vec{R}}(\vec{F}_g)$ to determine the value of α needed to offset the effect of gravity.

Solution.

$$
\begin{aligned}
\text{proj}_{\vec{R}}(\vec{F}_h) &= \frac{\vec{F}_h \cdot \vec{R}}{\vec{R} \cdot \vec{R}} \vec{R} \\
&= \frac{\begin{bmatrix} \alpha \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0.2 \end{bmatrix}}{\begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0.2 \end{bmatrix}} \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= \frac{\alpha \cdot 3 + 0 \cdot 0.2}{3 \cdot 3 + 0.2 \cdot 0.2} \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= 0.331858\alpha \begin{bmatrix} 3 \\ 0.2 \end{bmatrix} \\
&= \begin{bmatrix} 0.995575\alpha \\ 0.066372\alpha \end{bmatrix}
\end{aligned}
$$

Comparing $\vec{F}_R$ and $\text{proj}_{\vec{R}}(\vec{F}_h)$, in order to have $\text{proj}_{\vec{R}}(\vec{F}_h) = -\vec{F}_R$, we need $0.995575221238938\alpha = 9.75663716814159$, which yields $\alpha = 9.8$.

If we look back at our calculations, we might see that we are solving

$$
\text{proj}_{\vec{R}}(\vec{F}_g) = -\text{proj}_{\vec{R}}(\vec{F}_h)
$$

which means we are finding

$$\frac{\vec{F}_g \cdot \vec{R}}{\vec{R} \cdot \vec{R}} \vec{R} = -\frac{\vec{F}_h \cdot \vec{R}}{\vec{R} \cdot \vec{R}} \vec{R}.$$

Note that the only difference between the two sided of that equation is the use of $\vec{F}_g \cdot \vec{R}$ on the left and $-\vec{F}_h \cdot \vec{R}$ on the right. So, in effect, we are solving $\vec{F}_g \cdot \vec{R} = -\vec{F}_h \cdot \vec{R}$.

This equation becomes $3\alpha = 0.2 \cdot 147$, which gives $\alpha = (0.2 \cdot 147/3) = (1/5)(147/3) = 147/15 = 9.8$.

4.10 Vectors and Inverse Trigonometric Functions

In this activity, we explore the interpretation of vectors as quantities that have a **magnitude** and a **direction**, and we investigate how to move back and forth between this interpretation and the interpretation as a list of coordinates.

4.10.1 From Direction and Magnitude to Coordinates

Checkpoint 45 Investigate the following questions using Figure 46.

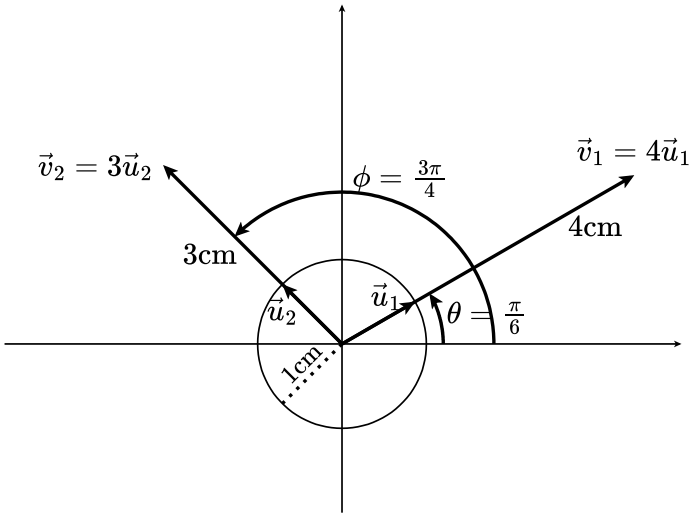


Figure 46 Vectors and the Unit Circle

(a) **(i)** The vector $\vec{u}_1$ has length 1cm and makes an angle of $\theta = \frac{\pi}{6}$radians ("theta" radians) with the positive x-axis. Find its coordinates.

Hint. The tip of $\vec{u}_1$ is on the unit circle.

(ii) The vector $\vec{v}_1$ has length 4cm and makes an angle of $\frac{\pi}{6}$radians with the positive x-axis. Find its coordinates.

Hint. How is $\vec{v}_1$ related to $\vec{u}_1$?

(b) **(i)** The vector $\vec{u}_2$ has length 1cm and makes an angle of $\phi = \frac{3\pi}{4}$radians ("phi" radians) with the positive x-axis. Find its coordinates.

(ii) The vector $\vec{v}_2$ has length 3cm and makes an angle of $\frac{3\pi}{4}$radians with the positive x-axis. Find its coordinates.

(c) **(i)** Find the coordinates of the unit vector, $\vec{u}_3$ that makes an angle of $\alpha = \frac{5\pi}{4}$radians ("alpha") with the positive x-axis.

(ii) Find the coordinates of the vector, $\vec{v}_3$ that has length $5\sqrt{2}$ and makes an angle of $\alpha = \frac{5\pi}{4}$radians with the positive x-axis.

(d) **(i)** Given any angle, β ("beta"), explain how to find the coordinates of the unit vector that makes that angle with the positive x-axis.

(ii) Given any angle, β, and any positive length, r, explain how to find the coordinates of the vector of that length that makes that angle with the positive x-axis.

(e) **(i)** The vector $\vec{u}_{1a}$ has length 1cm and makes an angle of $\theta = \frac{13\pi}{6}$radians ("theta" radians) with the positive x-axis. Find its coordinates.

(ii) The vector $\vec{v}_{1a}$ has length 4cm and makes an angle of $\frac{13\pi}{6}$radians with the positive x-axis. Find its coordinates.

(iii) Compare your results from Task 45.a and Task 45.e. What do you notice?

(f) Using a calculator, find (approximately) the coordinates of the vector $\vec{v}_4$ that has length 2 and makes an angle $\gamma = 2$rad ("gamma").

Checkpoint 47

(a) Let $\vec{V} = \begin{bmatrix} \sqrt{3} \\ -1 \end{bmatrix}$.

(i) What is the magnitude (length) of $\vec{V}$? (In other words, calculate $\left|\vec{V}\right|$.)

(ii) What are the coordinates of a unit vector, call it $\vec{U}$, that points in the same direction as $\vec{V}$?

(iii) Find an angle, ζ ("zeta") so that $\vec{U} = \begin{bmatrix} \cos(\zeta) \\ \sin(\zeta) \end{bmatrix}$.

(iv) Find at least one other valid answer for Task 47.a.iii.

Hint. What happens if you go completely around the unit circle before getting to where $\vec{U}$ is?

(v) Explain why, given coordinates for a vector, we cannot say "the angle" that it makes with the positive x-axis, but must instead say "an angle".

4.10.2 The Inverse Cosine Function

As we saw in Checkpoint 47 (particularly Task 47.a.v), given a value c with $-1 \leq c \leq 1$, we can't find *the* angle θ so that $\cos(\theta) = c$, be we can find *an* angle. This isn't really a huge problem in general, but it is a problem if we want to have a *function*, a sort of "un-cosine", that takes c as its input and gives an angle, because *a function must have one unique output for each input.* So, if we want to undo the cosine function, we need to make a decision on how to choose *the* angle.

4.10.2.1 The Principle Branch of Cosine

Figure 48 shows the graph $y = \cos(x)$, and a subsection of the graph is shown in bold, from $x = 0$ to $x = \pi$, so that every value of y from -1 to 1 occurs exactly once. By **restricting the domain** of cosine to $0 \leq x \leq \pi$ we get what is called the **principle branch** of cosine, and it is sometimes written with a capital C as $\text{Cos}(x)$.

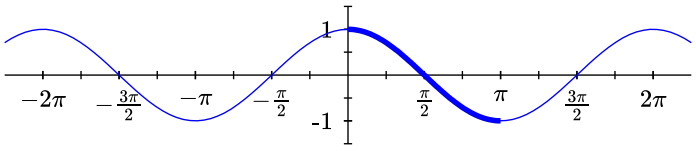


Figure 48 The Cosine Function and its Principle Branch

4.10.2.2 The "Un-Cosine"

Now, we can find a function that undoes this branch of cosine: the "un-cosine" of a number c is the *unique* angle θ so that: $\cos(\theta) = c$ ***and*** $0 \leq \theta \leq \pi$. For now, we will write $\text{uncos}(c)$ and give the following definition.

Definition 49 The "Un-Cosine" Function. For any value x with $-1 \leq x \leq 1$, saying $\text{uncos}(x) = \theta$ means that θ is the (unique) angle so that

(a) $\cos(\theta) = x$, and

(b) $0 \leq \theta \leq \pi$.

◆

Example 50 We will find $\text{uncos}(x)$ for several values of x using the unit circle.

1. To find $\text{uncos}\left(\frac{\sqrt{3}}{2}\right)$, we first locate points where the x-coordinate is $\frac{\sqrt{3}}{2}$, because x-coordinates correspond to the cosine. The points where the x-coordinate is $\frac{\sqrt{3}}{2}$ are $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ at $\theta = \frac{\pi}{6}$ and $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ at $\theta = \frac{11\pi}{6}$. Since $0 \leq \frac{\pi}{6} \leq \pi$,

$$\text{uncos}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}.$$

2. To find uncos $\left(-\frac{1}{2}\right)$, we first locate points where the x-coordinate is $-\frac{1}{2}$, because x-coordinates correspond to the cosine. The points where the x-coordinate is $-\frac{1}{2}$ are $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{2\pi}{3}$ and $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{4\pi}{3}$. Since $0 \leq \frac{2\pi}{3} \leq \pi$,

$$\text{uncos}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}.$$

▲

Checkpoint 51

(a) Find uncos $\left(\frac{\sqrt{2}}{2}\right)$.

(b) Find uncos $\left(-\frac{\sqrt{3}}{2}\right)$.

(c) Find uncos $\left(\frac{1}{2}\right)$.

4.10.2.3 Notation

The name we chose for our function, "un-cosine", is intuitive, but it's not what the standard notational conventions are. Much more common are the **arc-cosine**, $\arccos(x)$, named this because we finding the angle in the unit circle is the same as finding the arc, and **inverse cosine**, which is confusingly written as $\cos^{-1}(x)$.

4.10.2.4 Using Inverse Cosine

Example 52 We will use the inverse cosine/arc-cosine to find the angle θ in the third quadrant so that $\cos(\theta) = -0.3$.

Using a calculator, let $\theta_* = \arccos(-0.3) = 1.875488981$. We note that $\pi/2 \approx 1.570796327 < 1.875488981 < 3.141592654 \approx \pi$, so we see that θ_* is in the *second* quadrant, as shown in Figure 53. This is because $\arccos(x)$ always gives an angle in either the first or second quadrant.

Instead, to get an angle in the third quadrant with the same cosine, we need $-\theta_*$, or equivalently, $\theta = 2\pi - \theta_* = 4.407696326$.

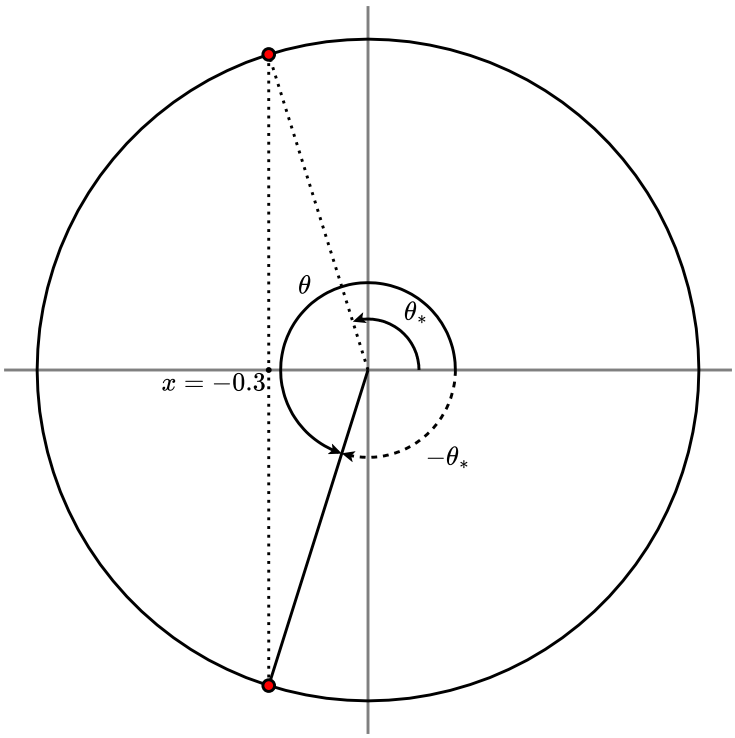


Figure 53 The Unit Circle Showing Points with $x = -0.3$.

We check with our calculator that $\cos(4.407696326) = -0.3$, and $\pi < 4.407696326 < 3\pi/2$, and thus is in the third quadrant, as desired. ▲

4.10.3 The Inverse Sine Function

Just like with cosine, we can't find *the* angle θ so that $\sin(\theta) = s$, be we can find *an* angle. So, to find an "un-sine" function that takes s between -1 and 1 as its input and gives an angle, we again need to find a **branch**.

4.10.3.1 The Principle Branch of Sine

Checkpoint 54 Use the graph of $y = \sin(x)$ in Figure 55 to find a subsection of the graph, $a \le x \le b$, that every value of y from -1 to 1 occurs exactly once in that range of x values.

Solution. One choice is $-\frac{1}{2}\pi \le x \le \frac{1}{2}\pi$. This is called the **principle branch** of sine, sometimes written with a capital S, $\text{Sin}(x)$.

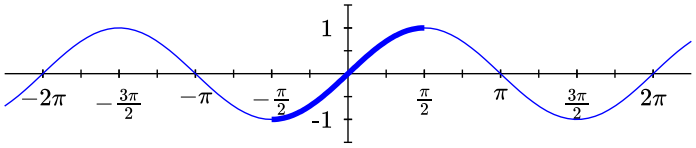


Figure 55 The Sine Function

4.10.3.2 The "Un-Sine"

Now, we can find a function that undoes this branch of sine: the "un-sine" of a number c is the *unique* angle θ so that: $\sin(\theta) = c$ ***and*** $0 \le \theta \le \pi$. For now, we will write $\text{unsin}(c)$ and give the following definition.

Definition 56 The "Un-Sine" Function. For any value x with $-1 \le x \le 1$, saying $\text{unsin}(x) = \theta$ means that θ is the (unique) angle so that

(a) $\sin(\theta) = x$, and

(b) $-\frac{1}{2}\pi \le \theta \le \frac{1}{2}\pi$.

◆

Example 57 We will find $\text{unsin}(x)$ for several values of x using the unit circle.

1. To find $\text{unsin}\left(\frac{\sqrt{3}}{2}\right)$, we first locate points where the y-coordinate is $\frac{\sqrt{3}}{2}$, because y-coordinates correspond to the sine. The points where the y-coordinate is $\frac{\sqrt{3}}{2}$ are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{\pi}{3}$ and $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{2\pi}{3}$. Since $0 \le \frac{\pi}{3} \le \pi$,
$$\text{unsin}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}.$$

2. To find $\text{unsin}\left(-\frac{1}{2}\right)$, we first locate points where the y-coordinate is $-\frac{1}{2}$, because y-coordinates correspond to the sine. The points where the y-coordinate is $-\frac{1}{2}$ are $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ at $\theta = \frac{11\pi}{6}$ and $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ at $\theta = \frac{7\pi}{6}$. Neither of these lie in the interval $\left[-\frac{1}{2}\pi, \frac{1}{2}\pi\right]$, though. Instead, we will look at negative angles. The point at $\theta = \frac{11\pi}{6}$ is also at $\theta = -\frac{\pi}{2}$. This is in the correct range, so
$$\text{unsin}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}.$$

▲

Checkpoint 58

(a) Find $\text{unsin}\left(\frac{\sqrt{2}}{2}\right)$.

(b) Find $\text{unsin}\left(-\frac{\sqrt{3}}{2}\right)$.

(c) Find $\text{unsin}\left(\frac{1}{2}\right)$.

4.10.3.3 Notation

The name we chose for our function, "un-sine", is intuitive, but it's not what the standard notational conventions are. Much more common are the **arc-sine**, $\arcsin(x)$, named this because we finding the angle in the unit circle is the same as finding the arc, and **inverse sine**, which is confusingly written as $\sin^{-1}(x)$.

4.10.3.4 Using the Inverse Sine

Example 59 We will find an angle in the second quadrant whose sine is 0.6.

We start by finding points on the unit circle with y-coordinate 0.6 as shown in Figure 60.

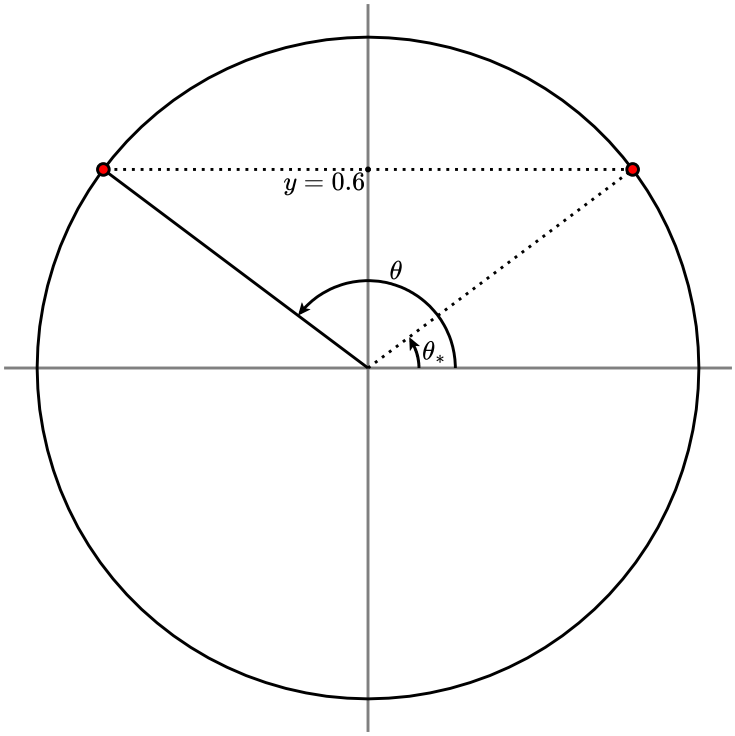


Figure 60 The Unit Circle Showing Points with $y = 0.6$.

The inverse sine always returns an angle between $-\pi/2$ and $\pi/2$, so when we calculate $\arccos(0.6)$ we get $\theta_* \approx 0.6435011$, which is in the first quadrant. This is our **reference angle**. The angle, θ, that we want will be $\pi - \theta_* \approx 2.49809$. We verify that $\sin(2.49809) = 0.6$, and $\pi/2 < 2.49809 < \pi$, so the angle is in the second quadrant, as desired. ▲

4.10.4 The Tangent and the Inverse Tangent

The tangent function is a little bit different from sine and cosine, because not only do values of the tangent repeat, but there are points where the tangent is not defined at all.

4.10.4.1 The Graph of the Tangent Function

As shown in Figure 61, the graph $y = \tan(x)$ is undefined with vertical asymptotes at $x = -\frac{3}{2}\pi, -\frac{1}{2}\pi, \frac{1}{2}\pi, \frac{3}{2}\pi$ and so forth. In fact for any whole number (or **integer**) k, $\tan(x)$ is undefined at $x = k\pi + \frac{1}{2}\pi = \frac{(2k+1)}{2}\pi$.

As we come up on one of these **discontinuities** from the left side, values of the tangent grow larger and larger. We formally say that as $x \to \frac{1}{2}\pi$, for example, from the left, the tangent *increases without bound*, and sometimes we write $\tan(x) \to \infty$ as $x \to \frac{1}{2}\pi^-$.

Similarly, as we come up on one of these discontinuities from the right side, values of the tangent grow smaller and smaller, or more and more negative, and we say that as $x \to \frac{1}{2}\pi$ from the right, the tangent *decreases without bound*, and sometimes we write $\tan(x) \to -\infty$ as $x \to \frac{1}{2}\pi^+$.

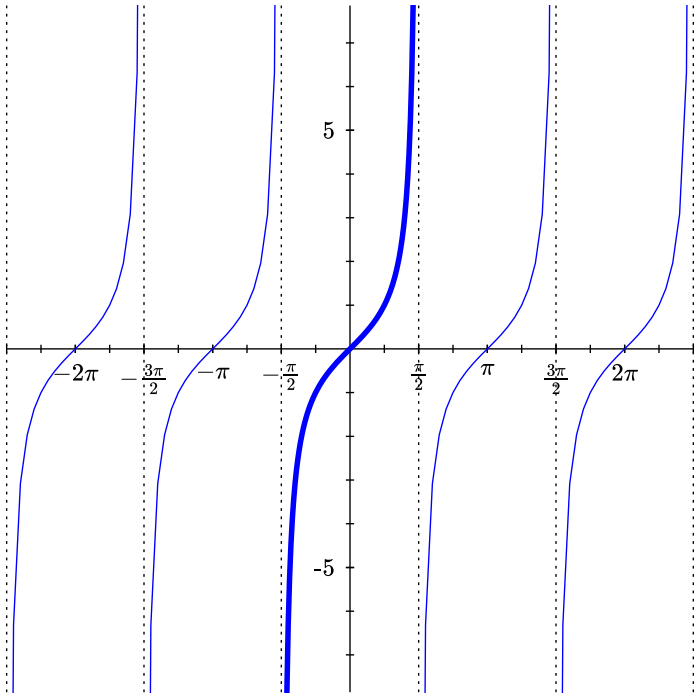


Figure 61 The Tangent Function

4.10.4.2 The Principle Branch of Tangent

Checkpoint 62 Use the graph of $y = \tan(x)$ in Figure 61 to find a subsection of the graph, $a < x < b$, that every real number value of y occurs exactly once in that range of x values.

Solution. One choice is $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$. This is called the **principle branch** of tangent, sometimes written with a capital T, $\text{Tan}(x)$.

4.10.4.3 The "Un-Tangent"

Now, we can find a function that undoes this branch of tangent: the "un-tangent" of a number c is the *unique* angle θ so that: $\tan(\theta) = c$ ***and*** $0 < \theta < \pi$. For now, we will write $\text{untan}(c)$ and give the following definition.

Definition 63 The "Un-Tangent" Function. For any real x, saying $\text{untan}(x) = \theta$ means that θ is the (unique) angle so that

(a) $\tan(\theta) = x$, and

(b) $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$.

◆

Example 64 We will find $\text{untan}(x)$ for several values of x using the unit circle.

1. To find $\text{untan}\left(\sqrt{3}\right)$, we first locate points where the slope is $\sqrt{3}$, because $\tan(\theta) = \frac{y}{x}$. The points where the slope is $\sqrt{3}$ are $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{\pi}{3}$ and $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ at $\theta = \frac{4\pi}{3}$. Since $0 < \frac{\pi}{3} < \pi$,

$$\text{untan}\left(\sqrt{3}\right) = \frac{\pi}{3}.$$

2. To find $\text{untan}\,(-1)$, we first locate points where the slope is -1. The points where the slope is -1 are $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ at $\theta = \frac{7\pi}{4}$ and $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ at $\theta = \frac{3\pi}{4}$. Neither of these lie in the interval$\left(-\frac{1}{2}\pi, \frac{1}{2}\pi\right)$, though. Instead, we will look at negative

angles. The point at $\theta = \frac{7\pi}{4}$ is also at $\theta = -\frac{\pi}{4}$. This is in the correct range, so

$$\operatorname{untan}(-1) = -\frac{\pi}{4}.$$

▲

4.10.4.4 Notation

The name we chose for our function, "un-tangent", is intuitive, but it's not what the standard notational conventions are. Much more common are the **arc-tangent**, $\arctan(x)$, named this because we finding the angle in the unit circle is the same as finding the arc, and **inverse tangent**, which is confusingly written as $\tan^{-1}(x)$.

4.10.4.5 Using the Inverse Tangent

Example 65 We will write the vector $\vec{v} = \begin{bmatrix} -4 \\ -1 \end{bmatrix}$ in terms of its magnitude and its angle.

Figure 66 shows the vector $\vec{v}$, the unit circle, and the unit vector $\vec{u}$, the unit vector that points in the direction of $\vec{v}$.

To find this unit vector, we calculate $\vec{u} = \frac{1}{|\vec{v}|}\vec{v}$. So,

$$\begin{aligned} |\vec{v}| &= \sqrt{(-4)^2 + (-1)^2} \\ &= \sqrt{16+1} \\ &= \sqrt{17}, \end{aligned}$$

and so

$$\begin{aligned} \vec{u} &= \frac{1}{\sqrt{17}} \begin{bmatrix} -4 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{-4}{\sqrt{17}} \\ \frac{-1}{\sqrt{17}} \end{bmatrix} \end{aligned}$$

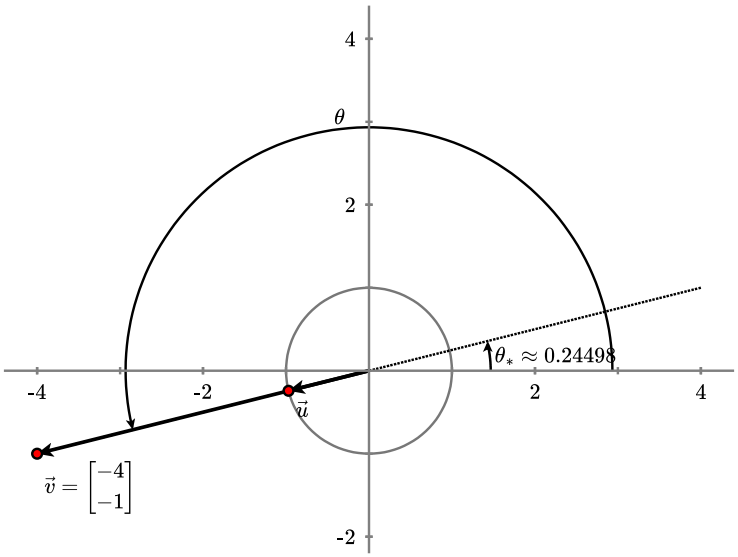


Figure 66 The vector $\vec{v}$ represented as a point and the unit circle.

Now we need to find the angle, θ, so that $\cos(\theta) = \frac{-4}{\sqrt{17}}$ and $\sin(\theta) = \frac{-1}{\sqrt{17}}$. We will do this by noting that $\tan(\theta) = \left(\frac{-1}{\sqrt{17}}\right) / \left(\frac{-4}{\sqrt{17}}\right) = (-1)/(-4) = 1/4$. It is worth noting for future reference that the tangent can be obtained directly from the original vector $\vec{v}$ by dividing the second component by the first.

We calculate $\theta_* = \arctan(1/4) \approx 0.24498$. This angle is in the first quadrant and is our **reference angle**. Because of the periodicity of tangent, the angle for our vector is $\theta = \pi + \theta_* \approx 3.38657$.

We check our answer by calculating

$$\begin{aligned}\sqrt{17}\begin{bmatrix}\cos(3.38657)\\ \sin(3.38657)\end{bmatrix} &= \begin{bmatrix}\sqrt{17}\cos(3.38657)\\ \sqrt{17}\sin(3.38657)\end{bmatrix}\\ &= \begin{bmatrix}-4\\ -1\end{bmatrix}\end{aligned}$$

▲

4.11 What Time Is It?

1. An (analog) clock has a minute hand that is 5in long. We begin observing at noon. We also have a digital clock that is synced with the analog clock.

(a) What time is showing on the digital clock (including seconds) when the tip of the minute hand is first (after noon) 3in horizontally to the left of the center of the clock? (That is, the tip of the minute hand is 3in to the left of the line through the 12 and the 6 on the clock.)

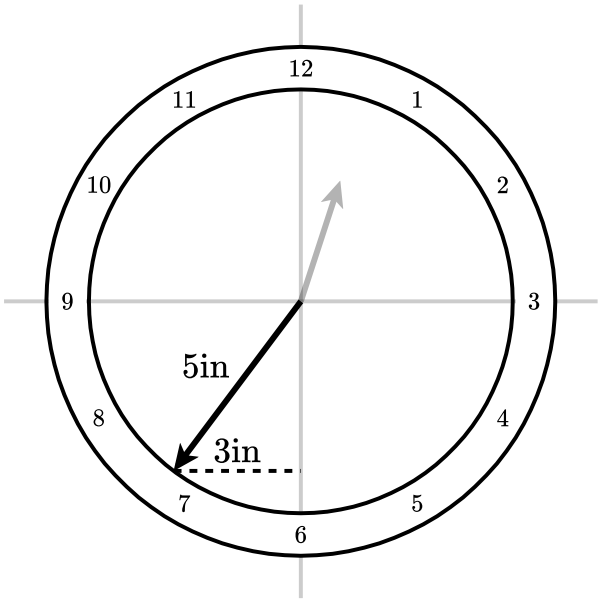


Figure 67 Minute Hand Position on an Analog Clock

(b) What time is showing on the digital clock (including seconds) when the tip of the minute hand is first (after noon) 2in vertically below the center of the clock? (That is, the tip of the minute hand is 2in below the line through the 9 and the 12 on the clock.)

4.12 Finding Forces for an Asymmetric Hanging Weight

4.12.1 Motivating Problem

A weight with a mass of 20kg is suspended from from the ceiling using two wires. One wire is 30cm long, the other is 50cm long, and the points on the ceiling where they are anchored are 60cm apart.

We want to find the magnitudes of the force along each wire.

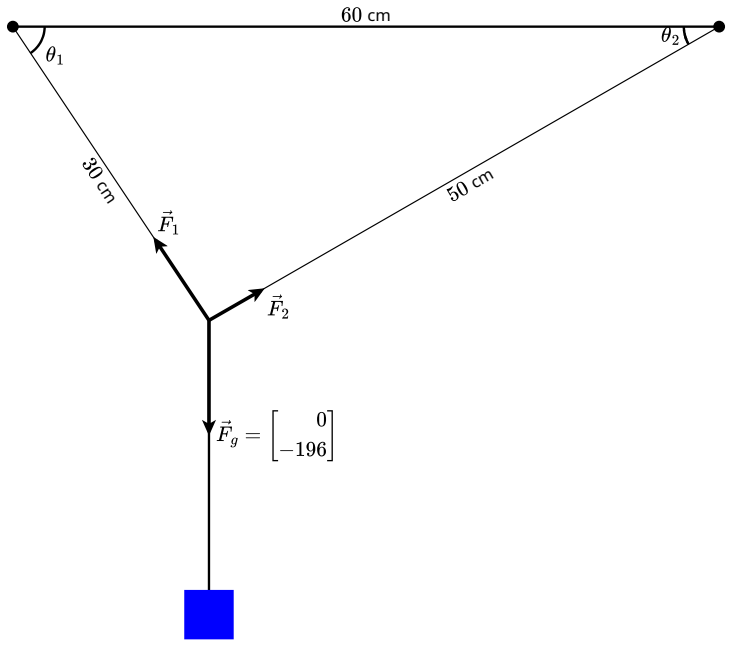


Figure 68 A Weight Suspended Asymmetrically

We could use the algebra and the trigonometric identities we have learned so far to solve this particular problem, but in mathematics there is a strong drive to solve a *category* of problem once rather than many problems that fit the category individually. Toward that end, we will now develop two important trigonometric identities: the **law of sines** and the **law of cosines**. Both apply to arbitrary triangles, such as the one shown in Figure 69 .

4.12.2 The Law of Sines

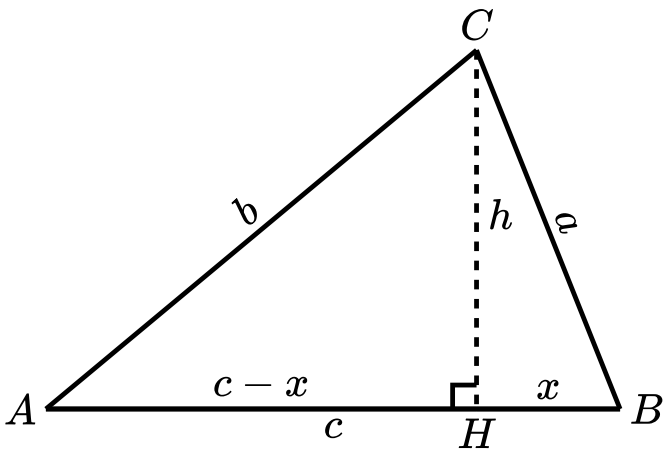


Figure 69 An Arbitrary Triangle

In $\triangle ABC$ shown in Figure 69, let the angles at points A, B, and C as α, β, and γ, respectively, and the lengths of the sides *opposite* those points as a, b, and c.

We draw the altitude from C to the line $\overline{AB}$ (that is, the segment from C perpendicular to $\overline{AB}$). We will refer to the length of the altitude as h, and we will let the altitude split $\overline{AB}$ into as segment of length x (between the altitude and B) and $c - x$ (between the altitude and A).

We will refer to the point where the altitude intersects $\overline{AB}$ as H.

Since the altitude is perpendicular to $\overline{AB}$, we have divided the original triangle into two right triangles, $\triangle AHC$ and $\triangle BHC$.

We know from applying trigonometry to right triangles that

$$\sin(A) = \frac{h}{b} \qquad \text{and} \qquad \sin(B) = \frac{h}{a}.$$

Solving each of these equations for h, we have

$$h = b\sin(A) \qquad \text{and} \qquad h = a\sin(B),$$

This means that $b\sin(A) = a\sin(B)$, which we can rearrange to give

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b}.$$

Note that by drawing a different altitude, we would also have $\frac{\sin(A)}{a} = \frac{\sin(C)}{c}$. This says that for any triangle, the ratio of the sine of an angle to the length of the opposite side will be a constant. More formally, this gives us the **law of sines**.

Theorem 70 The Law of Sines. *For any triangle with angles A, B, and C, and with side lengths opposite each angle given by a, b, and c, respectively,*

$$\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}. \tag{4.1}$$

4.12.3 The Law of Cosines

There is another important identity for general triangles called the *law of cosines*. It is simple to state, but it is a bit harder to derive, so we will state it first, set up the machinery we need to derive it, and then go through the whole derivation.

Theorem 71 The Law of Cosines. *For any triangle with angles A, B, and C, and with side lengths opposite each angle given by a, b, and c, respectively,*

$$c^2 = a^2 + b^2 - 2ab\cos(C). \tag{4.2}$$

In order to derive this identity, we need to recall a few identities we have already developed and apply them to the situation here.

First, we need the Pythagorean identity for sine and cosine: for any (real number) angle θ,

$$\sin^2(\theta) + \cos^2(\theta) = 1. \tag{4.3}$$

Note that this equation also means that $\cos^2(\theta) = \left(1 - \sin^2(\theta)\right)$ and $\sin^2(\theta) = \left(1 - \cos^2(\theta)\right)$.

Next, we will state the angle sum identities (although we will only use the identity for cosine). For any (real number) angles θ_1 and θ_2,

$$\cos(\theta_1 + \theta_2) = \cos(\theta_1)\cos(\theta_2) - \sin(\theta_1)\sin(\theta_2) \tag{4.4}$$
$$\sin(\theta_1 + \theta_2) = \cos(\theta_1)\sin(\theta_2) + \sin(\theta_1)\cos(\theta_2). \tag{4.5}$$

Also, recall that for any angle θ, $\cos(-1 \cdot \theta) = \cos(\theta)$ and $\sin(-1 \cdot \theta) = -1 \cdot \sin(\theta)$. Thus,

$$\begin{aligned} \cos(\theta_1 - \theta_2) &= \cos(\theta_1)\cos(-\theta_2) - \sin(\theta_1)\sin(-\theta_2) \\ &= \cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2) \end{aligned} \tag{4.6}$$

$$\begin{aligned} \sin(\theta_1 + \theta_2) &= \cos(\theta_1)\sin(-\theta_2) + \sin(\theta_1)\cos(-\theta_2). \\ &= -\cos(\theta_1)\sin(\theta_2) + \sin(\theta_1)\cos(\theta_2). \end{aligned} \tag{4.7}$$

Finally, a special case of the angle sum formula for cosine, using the facts that $\cos(\pi) = -1$ and $\sin(\pi) = 0$, for any angle θ

$$\begin{aligned} \cos(\pi - \theta) &= \cos(\pi)\cos(\theta) + \sin(\pi)\sin(\theta) \\ &= -1 \cdot \cos(\theta) + 0 \cdot \sin(\theta) \\ &= -\cos(\theta) \end{aligned} \tag{4.8}$$

Proof of Theorem 71. We need to deal with two different cases. In one case, we have an interior altitude, as shown in Figure 69, and in the other case, we have an exterior altitude, as shown in Figure 72.

Referring to Figure 69, we see that

$$\cos(A) = \frac{c - x}{b} \qquad \text{and} \qquad \cos(B) = \frac{x}{a},$$

so

$$\begin{aligned} c &= (c - x) + x \\ &= b\cos(A) + a\cos(B), \end{aligned}$$

which means that

$$\begin{aligned} c^2 &= (b\cos(A) + a\cos(B))^2 \\ &= b^2\cos^2(A) + 2b\cos(A)a\cos(B) + a^2\cos^2(B) \\ &= b^2\cos^2(A) + 2ab\cos(A)\cos(B) + a^2\cos^2(B). \end{aligned}$$

Using the (rearranged) Pythagorean identity twice, we have

$$\begin{aligned} c^2 &= b^2\cos^2(A) + 2ab\cos(A)\cos(B) + a^2\cos^2(B). \\ &= b^2\left(1 - \sin^2(A)\right) + 2ab\cos(A)\cos(B) + a^2\left(1 - \sin^2(B)\right). \\ &= b^2 - b^2\sin^2(A) + 2ab\cos(A)\cos(B) + a^2 - a^2\sin^2(B) \end{aligned}$$

Rearranging (4.4), we see that $\cos(\theta_1)\cos(\theta_2) = \cos(\theta_1 + \theta_2) + \sin(\theta_1)\sin(\theta_2)$ so $\cos(A)\cos(B) = \cos(A + B) + \sin(A)\sin(B)$. Substituting this into our equation for c^2 gives

$$\begin{aligned} c^2 &= b^2 - b^2\sin^2(A) + 2ab\left(\cos(A)\cos(B)\right) + a^2 - a^2\sin^2(B) \\ &= b^2 - b^2\sin^2(A) + 2ab\left(\cos(A + B) + \sin(A)\sin(B)\right) + a^2 - a^2\sin^2(B) \\ &= b^2 + 2ab\cos(A + B) + 2ab\sin(A)\sin(B) + a^2 - a^2\sin^2(B) - b^2\sin^2(A) \\ &= \left(a^2 + b^2 + 2ab\cos(A + B)\right) + \left(2ab\sin(A)\sin(B) - a^2\sin^2(B) - b^2\sin^2(A)\right) \end{aligned}$$

$$\begin{aligned}
&= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) - \left(a^2\sin^2(B) - 2ab\sin\left(A\right)\sin\left(B\right) - b^2\sin^2(A)\right) \\
&= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) - \left(\left(a\sin(B)\right)^2 - 2\left(a\sin\left(B\right)\right)\left(b\sin\left(A\right)\right) - \left(b\sin(A)\right)^2\right) \\
&= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) - \left(\left(a\sin(B)\right) - \left(b\sin(A)\right)\right)^2 .
\end{aligned}$$

Next, we note that $\sin(A) = \frac{h}{b}$ and $\sin(B) = \frac{h}{a}$, so $b\sin(A) = h$ and $a\sin(B) = h$, so we have

$$\begin{aligned}
c^2 &= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) - \left(\left(a\sin(B)\right) - \left(b\sin(A)\right)\right)^2 . \\
&= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) - \left(h - h\right)^2 \\
&= a^2 + b^2 + 2ab\cos\left(A + B\right) .
\end{aligned}$$

Finally, we know that the angle sum for any triangle is $180° = \pi$, so in particular, $A + B + C = \pi$, which means that $A + B = \pi - C$. Thus, using (4.8),

$$\begin{aligned}
c^2 &= \left(a^2 + b^2 + 2ab\cos\left(A + B\right)\right) \\
&= \left(a^2 + b^2 + 2ab\cos\left(\pi - C\right)\right) \\
&= a^2 + b^2 - 2ab\cos\left(C\right) .
\end{aligned}$$

In the case of an exterior altitude, the derivation is almost the same, but there are a couple of differences.

Referring to Figure 72, let B' be the measure of $\angle HBC$, so $B + B' = \pi$, and so $B' = \pi - B$. Then we see that $\cos(A) = \frac{c+x}{b}$ and $\cos(B') = \frac{x}{a}$, $c = (c + x) - x = b\cos(A) - a\cos(B')$. However, $\cos(B') = \cos(\pi - B) = -\cos(B)$. This means that $c = b\cos(A) - a\cos(B') = b\cos(A) - a(-\cos(B)) = b\cos(A) + a\cos(B)$, and the rest of the proof continues in the same way. ■

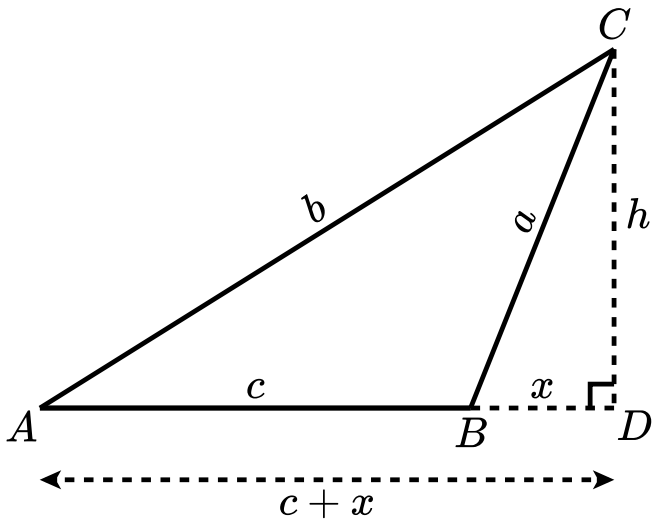


Figure 72 An Triangle with an Exterior Altitude

4.12.4 Solution to the Motivating Problem

4.12.4.1 Understanding the Problem

Looking again at Figure 68, let's label the figure and assign variables.

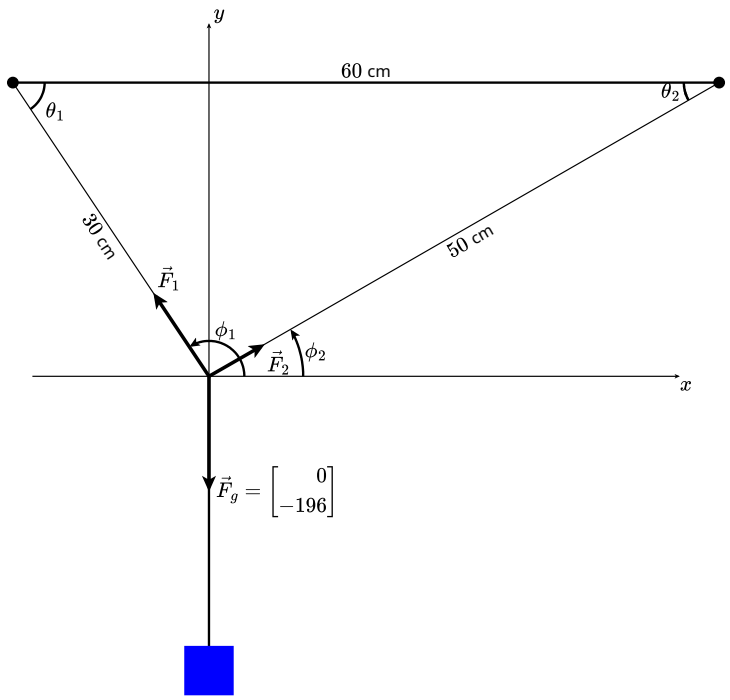


Figure 73 A Weight Suspended Asymmetrically with Axes Imposed

First, let's place our origin at the point where the two wires connect, let $\vec{F}_g$ be the vector representing the force due to gravity, let $\vec{F}_1$ be the force along the wire of length 30cm, and let $\vec{F}_2$ be the force along the wire of length 50cm.

Our goal is to find $\left|\vec{F}_1\right|$ and $\left|\vec{F}_2\right|$, given that $\vec{F}_g + \vec{F}_1 + \vec{F}_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, so let our unknowns be $X = \left|\vec{F}_1\right|$ and $Y = \left|\vec{F}_2\right|$.

Next, let ϕ_1 be the angle from the positive x axis to $\vec{F}_1$ and ϕ_2 be the angle from the positive x axis to $\vec{F}_2$. Then the unit vector $\vec{u}_1$ in the direction of $\vec{F}_1$ and the unit vector $\vec{u}_2$ in the direction of $\vec{F}_2$ are given by

$$\vec{u}_1 = \begin{bmatrix} \cos(\phi_1) \\ \sin(\phi_1) \end{bmatrix} \qquad \vec{u}_2 = \begin{bmatrix} \cos(\phi_2) \\ \sin(\phi_2) \end{bmatrix},$$

and thus $\vec{F}_1 = X\vec{u}_1$ and $\vec{F}_2 = Y\vec{u}_2$.

Let θ_1 be the (acute) angle that $\vec{F}_1$ makes with the ceiling, and let θ_2 be the (acute) angle that $\vec{F}_2$ makes with the ceiling. We note that because our x-axis is parallel to the ceiling, by alternate interior angles $\theta_2 = \phi_2$. Furthermore, the angle between $\vec{F}_1$ and the negative x-axis is $\pi - \phi_1$, and so by alternate interior angles again, $\theta_1 = \pi - \phi_1$, and so $\phi_1 = \pi - \theta_1$.

4.12.4.2 Making a Plan

Because $X = \left|\vec{F}_1\right|$ and $Y = \left|\vec{F}_2\right|$, $\vec{F}_1 = X\vec{u}_1$ and $\vec{F}_2 = Y\vec{u}_2$.

Because our mass is 20kg, acceleration due to gravity is $9.8\frac{\text{m}}{\text{s}^2}$, and because gravity is acting in downward (so in the direction of the unit vector $\vec{u}_g = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$), we have

$$\vec{F}_g = \begin{bmatrix} 0\text{N} \\ -196\text{N} \end{bmatrix} \qquad \text{and} \qquad \vec{F}_1 + \vec{F}_2 + \begin{bmatrix} 0\text{N} \\ -196\text{N} \end{bmatrix} = \begin{bmatrix} 0\text{N} \\ 0\text{N} \end{bmatrix}.$$

Therefore,

$$\begin{aligned} \begin{bmatrix} 0\text{N} \\ 0\text{N} \end{bmatrix} &= \vec{F}_1 + \vec{F}_2 + \begin{bmatrix} 0\text{N} \\ -196\text{N} \end{bmatrix} \\ &= X\vec{u}_1 + Y\vec{u}_2 + \begin{bmatrix} 0\text{N} \\ -196\text{N} \end{bmatrix} \end{aligned}$$

$$
\begin{aligned}
&= X\begin{bmatrix}\cos(\phi_1)\\ \sin(\phi_1)\end{bmatrix} + Y\begin{bmatrix}\cos(\phi_2)\\ \sin(\phi_2)\end{bmatrix} + \begin{bmatrix}0\text{N}\\ -196\text{N}\end{bmatrix}\\
&= \begin{bmatrix}X\cos(\phi_1) + Y\cos(\phi_2) + 0\text{N}\\ X\sin(\phi_1) + Y\sin(\phi_2) + -196\text{N}\end{bmatrix}\\
&= \begin{bmatrix}X\cos(\phi_1) + Y\cos(\phi_2)\\ X\sin(\phi_1) + Y\sin(\phi_2) - 196\text{N}\end{bmatrix}.
\end{aligned}
$$

which means that we have two equations:

$$X\cos(\phi_1) + Y\cos(\phi_2) = 0 \tag{4.9}$$
$$X\sin(\phi_1) + Y\sin(\phi_2) = 196\text{N} \tag{4.10}$$

Now, our plan is to first calculate $\cos(\phi_1)$, $\sin(\phi_2)$, $\cos(\phi_1)$, and $\sin(\phi_2)$, and then substitute those values into (4.9) and (4.10). Then, we can solve those equations for X and Y.

4.12.4.3 Carrying Out the Plan

We will use the law of cosines to learn more about the angles θ_1 and θ_2.

If we let θ_1 take the role of the angle C in the law of cosines, the we have $c = 50\text{cm}$. It doesn't matter which of the other two sides of the triangle is a and which is b, so let's let $a = 30\text{cm}$ and $b = 60\text{cm}$. Then $c^2 = a^2 + b^2 - 2ab\cos(C)$ becomes

$$(50\text{cm})^2 = (30\text{cm})^2 + (60\text{cm})^2 - 2(30\text{cm})(60\text{cm})\cos(\theta_1).$$

To make the next step easier, let's replace $\cos(\theta_1)$ by w. Then we have

$$(2500\text{cm}^2) = (900\text{cm}^2) + (3600\text{cm}^2) - (3600\text{cm}^2)w.$$

Solving this equation for w gives

$$
\begin{aligned}
w &= \frac{(900\text{cm}^2) + (3600\text{cm}^2) - (2500\text{cm}^2)}{(3600\text{cm}^2)}\\
&= \frac{(2000\text{cm}^2)}{(3600\text{cm}^2)}\\
&= \frac{5}{9}.
\end{aligned}
$$

Thus, we now know that $\cos(\theta_1) = \frac{5}{9}$.

If we let θ_2 take the role of the angle C in the law of cosines, the we have $c = 30\text{cm}$. It doesn't matter which of the other two sides of the triangle is a and which is b, so let's let $a = 50\text{cm}$ and $b = 60\text{cm}$. Then $c^2 = a^2 + b^2 - 2ab\cos(C)$ becomes

$$(30\text{cm})^2 = (50\text{cm})^2 + (60\text{cm})^2 - 2(50\text{cm})(60\text{cm})\cos(\theta_2).$$

To make the next step easier, let's again replace the cosine by w. Then we have

$$(900\text{cm}^2) = (2500\text{cm}^2) + (3600\text{cm}^2) - (6000\text{cm}^2)w.$$

Solving this equation for w gives

$$
\begin{aligned}
w &= \frac{(2500\text{cm}^2) + (3600\text{cm}^2) - (900\text{cm}^2)}{(6000\text{cm}^2)}\\
&= \frac{13}{15}.
\end{aligned}
$$

Thus, we now know that $\cos(\theta_2) = \frac{13}{15}$.

Using the Pythagorean identity for sine and cosine, we find

$$
\begin{aligned}
\sin(\theta_1) &= \sqrt{1 - \cos^2(\theta_1)}\\
&= \sqrt{1 - \left(\frac{5}{9}\right)^2}
\end{aligned}
\qquad\qquad
\begin{aligned}
\sin(\theta_2) &= \sqrt{1 - \cos^2(\theta_2)}\\
&= \sqrt{1 - \left(\frac{13}{15}\right)^2}
\end{aligned}
$$

$$= \sqrt{\frac{81}{81} - \frac{25}{81}} \qquad = \sqrt{\frac{225}{225} - \frac{169}{225}}$$
$$= \sqrt{\frac{56}{81}} \qquad = \sqrt{\frac{56}{225}}$$
$$= \frac{2\sqrt{14}}{9} \qquad = \frac{2\sqrt{14}}{15}$$

Realizing that $\cos(\pi - \theta) = -\cos(\theta)$ and $\sin(\pi - \theta) = \sin(\theta)$, we have

$$\cos(\phi_1) = -\cos(\theta_1) \qquad \cos(\phi_2) = \cos(\theta_2)$$
$$= -\frac{5}{9} \qquad = \frac{13}{15}$$
$$\sin(\phi_1) = \frac{2\sqrt{14}}{9} \qquad \sin(\phi_2) = \frac{2\sqrt{14}}{15}.$$

Substituting these values into (4.9) and (4.10) gives us

$$\frac{-5}{9}X + \frac{13}{15}Y = 0 \tag{4.11}$$
$$\frac{2\sqrt{14}}{9}X + \frac{2\sqrt{14}}{15}Y = 196\text{N}. \tag{4.12}$$

There are several ways to solve (4.11) and (4.12), including solving one equation for one of the variables and substituting it into the other, or using a technique called elimination. However, in this particular case, because the right hand side of the first equation is 0, we can find any values of X and Y that make the first equation true, and if we multiply both of those values by any constant, we will still have a solution. We can then use the solution with an arbitrary constant to solve the second equation.

For example, it is pretty quickly clear that if we set $X = \frac{13}{15}$ and $Y = \frac{5}{9}$we get

$$\begin{aligned}\frac{-5}{9}X + \frac{13}{15}Y &= \frac{-5}{9}\frac{13}{15} + \frac{13}{15}\frac{5}{9} \\ &= \frac{-65}{135} + \frac{65}{135} \\ &= 0.\end{aligned}$$

On the other hand, if we set $X = 3\frac{13}{15} = \frac{39}{15}$ and $Y = 3\frac{5}{9} = \frac{15}{9}$we get

$$\begin{aligned}\frac{-5}{9}X + \frac{13}{15}Y &= \frac{-5}{9}\frac{39}{15} + \frac{13}{15}\frac{15}{9} \\ &= \frac{-195}{135} + \frac{195}{135} \\ &= 0.\end{aligned}$$

In general, if we set $X = k\frac{13}{15}$ and $Y = k\frac{5}{9}$, we get

$$\begin{aligned}\frac{-5}{9}X + \frac{13}{15}Y &= \frac{-5}{9}k\frac{13}{15} + \frac{13}{15}k\frac{5}{9} \\ &= -k\frac{65}{135} + k\frac{65}{135} \\ &= k\left(-\frac{65}{135} + \frac{65}{135}\right) \\ &= k \cdot 0 \\ &= 0.\end{aligned}$$

Substituting $X = k\frac{13}{15}$ and $Y = k\frac{5}{9}$ into (4.12), we have

$$196\text{N} = \frac{2\sqrt{14}}{9}X + \frac{2\sqrt{14}}{15}Y$$

$$\begin{aligned}
&= k\frac{13}{15}\frac{2\sqrt{14}}{9} + k\frac{5}{9}\frac{2\sqrt{14}}{15} \\
&= k\frac{26\sqrt{14}}{135} + k\frac{10\sqrt{14}}{135} \\
&= k\left(\frac{26\sqrt{14}}{135} + \frac{10\sqrt{14}}{135}\right) \\
&= k\left(\frac{36\sqrt{14}}{135}\right) \\
&= k\left(\frac{4\sqrt{14}}{15}\right).
\end{aligned}$$

Solving $k\left(\frac{4\sqrt{14}}{15}\right) = 196\text{N}$ for k gives

$$\begin{aligned}
k &= 196\text{N} \cdot \frac{15}{4\sqrt{14}} \\
&= 196\text{N} \cdot \frac{15\sqrt{14}}{4 \cdot 14} \\
&= \frac{105\sqrt{14}}{2}\text{N}.
\end{aligned}$$

Thus, we have

$$\begin{aligned}
\left|\vec{F}_1\right| &= X \\
&= \frac{13}{15}k \\
&= \frac{13}{15}\frac{105\sqrt{14}}{2}\text{N} \\
&= \frac{91\sqrt{14}}{2}\text{N} \\
&\approx 170.2454\text{N}
\end{aligned}$$

$$\begin{aligned}
\left|\vec{F}_2\right| &= Y \\
&= \frac{5}{9}k \\
&= \frac{5}{9}\frac{105\sqrt{14}}{2}\text{N} \\
&= \frac{175\sqrt{14}}{6}\text{N} \\
&\approx 109.1317\text{N}.
\end{aligned}$$

Interpreting these results in the context of the problem, we have that the force along the short wire is approximately 170.2454N and the force along the long wire is approximately 109.1317N.

4.12.4.4 Looking Back

Looking back on this problem, we see that the case where the wires are the same length is a special case where $\theta_1 = \theta_2 = \theta$. Then our first equation becomes $0 = -X\cos(\theta_1) + Y\cos(\theta_2) = (-X + Y)\cos(\theta)$, so we have to have $X = Y$, and then our second equation will be $X\sin(\theta) + X\sin(\theta) = 2X\sin(\theta) = W$ where W is the magnitude of the force due to gravity, which yields $X = \frac{W}{2\sin(\theta)}$. Also, from the co-function identities, $\cos(\pi/2 - \theta) = \sin(\theta)$, so if ϕ is the complementary angle to θ then we have $X = \frac{W}{2\cos(\phi)}$.

Another thing we might notice is that another way to obtain the values for $\cos(\theta_1)$, $\sin(\theta_1)$, $\cos(\theta_2)$, and $\sin(\theta_2)$ would be for the angles to be given. In this case, we would not need to use the law of cosines.

4.12.4.5 Using the Law of Sines

Example 74

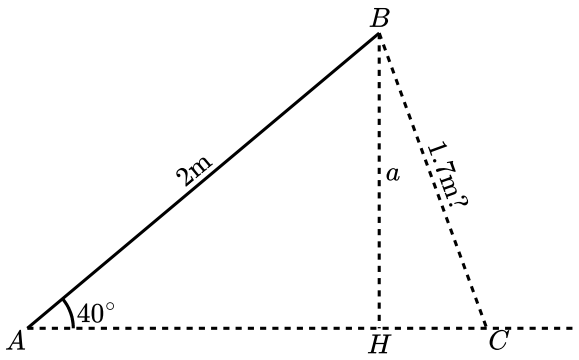


Figure 75 Is This Triangle Possible?

Figure 75 shows a triangle with one side length and one angle known. The side, side AB in the figure, is 2m long, and one of the angles adjacent to that side is 40°, at the point labeled A in the figure.

We want to know:

- Is it possible for the side opposite the 40° angle, that is, side BC, to be 1.7m long? If it is possible,
- what will the length of the third side be? And
- what will the measures of the other two angles be?

To answer the first question, we can find out what the shortest possible length is for the side opposite the 40° angle and see if it is less than 1.7m.

The shortest possible side opposite the 40° angle would be a segment perpendicular to the dashed line at the bottom. Let's say that the length of that side is a and the point where it intersects the horizontal side is H. Then because we have a right triangle, we would have

$$\sin(40^\circ) = \frac{a}{2\text{m}}.$$

Solving for m, we have

$$\begin{aligned} a &= 2\text{m} \cdot \sin(40^\circ) \\ &= 2\text{m} \cdot \sin\left(40^\circ \frac{\pi}{180^\circ}\right) \\ &= 2\text{m} \cdot \sin\left(\frac{2\pi}{9}\right) \\ &\approx 2\text{m} \cdot \sin(0.698131700798) \\ &\approx 1.28557521937\text{m}. \end{aligned}$$

Since $a > 1.7\text{m}$, it is possible to have the side opposite the $40°$ angle be 1.7m in length. However, the picture is a little misleading, because the point where the 1.7m side touches the third side could be on either side of the perpendicular segment, so we will need to make sure we consider both possibilities.

From the Law of Sines, if we label the angle at C as θ, then

$$\frac{\sin(\theta)}{2\text{m}} = \frac{\sin(40°)}{1.7\text{m}},$$

so

$$\begin{aligned}\sin(\theta) &= \frac{2\text{m}}{1.7\text{m}}\sin(40°)\\ &\approx 0.756220717278282.\end{aligned}$$

This is where we have to be careful. We know we are dealing with triangles, and so we know that $0 \le \theta \le \pi$, but there are two angles in that range whose sine is $\approx 0.756220717278282$, one in the first quadrant and one in the second quadrant.

If the side is similar to what is drawn in the picture, with the corner of the triangle on the far side of H from A, then the angle at that point will be acute, and so we have

$$\begin{aligned}\theta &= \arcsin\left(\frac{2}{1.7}\sin(40°)\right)\\ &\approx 0.857517750568782\\ &\approx 49.1321479651432°.\end{aligned}$$

In this case, the third angle will be $180° - 40° - 49.1321479651432° \approx 90.8678520348568°$.

Then, again using law of sines, if we say the length of AC is c, then

$$\begin{aligned}c &= \sin(90.8678520348568°)\frac{2\text{m}}{\sin(40°)}\\ &\approx 2.64442712406576\text{m}.\end{aligned}$$

On the other hand, if the corner of the triangle on the near side of H from A, then the angle at that point will be obtuse, and so we will have

$$\begin{aligned}\theta &= \pi - \arcsin\left(\frac{2}{1.7}\sin(40°)\right)\\ &\approx 2.284074903\\ &\approx 130.867852°.\end{aligned}$$

In this case, the third angle will be $180° - 40° - 130.867852° \approx 9.132147965°$.

Then, again using law of sines, if we say the length of AC is c, then

$$\begin{aligned}c &= \sin(9.132147965°)\frac{2\text{m}}{\sin(40°)}\\ &\approx 0.4197506484\text{m}.\end{aligned}$$

▲

Chapter 5

Progress Checks

5.1 Progress Check 1

1. The London Eye is a "cantilevered observation wheel" (not quite the same as a Ferris wheel, but close) that stands on the South Bank of the Thames in London, England. Here are some facts about the London Eye, and some questions for you to answer.



Figure 76 A Clipart image of the London Eye

- The London Eye is designed so that it turns slowly enough that riders can get on and off without the wheel ever having to stop.
- The London Eye has 32 passenger "pods", numbered 1–12, 14–33, skipping "unlucky" 13.
- The London Eye has a diameter of 135 meters.
- The London Eye rotates so that the speed with which the pods move around the Eye is 26cm/s.

(a) How long does it take the wheel to make one complete revolution? Give your answer in seconds and in terms that are meaningful for the average person (for example, the average person would not say that it takes them 3600 seconds to drive to Kalamazoo).

Solution. The circumference is 135π meters, and the speed the wheel travels is $26\text{cm}/s = 0.26\text{m/s}$, so the time it takes for one revolution is

$$\begin{aligned}\frac{135\pi\text{m}}{0.26\text{m/s}} &\approx 1631.21157\text{s} \\ &\approx 27.187\text{min}\end{aligned}$$

(b) If you miss one pod, how long do you have to wait for the next one?

Solution. Because there are 32 pods, the distance between one pod and the next is $\frac{1}{32}$ times the circumference. That means that it will take $\frac{1}{32}$ of the period to get from one pod to the next. Thus, the time between pods is

$$\frac{1631.21157\text{s}}{32} \approx 50.98\text{s}$$

(c) Assume that you get on the Eye at time $t = 0$s.

(i) Find a sinusoidal model of the form $h(t) = a\sin(\omega(t-\phi)) + d$, that gives your height, in meters, as a function of time, in seconds.

Solution. We know that the London Eye is 135m tall, and we can assume the bottom is at ground level, so max = 135 and min = 0, which gives us

$$\begin{aligned} a &= \frac{\max - \min}{2} \\ &= \frac{135 - 0}{2} \\ &= 67.5 \end{aligned} \qquad \begin{aligned} d &= \frac{\max + \min}{2} \\ &= \frac{135 + 0}{2} \\ d &= 67.5 \end{aligned}$$

We also know that the period is $P = 1631.21157$s, so

$$\begin{aligned} \omega &= \frac{2\pi}{P} \\ &= \frac{2\pi}{1631.21157} \\ &\approx 0.00385185185 \end{aligned}$$

Finally, we want the sinusoid to start at its minimum (0m), but sine starts at the midline (increasing). To move the minimum to $t = 0$, we need to either shift left by 3/4 of the period or shift right by 1/4 of the period. Thus, we can choose either $\phi = P/4$ or $\phi = -3P/4$. We will use $\phi = P/4 \approx 407.8028925$. So, our sinusoid is

$$h(t) = 67.5\sin\big(0.00385185185(t - 407.8028925)\big) + 67.5$$

(ii) How high will you be after 5 minutes? After 10 minutes? After 15 minutes?

Solution. First, we note that 5min = 300s, 10min = 600s, and 15min = 900s. So, we can evaluate our function, $h(t)$ at each of these times to get the height:

$$h(300\text{s}) = 40.269808\text{m} \qquad h(600\text{s}) = 113.03012\text{m} \qquad h(900\text{s}) = 131.46484\text{m}$$

2. You can see a plane in the air coming toward you. You know (somehow) that the plane is flying at a cruising altitude of 5km = 5000m. Figure 77 depicts the plane at two different times along with angles of elevation for each position.

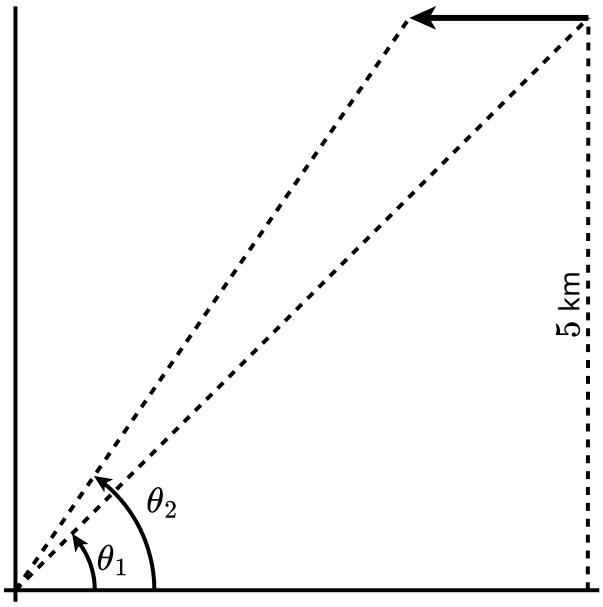


Figure 77 A Plane at Height 5km at Two Different Times

(a) When you first notice the plane, it has an angle of elevation of $\theta_1 = 0.785$ radians. How far away, horizontally, is the plane from you at that time?

Solution. Let x_1 be the horizontal distance for the first measurement. Then from the unit circle and similar triangles, we know that

$$\frac{\cos(\theta_1)}{x_1} = \frac{\sin(\theta_1)}{5000\text{m}}$$

so,

$$\begin{aligned} x_1 &= \frac{\cos(\theta_1)}{\sin(\theta_1)} 5000\text{m} \\ &= \frac{\cos(0.785)}{\sin(0.785)} 5000\text{m} \\ &\approx 5003.98\text{m} \end{aligned}$$

(b) When you look at the plane five seconds later, it has an angle of elevation of $\theta_2 = 0.968$ radians. How far away, horizontally, is the plane from you at that time?

Solution. Let x_2 be the horizontal distance for the second measurement. Then from the unit circle and similar triangles, we know that

$$\frac{\cos(\theta_2)}{x_2} = \frac{\sin(\theta_2)}{5000\text{m}}$$

so,

$$\begin{aligned} x_2 &= \frac{\cos(\theta_2)}{\sin(\theta_2)} 5000\text{m} \\ &= \frac{\cos(0.968)}{\sin(0.968)} 5000\text{m} \end{aligned}$$

$$\approx 3441.25\text{m}$$

(c) Based on your previous calculations, how fast, in meters per second, is the plane flying?

Solution. The distance travelled was $5003.98\text{m} - 3441.25\text{m} = 1562.73\text{m}$, and the elapsed time was 5s, so the speed was $1562.73\text{m}/5\text{s} = 312.547\text{m/s}$.

3. Figure 78 shows a pendulum along with vectors representing forces acting on the pendulum bob. The vector $\vec{F}_g = \langle 0, -32 \rangle$ represents the force due to gravity of 32 pounds in the downward direction, while the vectors $\vec{F}_R$ and $\vec{F}_T$ represent the radial and tangential components of $\vec{F}_g$, respectively. That is, $\vec{F}_R$ represents the part of the force from gravity that pulls against the pendulum string (and is balanced by an equal and opposite force that the string exerts) and $\vec{F}_T$ is the part of the force from gravity that tends to move the pendulum along its path, and $\vec{F}_T + \vec{F}_R = \vec{F}_g$.

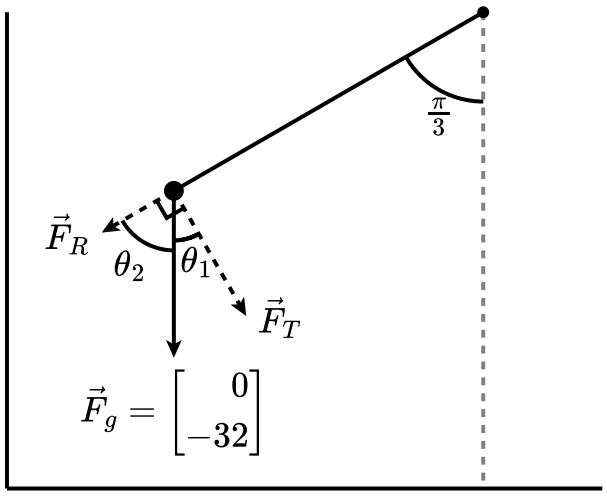


Figure 78 Force Vectors Acting on a Pendulum

(a) Use geometry and the fact that $\vec{F}_R$ and $\vec{F}_T$ are perpendicular to find angles θ_1 and θ_2.

Solution. Because $\vec{F}_R$ is pointing in the same direction as the pendulum string, we know that $\theta_2 = \pi/3$, and because $\vec{F}_R$ and $\vec{F}_T$ are perpendicular to each other, $\theta_1 + \theta_2 = \pi/2$, so we know that $\theta_2 = \pi/6$.

(b) Draw a copy of $\vec{F}_T$ with its tail at the head of $\vec{F}_R$ to form a right triangle and use trigonometry to find the lengths of $\vec{F}_T$ and $\vec{F}_R$. Call these lengths ℓ_T and ℓ_R.

Solution. We can see that $\frac{\ell_R}{\cos(\theta_2)} = \frac{32}{1}$, so $\ell_R = 32\cos(\pi/3) = 32 \cdot \frac{1}{2} = 16$, and similarly, $\frac{\ell_T}{\sin(\theta_2)} = \frac{32}{1}$, so $\ell_T = 32\sin(\pi/3) = 32\sqrt{3}/2 = 16\sqrt{3} \approx 27.7128$.

(c) Use your knowledge of the unit circle to find a vector of length 1 pointing in the direction of $\vec{F}_R$. Call this vector $\vec{u}_R$.

Hint. You can draw the unit circle with its center at the pendulum bob.

Solution. If we put the origin at the pendulum bob, then the angle from the positive x-axis to the $\vec{F}_R$ is $\pi/2 + \pi/3 = 5\pi/6$, but the angle is clockwise, so the point on the unit circle at that angle would be

$$\vec{u}_R = \begin{bmatrix} \cos\left(\frac{-5\pi}{6}\right) \\ \sin\left(\frac{-5\pi}{6}\right) \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-\sqrt{3}}{2} \\ \frac{-1}{2} \end{bmatrix}$$

(d) Use geometry and your knowledge of the unit circle to find a vector of length 1 pointing in the direction of $\vec{F}_T$. Call this vector $\vec{u}_T$.

Solution. If we again put the origin at the pendulum bob, then the angle from the positive x-axis to the $\vec{F}_T$ is $-\pi/2 + \pi/6 = -\pi/3$, so the point on the unit circle at that angle would be

$$\vec{u}_T = \begin{bmatrix} \cos\left(\frac{-\pi}{3}\right) \\ \sin\left(\frac{-\pi}{3}\right) \end{bmatrix}$$
$$= \begin{bmatrix} \frac{1}{2} \\ \frac{-\sqrt{3}}{2} \end{bmatrix}$$

(e) Using the lengths ℓ_R and ℓ_T that you found in Part (b) and the unit vectors $\vec{u}_R$ and $\vec{u}_T$ that you found in Part (c) and Part (d), find $\vec{F}_R$ and $\vec{F}_T$. Be sure to verify that $\vec{F}_T + \vec{F}_R = \vec{F}_g$!

Solution.

$$\vec{F}_R = \ell_R \vec{u}_R$$
$$= 16 \begin{bmatrix} \frac{-\sqrt{3}}{2} \\ \frac{-1}{2} \end{bmatrix}$$
$$= \begin{bmatrix} -8\sqrt{3} \\ -8 \end{bmatrix}$$

$$\vec{F}_T = \ell_R \vec{u}_R$$
$$= 16\sqrt{3} \begin{bmatrix} \frac{1}{2} \\ \frac{-\sqrt{3}}{2} \end{bmatrix}$$
$$= \begin{bmatrix} 8\sqrt{3} \\ -24 \end{bmatrix}$$

We then verify that $\vec{F}_R + \vec{F}_T = \vec{F}_g$:

$$\vec{F}_R + \vec{F}_T = \begin{bmatrix} -8\sqrt{3} \\ -8 \end{bmatrix} + \begin{bmatrix} -8\sqrt{3} \\ -24 \end{bmatrix}$$
$$= \begin{bmatrix} -8\sqrt{3} + 8\sqrt{3} \\ -8 - 24 \end{bmatrix}$$
$$= \begin{bmatrix} 0 \\ -32 \end{bmatrix}$$
$$= \vec{F}_g$$

Chapter 6

Class Handouts

6.1 An Introduction to Problem Solving

In his book, *How To Solve It*, mathematician George Polya formulated four steps that apply to any type of problem solving. These steps are *Understand the Problem*, *Make a Plan*, *Carry Out the Plan*, and *Look Back*.

Sometimes it feels like math classes give almost all their attention to carrying out a plan with little attention paid to the other three steps. Often, instructors *give* students the plan rather than asking them to work on *understanding* and *formulating* a plan for themselves, and almost no time is given to *looking back*.

This handout gives more information on each of the four steps, and in future problems we will attempt to apply all four.

Understanding the Problem. ***First.*** you have to *understand* the problem.

What is the unknown? What are the data? What is the condition? Is it possible to satisfy the condition? Is the condition sufficient to determine the unknown? Or is it insufficient? Or redundant? Or contradictory?

Draw a figure. Introduce suitable notation. Separate the various parts of the condition. Can you write them down?

Devising a Plan. ***Second.*** Find the connection between the data and the unknown.

You may be obliged to consider auxiliary problems if an immediate connection cannot be found. You should obtain eventually a *plan* of the solution.

Have you seen it before? Or have you seen the same problem in a slightly different form?

Do you know a related problem? Do you know a theorem that could be useful? *Look at the unknown!* And try to think of a familiar problem having the same or a similar unknown.

Here is a problem related to yours and solved before. Could you use it? Could you use its result? Could you use its method? Should you introduce some auxiliary element in order to make its use possible?

Could you restate the problem? Could you restate it still differently? Go back to definitions.

If you cannot solve the proposed problem try to solve first some related problem. Could you imagine a more accessible related problem? A more general problem? A more special problem? An analogous problem? Could you solve a part of the problem? Keep only a part of the condition, drop the other part; how far is the unknown then determined, how can it vary? Could you derive something useful from the data? Could you think of other data appropriate to determine the unknown? Could you change the unknown or the data, or both if necessary, so that the new unknown and the new data are nearer to each other?

Did you use all the data? Did you use the whole condition? Have you taken into account all essential notions involved in the problem?

Carrying Out the Plan. ***Third.*** *Carry out* your plan.

Carrying out your plan of the solution, *check each step.* Can you see clearly that the step is correct? Can you prove that it is correct?

Looking Back. ***Fourth.*** *Examine* the solution obtained.

Can you *check the result?* Can you check the argument?

Can you derive the result differently? Can you see it at a glance?

Can you use the result, or the method, for some other problem?